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Assessing the Effectiveness of Large Language Model Support for Generating GDPR ROPA Documentation

Master Thesis at Ulm University

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Abstract

Article 30 of the General Data Protection Regulation (GDPR) requires every data controller to maintain a Record of Processing Activities (ROPA), a demanding task that presupposes legal expertise most small organizations lack. LLM-assisted tools have been proposed as a practical remedy, but evaluations have largely relied on illustrative use cases rather than controlled studies. This thesis evaluates ROPAgen, an LLM-powered ROPA tool offering three modes that differ in LLM involvement: Form (checkbox-driven), Ask (user-driven dialogue), and Chat (free-form conversation). In a within-subjects study, $N = 73$ novices completed tasks in all three modes under Latin Square counterbalancing, yielding usability, cognitive load, confidence, preferences, and completion times alongside 113 documents. Each document was scored against an expert-written reference (BLEU, ROUGE, METEOR, BERTScore, SBERT) and rated by an LLM-as-a-judge on completeness, faithfulness, legal correctness, and a holistic score. Form, the least LLM-involved mode, was rated most usable and most preferred. On document quality, Form led on lexical recall while Chat led on semantic fidelity and received the highest holistic verdict; Ask trailed on every reference-based measure. Perceived confidence tracked the judge's holistic scores but diverged on legal correctness, where Chat ranked highest in confidence yet lowest in judge-assessed correctness. The thesis concludes that novices need scaffolding, a collaborator, and a safety net, and no single mode provides all three. The resulting design assigns enumerable Article 30 fields to a structured interface and scenario-specific content to conversational extraction, treating the LLM as co-author rather than teacher.

Contents

1 Introduction

1.1 Context

The General Data Protection Regulation (GDPR) requires organizations to maintain a Record of Processing Activities (ROPA), a formal document under Article 30 GDPR that records how personal data is collected, processed, stored, and shared. The content of a ROPA is rigidly specified (controller details, purposes, lawful bases, categories of data and recipients, retention periods, and technical and organizational measures), yet its substance is project-specific prose that has to be written from a blank page and kept current as processing activities change. Maintaining a ROPA is therefore a sustained cost rather than a one-off task; it requires legal and data-protection expertise that small and medium-sized organizations rarely have, and falls disproportionately on those least equipped to bear it [30].

ROPAgen, the tool evaluated in this thesis, is a web-based application developed at the Institute of Databases and Information Systems (DBIS) at the University of Ulm to address exactly this gap. It uses Large Language Models (LLMs) to assist novice users in drafting GDPR-compliant ROPA documentation, offering three interaction modes, each with a different level of LLM support:

Form mode, in which the user fills structured fields directly, with LLM suggestions available on demand. **Ask** mode, in which the user fills a free-form text field per section while a chatbot explains the section and answers questions. **Chat** mode, in which the user describes their requirements conversationally, and the LLM autonomously drafts and populates the ROPA. The three modes are described in detail in Section 3.1; by comparing them, the thesis evaluates how different levels of LLM support affect user performance and trust.

Although the application provides a functional framework for assisted ROPA generation, its effectiveness, usability, and impact on documentation quality have not yet been systematically evaluated. To ensure that such a system can be confidently integrated into practical compliance workflows, both qualitative and quantitative assessments are needed. These evaluations determine whether users can efficiently and correctly create ROPA documents, how the varying levels of LLM support influence performance and trust, and whether the system produces outputs aligned with GDPR requirements.

1.2 Research Problem and Research Questions

The goal of this thesis is to conduct a comprehensive evaluation of the existing LLM-supported ROPA generation application. The evaluation asks how the three interaction modes shape the experience of novice users, how they shape the documents those users produce, and whether the two are aligned. RQ1 is broken into three sub-questions tied to the user-study instruments; RQ2 is broken into two sub-questions tied to the document-quality streams. RQ3 is a single cross-stream question that reads the two evaluation streams against each other; it is reserved for the Discussion chapter, where the cross-signal interpretation belongs.

RQ1 (User experience). *How do varying degrees of LLM support influence novice users during ROPA document creation?*

RQ1.1 How do novice users perceive the *usability* of the three interaction modes (Form, Ask, Chat)?

RQ1.2 How does *cognitive load* differ across the three modes for users with no prior GDPR knowledge?

RQ1.3 How do novice users subjectively evaluate each mode: in their perception of the LLM's support, in their perceived confidence in the AI's output (a two-item composite covering trust in the AI's suggestions and confidence in the legal-basis identification), and in the mode they ultimately prefer?

RQ2 (Document quality). *How do varying degrees of LLM support influence the quality of ROPA documents produced by novice users?*

RQ2.1 How does document quality differ across the three modes when measured by reference-based NLP similarity to an expert-authored ROPA (BLEU, ROUGE, METEOR, BERTScore, SBERT)?

RQ2.2 As a supplementary, reference-free signal, what additional GDPR-specific quality properties (completeness, scenario faithfulness, legal correctness, hallucination) does an LLM-as-a-judge surface across the three modes?

RQ3 (Confidence–quality alignment). *Is there a discrepancy between user confidence (RQ1.3) and the document quality actually achieved (RQ2.1, RQ2.2)?*

1.3 Thesis Structure

The remainder of this thesis is organized into five further chapters and three appendices.

Chapter 2 (Background) defines the GDPR and the Article 30 ROPA obligation and locates this thesis in the related work on LLM-supported legal-document generation. **Chapter 3** (Methodology) introduces ROPAgen and its three interaction modes, then describes the user study: design, instruments, sample sizes, and the analytical approach through which the tool is evaluated. **Chapter 4** (User Study) reports the empirical findings from the four user-study instruments, descriptively first and then inferentially. **Chapter 5** (Document Quality) reports the document-quality analyses: the reference-based NLP metrics at document and chapter levels and the supplementary LLM-as-a-Judge layer. **Chapter 6** (Discussion) answers RQ1, RQ2, and RQ3; it interprets the confidence–quality gap that emerges across the user-study and document-quality streams, draws the design implications for novice-targeting LLM-assisted tooling, and names the study's limitations, closing with **Chapter 7** (Conclusion) summarizing the contributions and their implications for the design of LLM-assisted document tools. **Appendices A–C** contain the ROPAgen backend prompts and judge configuration, the verbatim study materials, and the full inferential statistics referenced from Chapters 4 and 5.

2 Background

This chapter establishes the legal and methodological context for the thesis. The first two sections introduce the GDPR and the Article 30 obligation to maintain a Record of Processing Activities (ROPA), explaining why a ROPA is both legally required and practically demanding to produce. The third section surveys related work along four lenses (GDPR-specific compliance automation, structured legal-document generation, evaluation methodologies, and user-facing legal systems), and the chapter closes by identifying the gap that this thesis addresses.

2.1 General Data Protection Regulation (GDPR)

The General Data Protection Regulation (GDPR), in force since 25 May 2018, is the European Union's framework for personal-data protection [10]. Its scope is extraterritorial: any organization that processes data on individuals within the EU is subject to it, regardless of its own location. The regulation rests on privacy as a fundamental right under the 1950 European Convention on Human Rights and modernizes the 1995 Data Protection Directive in response to the growth of cloud services, social media, and large-scale data breaches. [10, 9]

Under the GDPR, personal data is any information relating to an identifiable individual, ranging from names and email addresses to biometric data, location information, and web cookies. The regulation distinguishes between the data subject (the individual), the data controller (the entity determining the purposes and means of processing), and the data processor (a third party handling data on behalf of a controller). All processing must adhere to seven principles: lawfulness, fairness and transparency; purpose limitation; data minimization; accuracy; storage limitation; integrity and confidentiality; and accountability. Accountability shifts the burden of proof to the organization, which must produce documented evidence of compliance [10]. The regulation mandates that processing is only legal if justified by one of six specific bases, most notably through consent, which must be freely given, specific, and unambiguous. Other justifications include contractual necessity, legal obligations, or a legitimate interest that does not override the subject's rights. Furthermore, the GDPR introduces the concept of data protection by design and by default, requiring that privacy considerations be integrated into the initial development phases of any product or service. Large-scale processors or public authorities are often required to appoint a Data Protection Officer (DPO) to oversee these internal processes and serve as a liaison with regulatory bodies [10].

The GDPR grants individuals a defined set of privacy rights, including the rights to be informed, of access, of rectification, and of erasure (often termed the "right to be forgotten"). Data subjects also hold the right to data portability and to protections against automated decision-making and profiling. Compliance is enforced through substantial administrative fines under Article 83 GDPR, which has placed data protection among the core legal and operational concerns of organizations that handle personal data [10].

2.2 Records of Processing Activities under Article 30 GDPR

Article 30 of the General Data Protection Regulation (GDPR) establishes the obligation for controllers and, where applicable, processors to maintain a Record of Processing Activities (ROPA). The ROPA functions as a core accountability instrument within the GDPR framework, operationalizing the principle that organizations must not only comply with data protection requirements but also be able to demonstrate such compliance to supervisory authorities. Its primary purpose is to provide a structured, transparent, and up-to-date overview of how personal data are processed within an organization, thereby enabling effective oversight, risk assessment, and regulatory control [9, 30].

From a regulatory perspective, the ROPA serves several interrelated objectives. First, it enhances organizational awareness of data processing operations by requiring a systematic mapping of purposes, data flows, and safeguards. Second, it supports data protection by design and by default, as organizations must explicitly consider data categories, retention periods, and security measures. Third, it enables supervisory authorities to efficiently assess compliance, particularly in the context of audits, investigations, or following personal data breaches. As such, the ROPA is not merely a formal record, but a living compliance document that reflects actual processing practices [9].

Article 30 distinguishes between the obligations of data controllers and those of data processors. Controllers are required to document processing activities carried out under their responsibility, while processors must maintain records of processing performed on behalf of controllers. These records must be maintained in writing, including in electronic form, and must be made available to the competent supervisory authority upon request. Although the GDPR provides a limited exemption for organizations employing fewer than 250 persons, this exemption does not apply where processing is non-occasional, involves special categories of personal data, concerns criminal convictions, or is likely to result in risks to the rights and freedoms of data subjects. Consequently, in practice, many small and medium-sized organizations remain subject to the ROPA requirement [9].

The content of a ROPA is explicitly defined by Article 30 GDPR and is intended to be concise, structured, and easily comprehensible. Excessive or irrelevant information should be avoided, as clarity and usability are essential for both internal governance and external supervision. At a minimum, a ROPA should include the following elements.

- Name and contact details of the data controller (and, where applicable, joint controllers, representatives, and the data protection officer).
- Purposes of the processing activities.
- Categories of data subjects.
- Categories of personal data processed.
- Categories of recipients, including recipients in third countries or international organizations.

- Details of transfers of personal data to third countries or international organizations, including safeguards where applicable.
- Time limits for the erasure of different categories of personal data.
- A general description of the technical and organizational security measures implemented pursuant to Article 32 GDPR.

Where an organization acts as a data processor, the ROPA must additionally reflect the processing carried out on behalf of each controller, including the identity of those controllers, the categories of processing involved, relevant international data transfers, and applicable security measures. In both cases, the ROPA should be sufficiently detailed to provide a faithful representation of processing activities, while remaining accessible to non-technical stakeholders [9].

The GDPR does not prescribe a fixed update cycle for ROPAs; instead, it requires that records remain accurate and up to date. Consequently, any substantive change in processing purposes, data categories, recipients, or safeguards should trigger a review and update of the ROPA. Many organizations delegate the responsibility for maintaining the ROPA to a data protection officer, where one is appointed, to ensure consistency, avoid duplication of effort, and reduce the risk of omissions [9, 25].

Finally, Article 30 emphasizes the external relevance of ROPA. Organizations must be prepared to present their records to supervisory authorities upon request, and such requests may be accompanied by demands for supplementary documentation, including privacy notices, consent records, or contractual arrangements. In this sense, ROPA operates as a foundational compliance artifact, anchoring broader data protection governance and demonstrating an organization's commitment to accountability under the GDPR [9, 25].

2.3 Related Work

Prior work relevant to this thesis can be grouped along four lenses that mirror the design choices of the present study. The first concerns *domain*: research that targets GDPR compliance specifically and the automated production of regulatory artifacts such as ROPA. The second concerns *task*: architectures for generating structured, template-bound legal documents in adjacent jurisdictions. The third concerns *evaluation*: how the legal-AI literature measures the quality of generated documents, increasingly through a combination of surface-overlap metrics, semantic-similarity metrics, and either expert or LLM-based judgment. The fourth concerns the *user*: how legal AI is presented to non-expert audiences and whether it is accessible to people without prior legal training. Reviewing the literature through these four lenses makes clear what has been studied, what has not, and where the present thesis sits.

2.3.1 GDPR Compliance Automation

The closest prior work to this thesis targets the automation of GDPR compliance documentation directly. von Schwerin et al. [30] address the generation of data categories (e.g., contact information, payment data) for Records of Processing Activities, observing that existing legal benchmarks such as LawBench or LegalBench primarily evaluate static knowledge recall rather than the ability to produce documents that adhere to the precise formatting and terminology required by the GDPR. Their study compares four open-source models (Qwen2-7B, Meta-Llama-3-8B, zephyr-7b-beta, and orca_mini_v7_7b) against GPT-4 across four experiments: basic performance, model-specific prompting, few-shot prompting, and Retrieval-Augmented Generation (RAG). Outputs are assessed using a weighted Model Utility Score covering latency, structure and style, grammar, validity, and contextual understanding. The finding that open-source models, particularly Qwen2-7B, can match or exceed GPT-4 on structured ROPA outputs validates the use of locally deployable models for privacy-sensitive compliance work, and is the empirical foundation on which ROPAgen builds. The study also documents persistent risks (ambiguous legal language, jurisdictional variation, hallucination, and the resulting need for a "human-in-the-loop") that motivate examining how real users behave when asked to produce ROPA documents with such tools.

Abualhaija et al. [1] approach the GDPR from a complementary angle, extracting technical software requirements for specific data subject rights (right of access, right to portability) from the regulation and supplementary legal sources. Their XTRAREG pipeline uses GPT-3.5 and GPT-4o together with RAG and prompting strategies (Zero-Shot Learning, Chain-of-Thought) to generate requirements, validated against a manual baseline of 108 expert-extracted reference requirements. The work confirms two patterns that recur throughout this thesis: RAG meaningfully improves both lexical and semantic alignment with expert references, and high BERTScores can coexist with low BLEU when the model captures the meaning of a requirement without matching its wording. The reported failure modes are incomplete coverage, perspective drift (writing as the data subject rather than as the system), and citation of non-binding recitals, all of which reinforce the case for evaluating LLM-generated GDPR artifacts against a fixed reference rather than trusting model output at face value.

Together, these two works establish GDPR documentation as a viable LLM target while leaving the user-side question unanswered: how do non-experts fare when asked to produce such documents themselves? ROPAgen, and this thesis, extends this line of work by testing the user-facing realization of an open-source GDPR pipeline rather than the model performance in isolation.

2.3.2 Structured Legal-Document Generation

A second body of work targets the broader problem of producing structurally coherent legal documents from limited user input, exploring the architectural strategies (retrieval, wrappers, fine-tuning) that constrain LLMs to follow rigid legal templates. Nigam et al. [24] address private legal-document drafting in the Indian system through their Model-Agnostic Wrapper (MAW), a two-phase framework in which the model first generates a structured list of section titles (which the user can manually refine) and then iteratively fills each section, using a vector database (ChromaDB) to summarize prior sections and maintain long-form coherence. The system is exposed through a Human-in-the-Loop interface and evaluated on the Vidhik-Dastaavej dataset of 489 anonymized documents using ROUGE, BLEU, METEOR, BERTScore, an automatic G-Eval LLM judge, and a Likert-scale expert assessment. The headline finding argues for orchestration over scale: direct fine-tuning on small datasets degrades quality, while a structured wrapper around a smaller open-source model can match GPT-4o.

Lin and Cheng [21] take the opposite architectural route, fine-tuning an open-source BLOOM-560M model locally on 74,823 unlabeled Traditional Chinese fraud verdicts (Taiwan Judicial Yuan, 2011–2021) without Chinese word segmentation. Their structural-correctness metric defines a fixed order of legal constitutive elements (Subject of Crime, Subjective Legal Element, Act, Victim, Causation, Result of Hazard) and checks whether the generated draft follows that order, supplemented by perplexity and training/validation loss. The work demonstrates that domain-specific structured drafting is feasible without cloud infrastructure on consumer-grade hardware, addressing the same privacy concerns that motivate locally deployed GDPR tooling. Li et al. [19] push the structural framing further with CaseGen, a benchmark that decomposes Chinese case-document drafting into four sequential stages (defense statements, trial facts, legal reasoning, and judgment results) over 500 expert-annotated cases. Their finding that general-domain LLMs (GPT-4o, Qwen2.5) outperform legal-specific models like ChatLaw or LexiLaw, while still scoring below 6 out of 10 overall, suggests that careful staging of the generation task matters at least as much as legal-domain pretraining. Together, these three works frame structured legal-document generation as primarily an architectural problem and explore wrappers, fine-tuning, and multi-stage decomposition as alternative solutions. ROPAgen takes a related but distinct stance: rather than comparing architectures, it fixes the architecture and varies the *interaction modality* (Form, Ask, Chat), asking what the user-facing surface contributes once the underlying model is held constant.

2.3.3 Evaluation Methodologies for Legal AI

A third strand of the literature concerns how generated legal documents are assessed, and a clear methodological pattern is emerging: dual evaluation that combines automated NLP metrics with either expert annotation or LLM-as-a-Judge scoring. Su et al. [33] formalize this for Chinese judgment-document generation through JuDGE, a benchmark of 2,505 fact-judgment pairs supplemented by over 55,000 statutory articles and 103,000 past judgments. The framework evaluates outputs along four dimensions: Penalty Accuracy (normalized absolute difference on prison terms and fines), Convicting Accuracy (Recall, Precision, F1 over charges), Referencing Accuracy (correctness of statutory citations), and Documenting Similarity (METEOR and BERTScore against ground truth). Su et al. explicitly argue that lexical metrics alone cannot capture the legal reasoning that distinguishes a sound judgment from a plausible-sounding one. Their results across direct generation, in-context learning, supervised fine-tuning, and Multi-source RAG show that even the strongest pipelines remain visibly short of professional ground truth, and that semantic-similarity metrics behave as proxies rather than ceilings.

Vayadande et al. [35], viewed from the methodological angle, illustrate the second template: comparative quantitative benchmarking against existing market tools (Kira Systems, Evisort) using accuracy percentages, time-efficiency in hours, and document-processing speed, supplemented by a debugging phase in which legal experts contribute balanced comments that refine the system. The reported 93–97 % accuracy and reduction of document time from 20 to 5 hours sit alongside an acknowledgment of remaining failures on judicial nuance and on input-quality dependencies, foreshadowing the same tension between aggregate metrics and qualitative caveats that emerges in this thesis. Together, JuDGE and Vayadande et al. converge on a shared template: surface-overlap metrics (BLEU, ROUGE, METEOR), semantic-similarity metrics (BERTScore, sentence embeddings), and a complementary qualitative judgment layer (human expert or LLM judge) that catches what the lexical and semantic metrics miss. This is the methodological template adopted in the present study for evaluating ROPAgen's three modes against a single expert reference document.

2.3.4 User-Facing Legal AI Systems

The fourth lens concerns how legal AI is presented to non-expert users, where prior work is markedly thinner. Surya et al. [34] target the accessibility gap directly with an AI-powered interactive legal chatbot for the Indian Department of Justice, aiming to democratize legal information for citizens facing complex jargon and the cost of professional counsel. Their three-tier architecture (a JavaScript single-page frontend, a Python Flask orchestrator handling translation, and an LLM service running LLaMA 3.1 via Ollama) supports real-time legal queries, document summarization (e.g., First Information Reports), and multilingual interaction by translating Hindi queries to English for the LLM and back. The system also surfaces citation links to government portals, addressing the trust problem that arises whenever non-experts consume LLM output on a high-stakes topic. Notably, however, the paper evaluates these features through illustrative use cases rather than a controlled user study, leaving open how non-expert users actually interact with such a system in practice.

Vayadande et al. [35], considered from the user-facing angle rather than the methodological one, fit the same pattern. Their assistant integrates ChatGPT 3.5 with OpenAI embeddings, LangChain, and a vector retrieval stack to support legal-document generation, comprehension, and abnormality detection, framing the tool as a means of reducing the practical effort of legal work. The system is again validated through accuracy and efficiency comparisons rather than through evidence of how non-expert users navigate the interface, interpret the output, or judge their own confidence in it. Both works establish that accessibility, multilingual support, and citation transparency are necessary properties of user-facing legal systems, but stop short of empirically measuring whether the resulting tools serve the non-experts they target.

2.4 Summary

Under the GDPR, accountability is a documented obligation rather than a stated principle: organizations must demonstrate compliance to supervisory authorities, and the Record of Processing Activities under Article 30 is one of the principal artifacts through which that demonstration is performed. The content of a ROPA is rigidly specified (controller details, purposes, lawful bases, categories of data and recipients, retention periods, and technical and organizational measures), yet its substance is project-specific prose that has to be written from a blank page and kept current as processing activities change. Maintaining a ROPA is therefore a sustained cost rather than a one-off task, and one that falls disproportionately on small and medium organizations that lack dedicated legal or data-protection staff. This is the practical gap into which an LLM-assisted ROPA tool such as ROPAgen is built.

Across the four lenses surveyed in the related-work section, *domain* (GDPR-specific compliance automation), *task* (structured legal-document generation more broadly), *evaluation methodology* (the emerging dual-evaluation template of NLP metrics paired with expert or LLM-as-a-Judge scoring), and *user* (legal AI presented to non-expert audiences), the literature converges on three established points. LLM-based legal-document generation is technically feasible. Architectural choices (retrieval-augmented generation, model wrappers, multi-stage decomposition, local fine-tuning) measurably improve output. And dual evaluation combining surface-form and semantic NLP metrics with either expert annotation or LLM-as-a-Judge scoring has become the de facto template for assessing the resulting documents.

What remains absent is the intersection of three concerns. No prior work simultaneously (a) targets GDPR-specific structured output, (b) exposes the system to *novice users* without legal training, and (c) measures the gap between user confidence and actual document quality under varying levels of LLM support. This thesis fills that gap by holding the underlying ROPAgen pipeline constant, varying the interaction modality across Form, Ask, and Chat, and applying the dual-evaluation template from the legal-AI literature to a within-subjects user study against a single shared scenario. The instruments, sample, and analysis plan through which that gap is closed are described next in Chapter 3.

3 Methodology

This chapter is divided into two complementary parts. The first introduces ROPA-gen, the tool under investigation, covering each of its three interaction modes and the corresponding prompts submitted to the LLM. The second details the design of the user study, including participants, instruments, sample sizes, and the analytical approach through which the tool is evaluated. Together, these sections establish the scope of what is measured and the means by which it is assessed, prior to any presentation of results.

3.1 ROPAgen

The tool evaluated in this study is ROPAgen [15], a web-based application that assists users in drafting GDPR Article 30 Records of Processing Activities through three different levels of LLM support. ROPAgen was developed at the Institute of Databases and Information Systems (DBIS), University of Ulm, prior to the present evaluation; this section describes the tool as it stood at the time of the study. ROPA-gen exists to ease the burden of ROPA drafting: the document has rigid structural requirements (Art. 30 GDPR) but most of its substance is project-specific prose that has to be written from a blank page. The tool offers three interaction modes (Form, Ask, and Chat), which offer different levels of LLM support;

Form mode places the user in full control: they select pre-defined options and fill in structured fields, with LLM support available only on demand via a suggestion button. **Ask** mode is a hybrid: users fill in free-form text fields, while an LLM chatbot explains the requirements of each section and answers user questions, but does not directly populate any structured fields. **Chat** mode is fully autonomous: the user describes their requirements in free-form conversation, and the LLM independently determines which structured fields to populate in the background and drafts the ROPA content on its own. This thesis evaluates the three modes: how the level of LLM support shapes user experience and how it shapes the documents that are actually produced. The remainder of this section describes the internal representation of a ROPA document in the tool, walks through the three modes from the user's perspective, and details the LLM backend that all three modes share.

Mode	LLM's role in the section	Predefined booleans set by	Free-text fields written by
Form	On-demand suggestion only (the *AI Suggest* button)	User (clicks), or LLM if *AI Suggest* is used	User, or LLM if *AI Suggest* is used
Ask	Explainer only; emits no structured update	Nobody (left at the default `false`)	User only
Chat	Conversational extractor; emits `DATA_UPDATE` blocks merged into `DocumentData`	LLM (via `DATA_UPDATE`)	LLM (via `DATA_UPDATE`)

Table 3.1: The three ROPAgen interaction modes at a glance, described in terms of who populates which part of the shared `DocumentData` object that the final-generation step (`finalropa`) receives.

3.1.1 ROPA Documents in ROPAgen

Internally, ROPAgen represents a ROPA as a single structured `DocumentData` object organized around eight thematic groups that together cover the disclosures required under Art. 30(1) GDPR:

1. **Organization and controller** — name, postal address, e-mail, phone number, and the contact details of the data protection officer and any external representative.
2. **Legal basis** — Boolean flags for each lawful-basis option in Art. 6(1) GDPR (consent, contract, legal obligation, legitimate interest), the German national-law bases relevant to the study scenario (§ 26 BDSG, § 28b IfSG, § 257/239 HGB, § 147 AO), and a free-text fallback for other grounds.
3. **Data categories** — nine grouped categories (personal master data, employment, finance, health, legal, behavioral, documentation, vehicle, and other)

each with predefined Boolean sub-fields such as surname, date of birth, working hours, or holiday entitlement.

4. **Data sources** — Boolean flags for common upstream systems (Office 365, Azure, Google Workspace, AWS, e-mail, etc.) plus a free-text field for sources not covered by the predefined list.

5. **Person categories** — affected persons, internal recipients, external recipients within the EU, and authorized personnel, each populated through Boolean role tags with an optional free-text override.

6. **Retention periods** — Boolean flags for the standard retention triggers (after termination of employment, after termination of contract, upon revocation of consent), a free-text field for the concrete deletion period, and flags for legal-retention exceptions.

7. **Free-text fields** — three short prose fields covering the purpose of processing, the technical and organizational measures in place, and any additional information that does not fit elsewhere.

8. **Metadata** — the document identifier, title, and natural language in which the final ROPA should be generated.

This structured form is the tool's working representation while the user is drafting. The final ROPA itself is rendered as a Markdown document (with optional PDF export through a server-side rendering route) that the LLM produces in a single generation step from the assembled `DocumentData` object. A complete example of the resulting document serves as the reference baseline against which all participant outputs are scored by the NLP metric pipeline.[1]

3.1.2 The Three Interaction Modes

ROPAgen offers three generation modes, each with a different level of LLM support: Form has the least LLM support, Chat is fully autonomous, and Ask is a hybrid mode that sits in the middle. The subsections that follow describe each mode in detail; Table 3.1 summarizes them at a glance. All three act on the same internal

[1] Full reference ROPA in Appendix A.2.

DocumentData schema and, when the user clicks the final *Generate* button, hand that schema to the same prose-generation prompt (finalropa); the output ROPA is therefore produced by an identical generation step in all three conditions, and only the path the user takes to populate DocumentData differs between modes. The *Document Title, Organization* entry screen and the final *Generate Document* screen that frame this path are also identical across the three modes (Figures 3.1 and 3.2); only the mode-specific interaction between those two screens differs.

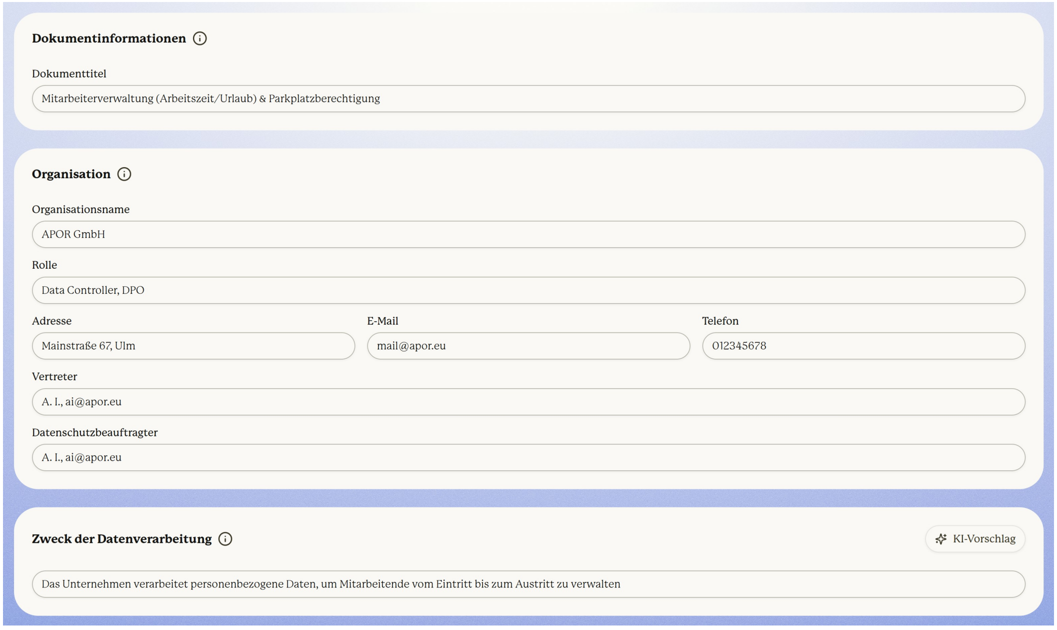


Figure 3.1: The *Organization* entry screen, shown to participants before the mode-specific interaction begins. Identical across all three modes.

Form Mode

Form mode offers the lightest LLM support: the user sees and does everything. Every field of the DocumentData schema is visible and edited by hand, laid out as a linear questionnaire of checkboxes and short text fields across the eight thematic groups (Figure 3.3). The LLM is invoked only when the user explicitly asks for it: each section offers an optional *AI Suggest* button that asks the LLM, given the document's current state, to propose values for that section's fields; the suggestions populate the form but remain freely editable, and the user is under no obligation to use them. Once the user is satisfied with the form, clicking *Generate* sends the assembled DocumentData object to the finalropa prompt, which synthesizes a

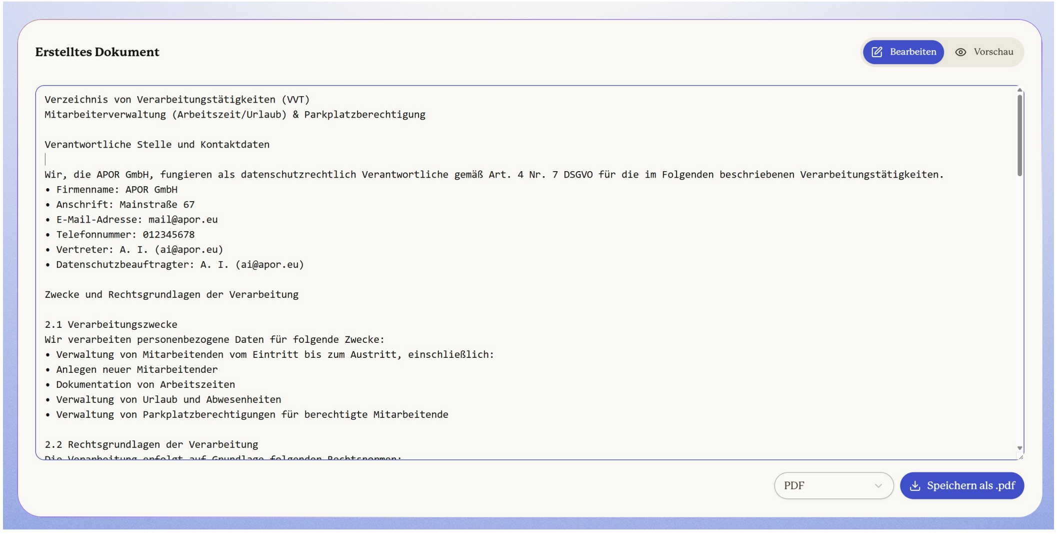


Figure 3.2: The final *Generate Document* screen, shown to participants after the mode-specific interaction concludes. Identical across all three modes; the `finalropa` prompt (Section A.1.1) runs at this point regardless of which mode was used.

complete ROPA document in formal prose. Form mode therefore exposes the entire structure of a ROPA upfront and uses the LLM only at the margins. In short, the user populates the predefined boolean scaffolding and the free-text fields directly, and the LLM is involved only as an optional per-section suggestion engine and at the final prose-synthesis step.

Ask Mode

Ask mode pairs each section with an LLM chat panel and a free-form text field; there are no checkboxes or pre-defined options. The user can ask the LLM, section by section, to explain what the section requires or to suggest how to fill it, but the user is responsible for the final wording entered into the section's text field. Each chat panel (Figure 3.4) opens with the LLM explaining the section's purpose and asking the user for the relevant information; subsequent turns maintain the conversation history for that section and ask follow-up questions. The model is given the current `DocumentData` state alongside the user's message so its explanations can be context-aware, but its reply is conversational only: it does not return a structured update block and no field is auto-populated. The predefined boolean flags of the `DocumentData` schema are not touched by either the user or the LLM in Ask mode. The final *Generate* step is identical to Form mode: the same `finalropa` prompt receives the same `DocumentData` object. The `DocumentData` object that reaches

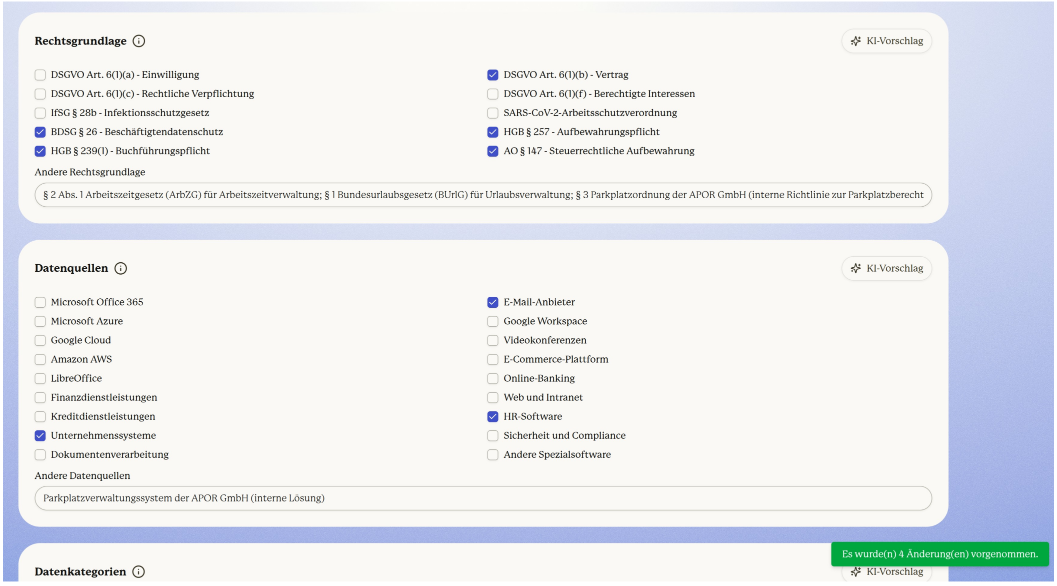


Figure 3.3: Form mode. The eight thematic groups of the `DocumentData` schema are exposed as a linear questionnaire of checkboxes and short text fields. Each section offers an optional *AI Suggest* button that calls the LLM with the section-specific Form-mode prompt (Section A.1.2).

the final-generation step in Ask mode therefore carries only the user's per-section free-text writing; the predefined boolean scaffolding is left at its default. Any section the user does not type into reaches the generator with no scaffolded content to work from. In short, the user only chats with the LLM and populates the free-text `other` field of each section themselves; the predefined boolean scaffolding is never touched, so the final-generation step has no scaffolded content to work from beyond the user's own typing.

Chat Mode

Chat mode hands control to the LLM. The user does not see or fill the structured fields at all; instead, after entering the organization's metadata (name, contact information, and title of the processing activity), the user describes their processing in their own words, and the LLM decides which structured fields to populate from those descriptions (Figure 3.5). The user navigates section by section through a chat interface only; the structured form is hidden from view. The LLM is explicitly instructed to extract structured data from the conversation and to emit it as a `DATA_UPDATE` block alongside its natural-language reply; the front-end parses that block and merges it into the shared `DocumentData` object. A mentioned vendor

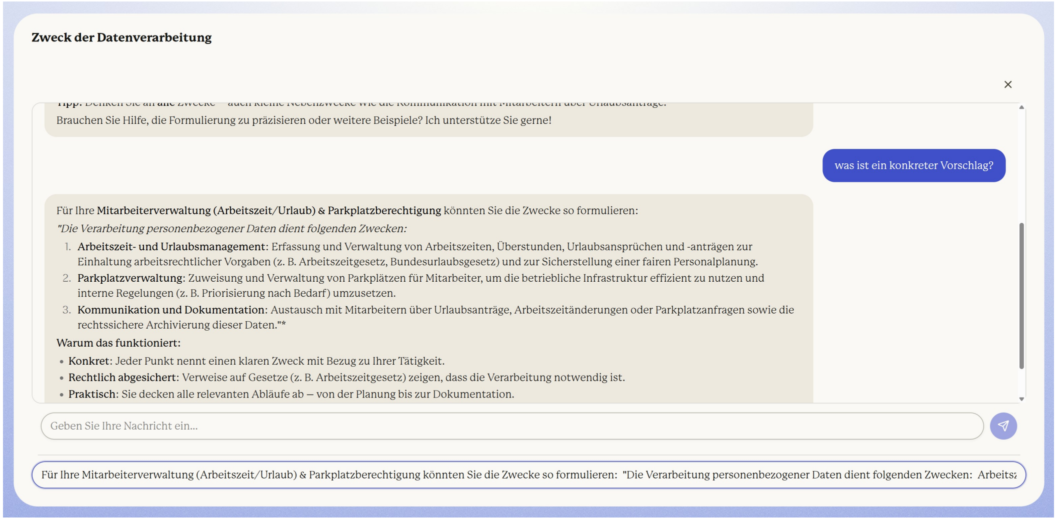


Figure 3.4: Ask mode. Each section presents a free-form text field (no checkboxes) paired with a per-section chat panel that explains the section's purpose and answers follow-up questions. The LLM's reply is conversational only; no field is auto-populated and the user types all section content manually.

name therefore flips the corresponding data-source boolean, a mentioned data category populates the corresponding boolean, and any specific item that does not match a predefined option lands in the section's `other` free-text field. The user therefore never directly interacts with the underlying `DocumentData` object, although the same object is being built up under the hood. The final *Generate* step is once again identical to the other two modes: the same `finalropa` prompt receives the same `DocumentData` object. Chat mode is therefore a conversational-first workflow in which both the scaffolded booleans and the free-text fields are populated by the LLM, and only the final generation step is shared with the other modes. In short, the user describes the processing in their own words and the LLM populates both the predefined boolean scaffolding and the free-text fields on the user's behalf; the user never directly interacts with the structured form.

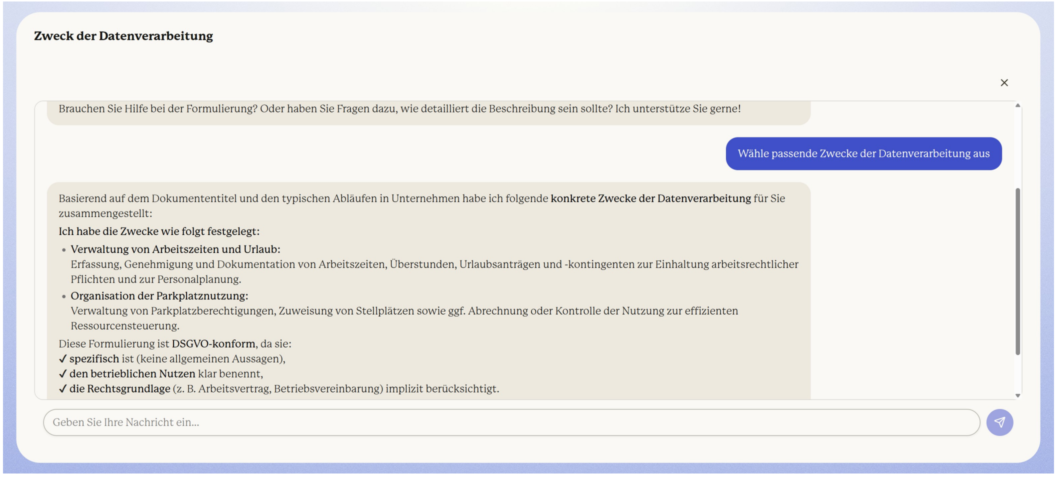


Figure 3.5: Chat mode. The structured form is hidden; participants describe their processing activity in free-form text and the LLM populates the underlying `DocumentData` object via silent `DATA_UPDATE` blocks emitted on every turn.

3.1.3 LLM Backend

All three modes are backed by the same model: Mistral Large 3 (model identifier `mistralai/mistral-large-2512`),[2] accessed through the OpenRouter API gateway via the Vercel AI SDK. The choice of Mistral rests on three considerations relevant to a GDPR-compliance tool. First, Mistral AI is a European model provider [23], which keeps the generation workload aligned, in spirit, with the data-protection regime ROPAgen is helping users to document. Second, Mistral AI has a public commitment to open-weight releases such as Mistral 7B [13] and Mixtral 8x7B [14]; although the Mistral Large 3 variant used here is the company's commercial flagship rather than an open release, this lineage offers a credible path to a fully self-hosted deployment. Third, the Mistral family has shown competitive performance on the long-form reasoning and structured-output tasks relevant to legal-document generation [13]. Two temperature regimes are used (Table 3.2). Conversational and suggestion calls (per-section LLM suggestions in Form mode, dialogue turns in Ask and Chat modes, and the silent data-extraction calls in Chat mode) use temperature 0.2 and top-p 0.5, which permits variation in phrasing while keeping the responses on-task. The final ROPA generation step, in contrast, uses temperature 0.05 and top-p 0.05, which makes the output close to deterministic given the same `DocumentData` input.

[2] Hugging Face model card: `https://huggingface.co/mistralai/Mistral-Large-3-675B-Instruct-2512`.

Call type	Temperature	top-p
Conversational / suggestion	0.2	0.5
Final ROPA generation	0.05	0.05

Table 3.2: ROPAgen LLM call parameters. The same model (`mistralai/mistral-large-2512`) is used for both regimes.

Crucially, the prompt used in the final generation step is fixed across all three modes: the only thing that differs between a Form-mode output and a Chat-mode output is the contents of the `DocumentData` object that has been built up along the way.[3]

Whether the differing paths to `DocumentData` produce documents of differing quality, and whether those quality differences are visible to the user filling out the form, are the questions taken up by the user study described next.

3.2 User Study

The user study is a structured survey: after each interaction mode, participants completed a fixed battery of questionnaire items, and after all three modes they completed a final ranking task and a demographics block. For consistency, the remainder of this thesis refers to the data collection as the *user study* rather than the survey.

3.2.1 Study Design

The study followed a within-subjects experimental design with counterbalanced ordering. The recruited sample consisted of $N = 73$ university students, all novice users with no prior GDPR experience, and each session lasted approximately 90 minutes per participant. Counterbalancing was implemented through a Balanced Latin Square: the three modes (Form, Ask, Chat) were arranged into six ordered sequences, and participants were randomly assigned to one of the six groups, with each participant completing all three modes in their group's predetermined order.

[3] Full text of the backend prompts in Appendix A.1, as the counterpart to the LLM-as-a-judge configuration recorded in Appendix A.3.

This design controls for order effects, learning effects, and fatigue. Demographic information was collected at the end of the session, after the three modes and the final ranking. Figure 3.6 situates the per-session flow within the broader project arc, from the development of ROPAgen through data collection to the analysis pipeline reported in the chapters that follow.

3.2.2 Task Scenario

A single standardized scenario was used across all participants and modes to ensure consistency. The scenario described a relatable workplace data processing activity at *APOR GmbH*, a fictional small software company in Ulm with approximately 50 employees: the two processing activities under consideration were the administration of employee working time and leave, and the management of parking authorizations for the company car park. The scenario covers all GDPR Article 30 fields and is accessible to participants without legal expertise.[4] The LLM backing all three modes (Section 3.1) was held constant throughout the study.

3.2.3 Participants

Recruitment yielded $N = 73$ university students. Participants were screened for minimal to no GDPR knowledge (expected and required), prior AI tool usage experience (documented), general technology comfort level (assessed), and the absence of any professional legal or compliance background. As part of onboarding, participants received a brief standardized introduction to GDPR and ROPA concepts before beginning the tasks.

3.2.4 Data Availability and Sample Sizes

Due to technical data loss and incomplete submissions during data collection, the effective sample sizes differ across analysis streams. The canonical sample sizes used throughout this thesis are summarized below.

[4] Full scenario, in both the German version shown to participants and an English translation, in Appendix B.1.

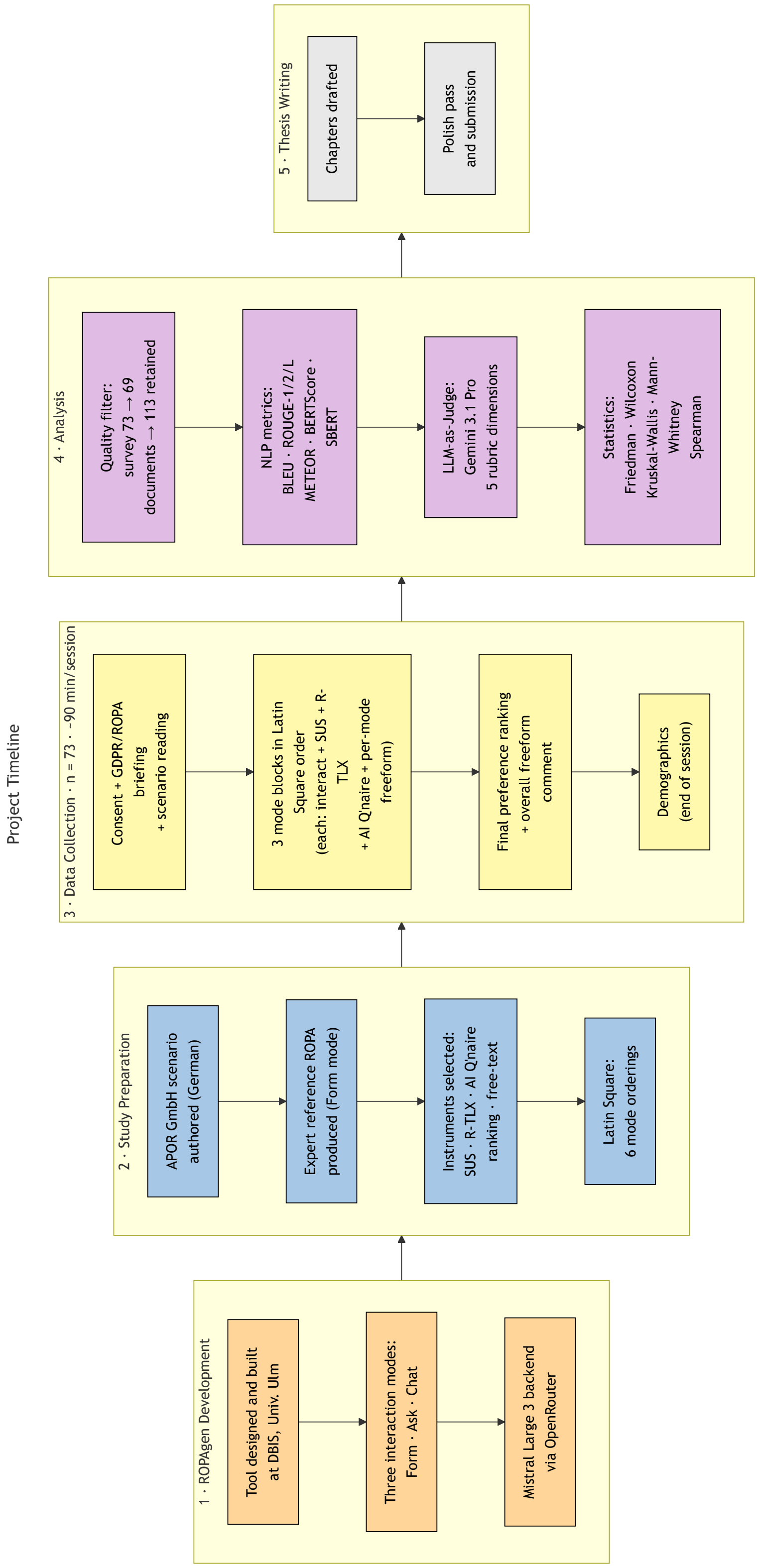


Figure 3.6: Project timeline. Five phases run left-to-right: ROPAgen development at DBIS; study preparation (scenario authoring, reference ROPA, instrument selection, Latin-Square construction); per-participant data collection in 90-minute sessions ($n = 73$); analysis (quality filter, NLP metric pipeline, LLM-as-Judge, statistical layer); and thesis writing. Demographics were administered at the end of each session rather than at the start.

- **User study, recruited N = 73 / valid N = 69** — 73 participants completed the study; 69 retained after dropping four records with invalid data. All 69 valid participants completed all three modes, so per-mode sample size, matched-triple user-study subset, and valid N are all **n = 69**. The matched-triple n = 69 is used for within-subjects Friedman and Wilcoxon signed-rank tests on *SUS*, *R-TLX*, confidence items, and the perceived AI support composite.
- **Documents, 113 generated ROPAs** — distributed across modes as 37 (Form), 37 (Ask), and 39 (Chat). The full $n = 113$ sample is used throughout: NLP metric scores and per-mode completion time are analyzed inferentially with Kruskal–Wallis and Bonferroni-corrected Mann–Whitney U as the between-groups family; LLM-as-a-judge scores are reported descriptively on the same sample.

The canonical figures used in the inferential analyses are **n = 113** for NLP metric and completion-time comparisons and **n = 69** for user-study instruments.

Descriptive statistics (means, medians, standard deviations) are computed on the full available N per mode for each analysis stream. Inferential tests on NLP metric and per-mode completion-time measures operate on the full $n = 113$ sample as a between-groups comparison; inferential tests on user-study instruments operate on the matched $n = 69$ within-subjects sample. The LLM-as-a-judge scores are reported descriptively only and are not subjected to inferential testing (Section 3.5).

3.2.5 Data Collection

Quantitative measures were collected after each mode. Table 3.3 summarizes the four user-study instruments, the post-experiment preference ranking, and the post-session demographics questionnaire.

The *System Usability Scale* (*SUS*) [5] was administered as ten standardized questions on a 5-point Likert scale; the *Raw Task Load Index* (*R-TLX*) [12, 11] captured six cognitive load dimensions on the standard NASA 21-point scale (1 = lowest demand, 21 = highest), as Raw TLX retains the unweighted subscale ratings on their original NASA scale.[5] In addition, participants completed a six-item *Perceived AI Support* instrument on a 5-point Likert scale:

[5]Full text of the 10 SUS items and the 6 NASA R-TLX subscale descriptions in Appendix B.2.

Instrument	What it captures
SUS	Perceived usability
R-TLX	Cognitive load (six subscales)
Perceived AI Support	LLM usefulness, trust, transparency, confidence
Free-text feedback	Qualitative comments per mode
Mode-preference ranking	Post-experiment overall preference
Demographics	Age, gender, degree, field, prior GDPR/AI familiarity

Table 3.3: Instruments collected during and after the user study.

1. *Perceived Usefulness:* “The AI was helpful in creating this document.” (*“Die KI war bei der Erstellung dieses Dokuments hilfreich.”*)
2. *Efficiency:* “The AI helped me to complete the document faster.” (*“Die KI hat mir geholfen, das Dokument schneller fertigzustellen.”*)
3. *Transparency:* “I understood why the AI suggested specific GDPR fields.” (*“Ich habe verstanden, warum die KI bestimmte DSGVO-Felder vorgeschlagen hat.”*)
4. *Trust:* “I trust that the AI made correct suggestions.” (*“Ich vertraue der KI, dass sie korrekte Vorschläge gemacht hat.”*)
5. *Task-Specific Confidence:* “I am confident that the AI identified the correct legal basis.” (*“Ich bin zuversichtlich, dass die KI die richtige Rechtsgrundlage identifiziert hat.”*)
6. *Intention to Use:* “I would use this mode again.” (*“Ich würde diesen Modus wieder verwenden.”*)

All study materials, including the scenario, the consent and onboarding text, the four user-study instruments, and the post-experiment ranking, were prepared in both German and English, and participants could choose the language in which they completed the session; the parenthetical text above reproduces the German wording, which the majority of the recruited cohort selected. Items 4 and 5 are averaged into a two-item *perceived confidence* composite that is treated as the **primary confidence measure** throughout the analysis. These are the only two items that ask participants to judge the substantive correctness of the LLM’s output: item 4 the correctness of the AI’s suggestions in general, item 5 the correctness of the legal-basis identification under Article 6 GDPR. The composite is therefore directly comparable to the document-quality signals from the NLP and judge pipelines, which themselves evaluate correctness at the whole-document level; the two items are reported individually as well so that the legal-basis dimension can be tracked on its own. The remaining four items, Item 1 (Perceived Usefulness), Item 2 (Efficiency), Item 3 (Transparency), and Item 6 (Intention to Use), capture facets of *perceived AI support* that are about user experience rather than output correctness, and are reported as a secondary composite (the *perceived AI support* mean across all six items). Free-text qualitative feedback was also collected after each mode. Two timing measurements were retained: total session duration from the user-study questionnaire (one value per participant) and per-mode completion time emitted by ROPAgen at the end of each session and embedded in every generated document (one value per participant per mode, in seconds). After completing all three conditions, participants ranked all three modes by overall preference in a post-experiment ranking task.

3.3 NLP Metrics

Following the dual-evaluation template adopted by Su et al. [33] and Vayadande et al. [35], both surface-form and semantic measures are reported alongside an LLM-as-Judge rubric. All participant-generated documents are compared against a single reference document using a suite of automated NLP metrics. After validation for errors and completeness, 113 documents were retained for analysis out of a theoretical maximum of 219 (73 participants $\times$ 3 modes). The reference document was created using Form mode (Section 3.1), applying the same standardized scenario presented to participants.[6] It is treated as the authoritative baseline, a root-of-trust representing a careful, complete application of the most structured interaction mode. This introduces an inherent framing effect: NLP metrics measure similarity to Form-style output, which may disadvantage the more conversational outputs of Ask and Chat modes independently of their substantive quality.

BLEU, ROUGE, METEOR, BERTScore, and SBERT are all computed at both the whole-document level and the per-chapter level, with ModernBERT-base as the embedding backbone for BERTScore and SBERT. At the chapter level, SBERT is ad-

[6]Full reference ROPA in Appendix A.2.

ditionally run with the smaller `all-MiniLM-L6-v2`; the MiniLM pass is chapter-only because its 512-token input limit excludes whole-document scoring. Results at every available granularity are aggregated by mode.

The five NLP metrics summarized in Table 3.4 (BLEU [26], ROUGE [20], METEOR [3], BERTScore [38], and SBERT [28]) are defined formally below. Each is computed against the single expert-authored reference document; interpretive framing of the resulting values is deferred to Chapter 5. Because the metrics differ sharply in the scale on which they report, the expected ranges and good-versus-poor bands for each are collected for reference in Appendix B.5.

Metric	What it measures
BLEU	n-gram precision with brevity penalty
ROUGE-1/2/L	Unigram / bigram / LCS overlap (F1)
METEOR	Precision–recall with synonym and stem matching
BERTScore P/R/F1	Token-level semantic similarity (ModernBERT)
SBERT cosine similarity	Sentence-embedding cosine similarity (ModernBERT; chapter-level MiniLM robustness check)

Table 3.4: NLP metrics used in the document-quality evaluation. Higher is better on all metrics.

3.3.1 BLEU

Bilingual Evaluation Understudy — n-gram precision with brevity penalty [26]. BLEU rewards exact word-sequence overlap with the reference: a high BLEU score means the candidate reuses the reference's wording almost verbatim, while a low BLEU score can equally indicate a legitimate paraphrase or genuinely missing content; the metric does not distinguish the two on its own. BLEU is the most pessimistic measure in the suite: even strong machine output rarely clears 0.3 to 0.4 against a single reference, and for content that is faithful but freely reworded, values in the 0.05 to 0.20 range are the expected outcome rather than a sign of failure [7]. BLEU computes the geometric mean of n-gram precisions p_n between candidate and reference, multiplied by a brevity penalty (BP) that discourages outputs much shorter than the reference. With uniform weights $w_n = 1/N$, candidate length c, and closest

[7] Appendix B.5

reference length r. This work uses $N = 4$ (matching unigrams through 4-grams):

$$\mathrm{BLEU} = \mathrm{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \quad \mathrm{BP} = \begin{cases} 1 & \text{if } c > r \\ e^{1-r/c} & \text{if } c \leq r. \end{cases} \tag{3.1}$$

3.3.2 ROUGE-1, ROUGE-2, ROUGE-L

Recall-Oriented Understudy for Gisting Evaluation — unigram, bigram, and longest-common-subsequence overlap (F1) [20]. ROUGE measures how much of the reference's vocabulary, bigrams, or longest common subsequence the candidate covers. A high ROUGE score signals strong lexical recall against the reference; a low score indicates that key reference content is absent from the candidate. ROUGE values run well above BLEU because single tokens and subsequences are easier to match than full n-grams: a ROUGE-1 in the 0.4 to 0.6 band reflects solid recall, while the bigram-based ROUGE-2 and subsequence-based ROUGE-L sit lower because contiguous matches are scarcer [8]. ROUGE reports the F1 of n-gram or longest-common-subsequence (LCS) overlap between candidate X and reference Y. ROUGE-1 counts unigrams, ROUGE-2 counts bigrams, and ROUGE-L counts the LCS. For ROUGE-N:

$$\begin{aligned} P_N &= \frac{|\mathrm{ngrams}(X) \cap \mathrm{ngrams}(Y)|}{|\mathrm{ngrams}(X)|}, \\ R_N &= \frac{|\mathrm{ngrams}(X) \cap \mathrm{ngrams}(Y)|}{|\mathrm{ngrams}(Y)|}, \\ F1_N &= \frac{2 P_N R_N}{P_N + R_N}. \end{aligned} \tag{3.2}$$

The same precision–recall–F1 form applies to ROUGE-2:

$$F1_{\text{ROUGE-2}} = \frac{2 P_2 R_2}{P_2 + R_2}. \tag{3.3}$$

[8] Appendix B.5

For ROUGE-L, n-gram counts are replaced by the length of $\mathrm{LCS}(X, Y)$. The reported ROUGE values are the F1 scores in each case:

$$\begin{aligned} P_L &= \frac{|\mathrm{LCS}(X,Y)|}{|X|}, \\ R_L &= \frac{|\mathrm{LCS}(X,Y)|}{|Y|}, \\ F1_L &= \frac{2P_L R_L}{P_L + R_L}. \end{aligned} \tag{3.4}$$

3.3.3 METEOR

Metric for Evaluation of Translation with Explicit ORdering — precision–recall balance with synonym matching [3]. METEOR balances precision and recall, credits synonym and stem matches, and penalizes fragmented overlap where shared words appear in scattered positions. A high METEOR score signals that the candidate covers the reference's content with vocabulary that need not match it verbatim; a low score indicates either missing content or surface forms that the synonym and stem matcher could not align. METEOR typically lands between BLEU and ROUGE-1, often in the 0.2 to 0.5 band for content-equivalent text, since synonym and stem credit lift it above raw n-gram precision while the fragmentation penalty holds it below a pure recall count [9].

METEOR is the weighted harmonic mean of unigram precision P and recall R (with recall up-weighted by α), multiplied by a fragmentation penalty that grows with the number of disjoint matching chunks ch relative to the number of matched unigrams m: Standard values $\alpha = 0.9$, $\beta = 3$, $\gamma = 0.5$ are used; matching is performed against exact, stemmed, and synonym/paraphrase matches via WordNet.

$$\mathrm{METEOR} = \underbrace{\frac{P \cdot R}{\alpha P + (1-\alpha)R}}_{F_{\mathrm{mean}}} \cdot \underbrace{\left(1 - \gamma\left(\frac{\mathrm{ch}}{m}\right)^{\beta}\right)}_{\text{penalty}}. \tag{3.5}$$

[9] Appendix B.5

3.3.4 BERTScore

Precision, Recall, F1 — token-level semantic similarity using ModernBERT embeddings [38]. BERTScore measures similarity between the contextual embeddings of candidate and reference tokens rather than their surface forms. *Precision* captures how many candidate tokens find a close meaning-match in the reference and therefore penalizes spurious or invented content; *recall* captures how many reference tokens are covered and therefore penalizes omission; *F1* is the harmonic mean and requires both to be reasonably high. Because the contextual embeddings of any two fluent texts are already broadly similar, BERTScore occupies a compressed high band: values below roughly 0.80 are rare, and the meaningful signal lives in differences of a few hundredths within the 0.85 to 0.95 range, so its absolute scores should not be read on the same intuition as the lexical metrics [10].

BERTScore replaces surface-form n-gram matching with cosine similarities between contextual token embeddings produced by a pretrained transformer. The embedding model used here is ModernBERT-base [2, 36], chosen for its long context window of 8192 tokens, which lets a whole ROPA be embedded without truncation, and for its popularity and positive reception in the Hugging Face community. With L2-normalized embeddings $\mathbf{x}_i$ for reference tokens and $\hat{\mathbf{x}}_j$ for candidate tokens:

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \tag{3.6}$$

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j, \tag{3.7}$$

$$F1_{\text{BERT}} = \frac{2 \cdot P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}. \tag{3.8}$$

3.3.5 SBERT Cosine Similarity

Sentence-BERT — document- and chapter-level semantic similarity using sentence embeddings [28]. SBERT compresses a passage into a single fixed-length embedding and reports the cosine similarity between the candidate and reference embeddings. Unlike BERTScore, which decomposes token-level matches into Precision,

[10] Appendix B.5

Recall, and F1, SBERT produces a single symmetric similarity score per comparison: the cosine of the angle between two compressed vectors does not distinguish candidate-into-reference coverage from reference-into-candidate coverage, so no separate precision and recall values are reported. SBERT is computed at both granularities: one whole-document score per ROPA against the full reference, and one chapter score per candidate chapter against the corresponding reference chapter. The primary encoder for both passes is ModernBERT-base [2], whose 8192-token context window accommodates a full ROPA without truncation. At the chapter level, a secondary pass with `all-MiniLM-L6-v2` [31] is run over the same six chapters as a robustness check on the primary ModernBERT pass; MiniLM has a 512-token input limit that excludes whole-document use. Both models were chosen for their popularity and positive reception in the Hugging Face community. Document-level cosine similarity is even more compressed than BERTScore, routinely exceeding 0.95 for any two ROPAs written against the same scenario; near-unity values are therefore the expected baseline rather than evidence of near-identical documents, and only the relative gaps between modes carry interpretable signal [11].

SBERT produces a single fixed-length embedding per passage by mean-pooling token embeddings from the underlying transformer (here, ModernBERT or MiniLM). Passage similarity is the cosine of the angle between two embeddings $\mathbf{e}(c_1)$ and $\mathbf{e}(c_2)$:

$$\mathrm{SBERT}(c_1, c_2) = \frac{\mathbf{e}(c_1)^\top \mathbf{e}(c_2)}{\|\mathbf{e}(c_1)\| \, \|\mathbf{e}(c_2)\|}. \tag{3.9}$$

3.3.6 Score Implications

The metrics above are read in combination, not in isolation, since each captures a different facet of similarity to the reference. Four pairings recur in Chapter 5. A high lexical score (BLEU, ROUGE, or METEOR) combined with a low semantic score (BERTScore F1 or chapter-level SBERT) indicates mechanical surface copying that does not align in meaning, whereas the reverse pattern (low lexical, high semantic) is the signature of a legitimate paraphrase that preserves content with different wording. Within BERTScore, high recall paired with low precision points to a verbose candidate that covers the reference but adds material the reference

[11] Appendix B.5

does not contain, while low recall paired with high precision points to a conservative candidate that omits required content without fabricating new material. A further reading concerns the two embedding-based metrics: because BERTScore matches at token granularity and SBERT at passage granularity, the two register different kinds of similarity, with BERTScore rewarding a candidate that reproduces reference vocabulary token by token and SBERT rewarding a candidate that preserves passage-level meaning under reworded phrasing. The two can therefore order the same documents differently even though both are nominally "semantic", and Chapter 5 reads the split as a granularity effect rather than as a contradiction. Finally, a high document-level score on the surface-form metrics, BERTScore, or SBERT paired with uneven chapter-level scores under the same metric signals that the aggregate is hiding per-section weakness; the chapter-level SBERT scores in particular serve as a per-section diagnostic alongside the whole-document SBERT summary. These heuristics are applied throughout Chapter 5.

3.4 LLM-as-a-Judge Evaluation

Documents are additionally evaluated using an LLM-as-a-judge approach, in which **Gemini 3.1 Pro Preview** (`google/gemini-3.1-pro-preview`, $T = 0$) serves as an automated evaluator assessing GDPR-specific quality. The judge is positioned as a *supplementary* evaluator: it is a single-pass, single-model assessment without inter-rater replication or legal-expert validation, and its scores are reported alongside, not in place of, the NLP metrics. The reference-based NLP metrics remain the primary RQ2 evidence; the judge complements them by providing a reference-free signal that captures GDPR-specific quality properties (legal correctness, hallucination, scenario faithfulness) which similarity-based metrics cannot directly observe. The judge evaluates each document across five metrics on a 1–5 scale:

1. **Completeness:** Coverage of all required Article 30(1) fields (controller identity, DPO contact, purposes, legal basis, data categories, recipients, retention periods, technical and organizational measures)
2. **Scenario Faithfulness:** Adherence to the specific details of the study scenario rather than generic or hallucinated content

3. **Legal Correctness:** Appropriate use of GDPR terminology and accurate citation of relevant articles
4. **Hallucination (inverse):** Absence of invented facts not present in the scenario (scored inversely: 5 = no hallucinations)
5. **Overall:** Holistic assessment of the document's usability as a real-world ROPA record

The judge evaluation is applied to all 113 retained documents.[12]

3.5 Statistical Analysis

All inferential tests are non-parametric: Shapiro–Wilk rejected normality on essentially every user-study, NLP, and timing measure, and the user-study instruments are ordinal Likert and workload scales for which non-parametric tests are the standard usability-research choice [29]. The test family is matched to the structure of each stream (Table 3.5): within-subjects on the user-study instruments (matched triples, $n = 69$), between-groups on the unbalanced document sample ($n = 113$), and goodness-of-fit on the mode-preference ranking ($n = 69$). The LLM-as-a-Judge scores are reported descriptively only. A significance threshold of $\alpha = 0.05$ is applied throughout; each three-way post-hoc family is Bonferroni-corrected to $\alpha' = 0.0167$. Effect-size magnitudes are read against a common convention for ε^2, Kendall's W, Cramér's V, and $|r|$: below 0.1 negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.

Analysis stream	Test	Effect size
User study (within, $n = 69$)	Friedman	Kendall's W
	Wilcoxon signed-rank	Rank-biserial r
NLP / timing (between, $n = 113$)	Kruskal–Wallis	ε^2
	Mann–Whitney U	Rank-biserial r
Mode preference ($n = 69$)	χ^2 goodness-of-fit	Cramér's V

Table 3.5: Statistical tests and their effect sizes by analysis stream. $\alpha = 0.05$ throughout; Bonferroni-corrected $\alpha' = 0.0167$ for post-hoc families.

[12] Full evaluation rubric, system prompt, and model configuration in Appendix A.3.

3.5.1 Shared Procedures

The *Shapiro–Wilk test* screens whether a sample is plausibly normal. With order statistics $x_{(i)}$ and tabulated constants a_i. It is applied as a normality screen to every user-study, NLP, and timing measure; normality is rejected on essentially all of them, which justifies the non-parametric pipeline that follows.

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}. \tag{3.10}$$

The *Bonferroni correction* divides the per-test threshold by the number of comparisons in a post-hoc family so the family-wise false-positive rate stays at α. Every post-hoc family compares three modes pairwise ($m = 3$), so $\alpha' = 0.05/3 = 0.0167$ throughout:

$$\alpha' = \alpha/m. \tag{3.11}$$

3.5.2 User-Study Instruments (Within-Subjects, $n = 69$)

Applied to *SUS*, *R-TLX*, the *perceived AI support* items, and the *perceived confidence* composite. Each participant rates every mode once, so the data are matched-pairs; Friedman with pairwise Wilcoxon signed-rank post-hocs is the standard non-parametric pipeline for ordinal Likert and workload data in repeated-measures usability studies [29].

The *Friedman test* [8] is the omnibus for $k \geq 3$ related samples. With n participants, k conditions, and rank sums R_j, asymptotically χ^2 distributed with $k - 1$ degrees of freedom. Its effect size is *Kendall's* W (coefficient of concordance) [17] on $[0, 1]$:

$$\chi_F^2 = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1), \tag{3.12}$$

$$W_{\text{Kendall}} = \chi_F^2/(n(k-1)). \tag{3.13}$$

When Friedman is significant, the *Wilcoxon signed-rank test* [37] is applied to each mode pair. Differences $d_i = x_i - y_i$ are ranked by absolute value, with the rank-biserial correlation as effect size on $[-1, +1]$. Positive r means the first-named mode ranks higher:

$$W = \min(W_+, W_-), \quad W_+ = \sum_{d_i>0} \mathrm{rank}(|d_i|), \quad W_- = \sum_{d_i<0} \mathrm{rank}(|d_i|), \tag{3.14}$$

$$r = (W_+ - W_-)/(W_+ + W_-). \tag{3.15}$$

3.5.3 Document-Quality Metrics and Completion Time (Between-Groups, $n = 113$)

Applied to BLEU, ROUGE-1/2/L, METEOR, BERTScore P/R/F1, SBERT, and per-mode completion time. The document sample is unbalanced across modes (Form 37, Ask 37, Chat 39); a between-groups family preserves all 113 submissions, where restricting to matched triples would discard valid data. Kruskal–Wallis with pairwise Mann–Whitney U post-hocs is the standard non-parametric counterpart to one-way ANOVA for unbalanced independent-groups designs [7]. The *Kruskal–Wallis H test* [18] is the omnibus. With N total observations, n_j in group j, and rank sums R_j, its effect size is ε^2 *(epsilon-squared)* [16] on $[0, 1]$, the share of total score variance explained by mode membership:

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1). \tag{3.16}$$

$$\varepsilon^2 = (H - k + 1)/(N - k). \tag{3.17}$$

When Kruskal–Wallis is significant, the *Mann–Whitney U test* [22] is applied to each mode pair. With group sizes n_1, n_2 and rank sum R_1, with rank-biserial r as effect size on the same $[-1, +1]$ scale as above. The same pipeline runs at both whole-document and per-chapter granularity, with Bonferroni correction applied within each (metric, chapter) family.

$$U = R_1 - \frac{n_1(n_1+1)}{2}, \tag{3.18}$$

3.5.4 Mode Preference Ranking ($n = 69$)

Each participant ranks the three modes 1st, 2nd, and 3rd at the end of the session. The inferential analysis collapses the ranking to its first-place column and tests whether the rank-1 vote distribution is uneven, which directly answers the headline question of whether a single dominant mode emerges. Reducing a preference ranking to its first-place counts for goodness-of-fit testing is a standard analytic choice when the question is whether one option dominates [29]; the 2nd- and 3rd-place ranks remain in the descriptive summary.

The *chi-square goodness-of-fit test* [27] compares observed counts O_i to expected counts E_i under a uniform null, its effect size is *Cramér's* V [6] on $[0, 1]$:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}. \tag{3.19}$$

$$V = \sqrt{\chi^2/(N(k-1))}. \tag{3.20}$$

With the tool, the study design, the instruments, the sample sizes, and the statistical machinery now in place, Chapter 4 reports the empirical findings from the user study (participant demographics, the four self-report instruments, the post-experiment ranking), and Chapter 5 reports the document-quality analyses (NLP metrics at document and chapter levels, followed by the supplementary LLM-as-a-Judge signal).

4 Qualitative Evaluation: User Study

This chapter reports the empirical findings of the user study. It opens with a descriptive demographics section that profiles the participant sample on which every result that follows rests. The four user-study instruments are then reported in the order in which they were administered after each interaction mode: *System Usability Scale*, *NASA Raw Task Load Index*, *Perceived AI Support*, and the post-experiment mode preference ranking together with the open-ended free-text comments.

4.1 Participant Demographics

This section presents the participant demographics. The study, set out in full in Chapter 3, recruited $N = 73$ participants who self-reported as ROPA novices for a within-subjects experiment in which every participant used all three ROPAgen interaction modes (Form, Ask, and Chat; see Section 3.1). Of the 73 recruited participants, four screened records were excluded for incomplete or invalid participation, yielding the matched within-subjects analysis sample of $n = 69$ on which every user-study result in this chapter rests. Mode order was counterbalanced across a six-group Balanced Latin Square so that learning, order, and fatigue effects could be distributed evenly across interaction modes rather than absorbed into any one of them. The task was designed around a single standardized GDPR scenario, the APOR GmbH employee and parking-management case at a fictional Ulm software company, as described in Section 3[1]. After interacting with each mode, participants completed the *SUS*, the *NASA R-TLX*, and the *Perceived AI Support* items, which are evaluated in the remaining sections of this chapter; at the end of the session, they ranked the three modes by overall preference and supplied the demographic

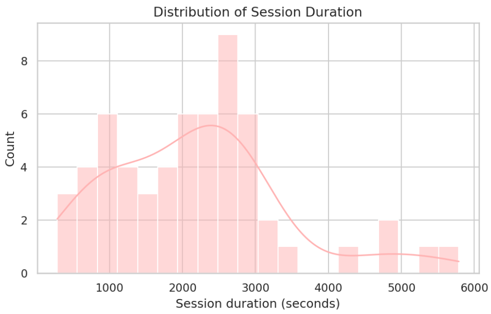
[1] Full scenario described in B.1

information profiled below. The demographic variables are reported descriptively only on the matched within-subjects sample of $n = 69$ participants.[2]

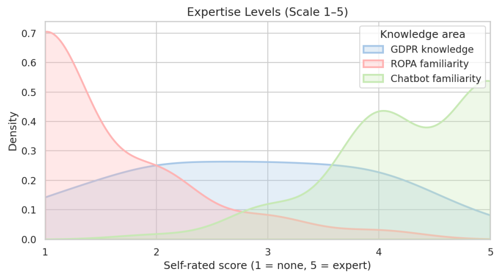
The mean session duration was 1866 s (≈ 31 min) per participant across the full session, see Figure 4.1a. Figure 4.1b shows the reported ROPA familiarity with a mean of 1.45. Chatbot familiarity is correspondingly high, the mean being 4.36, while GDPR knowledge sits near the scale midpoint (mean 2.81). Figure 4.1b makes the contrast vivid: chatbot familiarity concentrates at the high end, ROPA familiarity at the low end, and GDPR knowledge spreads broadly across the middle. In addition, the sample is dominated by Bachelor-level students (Figure 4.1e) studying or working in IT or Computer Science (Figure 4.1g). Of the participants who reported gender (Figure 4.1d), $62.1\,\%$ identified as male. The underlying age distribution (Figure 4.1c) clusters tightly around 24–26 years, with a thin right tail of older participants.

This profile (IT-leaning, chatbot-fluent, ROPA-novice) frames the results reported in the remainder of this chapter and in Chapter 5, and is taken up again in the Discussion chapter when generalizability is addressed.

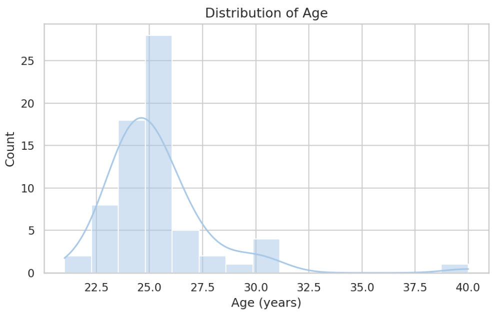
[2] Descriptive statistics for session duration and prior expertise, along with counts and inferential tests for the categorical demographic variables, are reported in Appendix B.3.



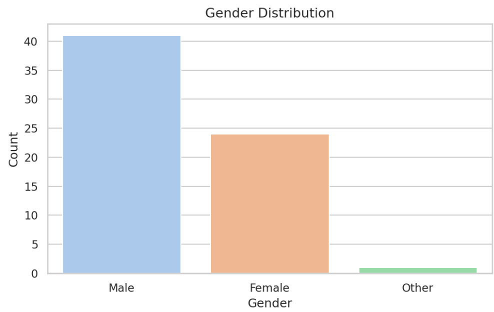
(a) Session duration (seconds).



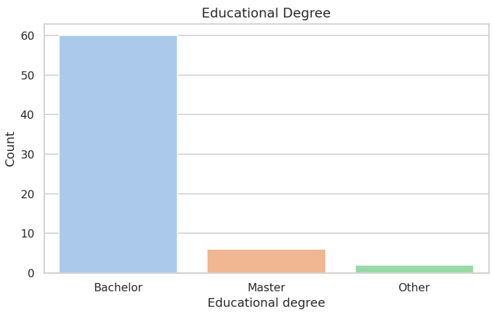
(b) Self-reported prior expertise (GDPR knowledge, ROPA familiarity, chatbot familiarity) on a 5-point Likert scale (1 = no familiarity, 5 = expert).



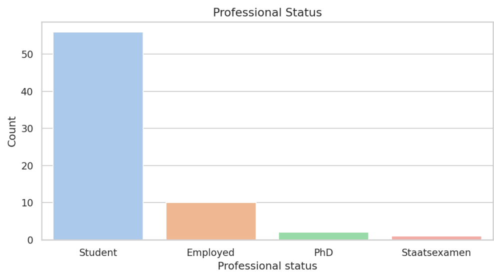
(c) Age distribution.



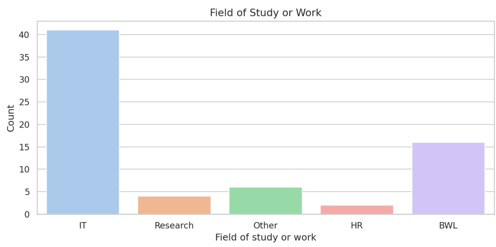
(d) Self-reported gender ($n = 66$; not all participants answered this item).



(e) Highest completed degree ($n = 68$; not all participants answered this item).



(f) Current professional status.



(g) Field of study or work.

Figure 4.1: Participant demographics of the $n = 69$ analysis sample: session duration, self-reported prior expertise on a 5-point Likert scale, and five categorical variables (age, gender, highest completed degree, professional status, and field of study or work).

4.2 ROPAgen Usability — System Usability Scale

The *System Usability Scale* (*SUS*) [5] captures how usable each interface felt to participants immediately after they finished interacting with it. It consists of ten Likert items on a five-point scale, with item polarity alternating (odd items: higher is better; even items: lower is better).[3]

The *SUS* yields a single composite score per mode on a 0–100 points scale, computed from the ten items using the standard scoring procedure of Brooke [5]. A score of 68 is widely used as the threshold above which a system is judged to have above-average usability [4]. Figure 4.2 reports the per-mode composite scores against this reference line. All three modes sit numerically above the threshold: Form scores highest at a mean of 78.62, while Ask and Chat sit closely together at 72.97 and 72.93 respectively. A one-sample Wilcoxon signed-rank test confirms that Form's composite is significantly above the 68 threshold, with a large effect.[4] Form leads the *SUS* evaluation, with Chat and Ask following closely behind; this pattern will become clearer across the user-study sections that follow. Item by item, Table 4.1 tells a Form-leading story.

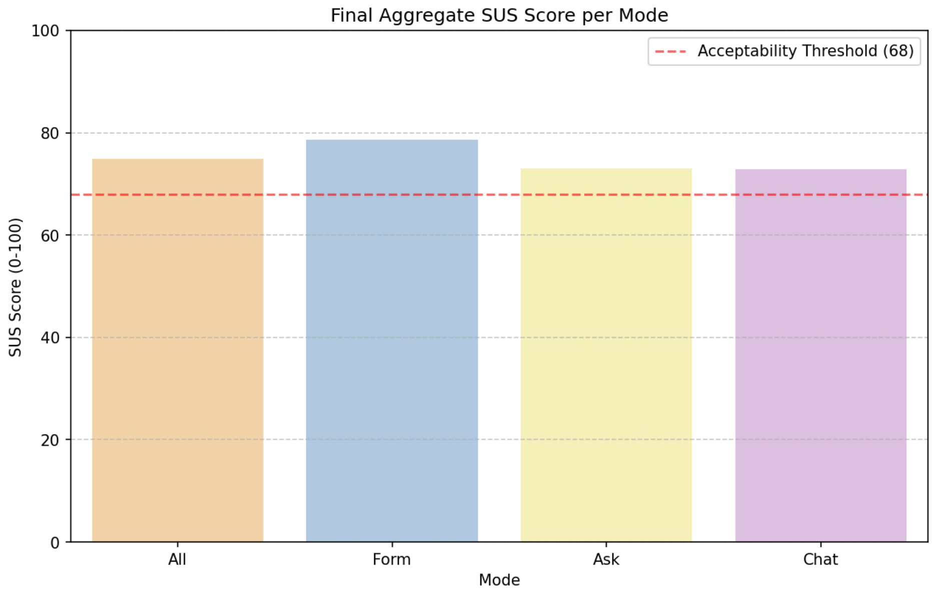


Figure 4.2: Mean SUS composite score per mode ($n = 69$ per mode) on the standard 0–100 scale. The dashed horizontal line marks the acceptability threshold of 68 [4].

[3] Full item text in Appendix B.2; per-item distributions and box plots in Appendix B.4.1.
[4] $W = 449$, $p < 0.001$, matched-pairs $r = +0.63$ ($n = 69$). Full statistics in Appendix C.

#	Item	Aggregated	Form	Ask	Chat
1	I think that I would like to use this system frequently.	3.32 (1.14)	**3.52** (1.09)	3.22 (1.08)	3.22 (1.24)
2	I found the system unnecessarily complex.	2.00 (1.12)	**1.64** (0.92)	2.25 (1.22)	2.12 (1.12)
3	I thought the system was easy to use.	4.13 (0.97)	**4.38** (0.91)	4.01 (0.81)	3.99 (1.13)
4	I think that I would need the support of a technical person to be able to use this system.	1.52 (0.89)	1.52 (0.85)	**1.49** (0.80)	1.55 (1.01)
5	I found the various functions in this system were well integrated.	3.77 (1.05)	**4.00** (0.92)	3.67 (1.02)	3.64 (1.16)
6	I thought there was too much inconsistency in this system.	2.07 (1.00)	2.16 (0.96)	2.04 (1.06)	**2.01** (0.98)
7	I would imagine that most people would learn to use this system very quickly.	4.11 (0.98)	**4.30** (0.96)	4.00 (0.95)	4.03 (1.01)
8	I found the system very cumbersome to use.	2.03 (1.13)	**1.65** (0.94)	2.16 (1.12)	2.29 (1.23)
9	I felt very confident using the system.	3.86 (1.04)	3.83 (1.04)	3.84 (0.93)	**3.90** (1.15)
10	I needed to learn a lot of things before I could get going with this system.	1.61 (0.90)	**1.61** (0.86)	**1.61** (0.81)	1.62 (1.03)

Table 4.1: System Usability Scale, mean (SD) per item by mode ($n = 69$ per mode), five-point Likert scale. The Aggregated column averages the three per-mode means. Odd items: higher is better; even items: lower is better.

Form leads on six of the ten items, and its lead is widest on the items that capture interaction overhead. Cumbersomeness shows the largest spread of any SUS item (Form 1.65 vs Chat 2.29, a 0.64-point gap), and complexity is close behind (Form 1.64 vs Ask 2.25). Form also leads on ease of use, perceived integration, willingness to use the system frequently, and learnability for others, each by roughly 0.30 to 0.40 Likert points. These are the items that capture the moment-to-moment feel of operating the system, and they are where the cost of LLM-driven interaction lands most directly on the user.

Chat takes two items by very thin margins: perceived inconsistency (2.01 vs Form 2.16) and within-system confidence (3.90 vs Form 3.83). Ask takes one, also narrow: whether the system would need a technical person's help to use (1.49 vs Form 1.52). Prior-learning (item 10) is effectively a three-way tie at the scale floor (Form and Ask both 1.61, Chat 1.62). On these four items the choice of interaction style does not move the rating.

Two Form-focused pairwise Wilcoxon signed-rank tests show Form significantly higher than both Ask and Chat with medium effects; Ask versus Chat is not tested. The Form lead holds inferentially against both Ask and Chat even though the three-way omnibus Friedman test on the per-mode SUS composite narrowly misses significance.[5]

The *SUS* results show Form as the clear leader, supported by three independent findings: Form is rated highest on six of ten items; Form's composite is significantly above the 68 acceptability threshold; and Form rated is significantly higher than both Ask and Chat on the Form-focused pairwise comparisons. The lead concentrates on items that capture the moment-to-moment feel of using the system; whether it carries to perceived workload is the question for Section 4.3.

[5] Bonferroni $\alpha = 0.025$ for two comparisons. Pairwise Ask $W = 578.5$, $p = 0.013$, $r = +0.37$; Chat $W = 549.5$, $p = 0.018$, $r = +0.36$; omnibus $\chi^2(2) = 5.51$, $p = 0.064$, $W = 0.040$. Full statistics in Appendix C.

4.3 Cognitive Load — NASA Raw Task Load Index

Where the *SUS* measured how usable the interfaces felt, the *Raw Task Load Index* (*RTLX*) [12] decomposes the participant's effort along six interpretable axes (Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, and Frustration), each rated on the standard NASA 21-point scale (1 = lowest demand, 21 = highest). Performance is reported in the positive-coded direction (higher equals worse perceived performance) for internal consistency with the other subscales. The overall *R-TLX* in this study is the unweighted mean of the six subscales on the same 1–21 scale (no rescaling is applied).[6] A short supplementary paragraph at the end of this section reports the objective per-mode completion time alongside the perceptual subscales, as a behavioral counterpart to Temporal Demand.

Figure 4.3 reports the composite R-TLX score per mode against the scale midpoint of 10.5, which marks the boundary between the lower and upper halves of the 1–21 range. All three modes sit comfortably in the lower half: Form scores lowest with a composite of 5.50, while Ask and Chat are closely tied at 6.33 each. As with the *SUS*, Form is the least demanding of the three at the aggregate level, with Ask and Chat following close behind by roughly 0.8 scale points.

Subscale	Aggregated	Form	Ask	Chat
Mental Demand	6.31 (4.75)	**5.77** (4.26)	6.80 (4.63)	6.36 (5.31)
Physical Demand	3.19 (3.40)	**2.77** (2.71)	3.54 (3.92)	3.26 (3.46)
Temporal Demand	4.89 (4.28)	**4.16** (3.82)	4.97 (4.15)	5.54 (4.78)
Effort	6.13 (4.57)	**5.68** (4.56)	6.39 (4.41)	6.32 (4.77)
Frustration	6.86 (5.24)	**5.88** (4.84)	7.58 (5.34)	7.13 (5.44)
Performance (pos.)	8.93 (5.82)	8.71 (5.48)	**8.68** (5.64)	9.39 (6.36)

Table 4.2: NASA R-TLX subscale mean (SD) by mode ($n = 69$ per mode), 1–21 scale. The Aggregated column averages the three per-mode means.

Subscale by subscale, Table 4.2 tells a Form-leading story with one exception. Form leads on five of the six subscales. The two where the lead reaches inferential significance are Temporal Demand and Frustration, and on both the gap widens monotonically as LLM support increases (Temporal Demand: Form 4.16, Ask 4.97, Chat 5.54; Frustration: Form 5.88, Chat 7.13, Ask 7.58). Form also leads on Mental

[6] Full subscale descriptions in Appendix B.2; per-subscale distributions and box plots in Appendix B.4.2.

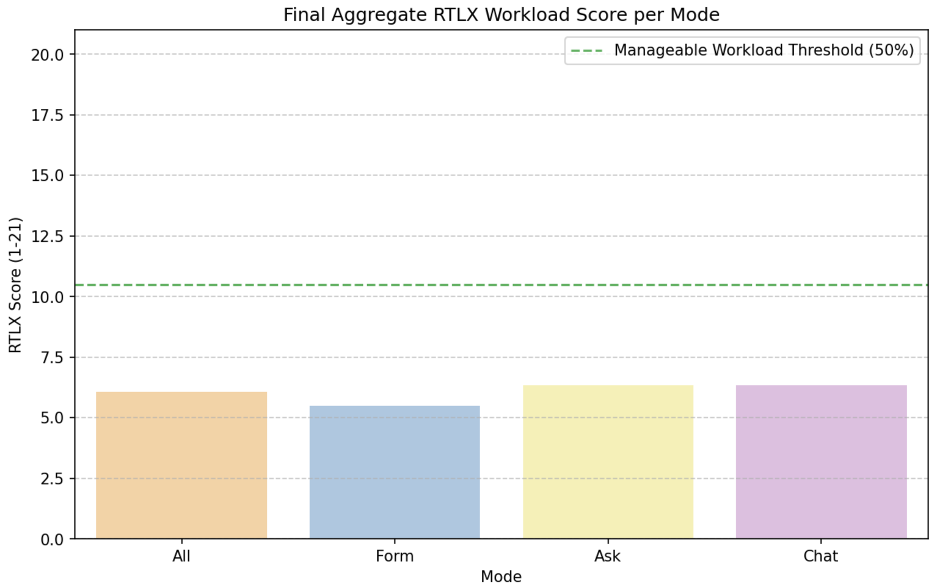


Figure 4.3: Mean R-TLX composite workload score per mode ($n = 69$), unweighted average of the six subscales on the 1–21 scale. The dashed horizontal line marks the scale midpoint (10.5), shown for visual reference.

Demand, Effort, and Physical Demand by roughly 0.6 to 1.0 points, though Physical Demand sits near the floor of the scale in all three modes since the task is not physically taxing. These are the workload dimensions where LLM-driven interaction measurably costs the user.

Performance is the one subscale where the Form lead disappears. Participants rated their own success at the task as essentially identical between Form (8.71) and Ask (8.68), with Chat marginally worse (9.39, higher equals worse perceived performance). Mode does not move how successfully participants think they completed the task; the slight self-doubt blip in Chat is the only deviation.

Two subscales reach significance individually: Temporal Demand and Frustration. Pairwise Wilcoxon signed-rank tests with Bonferroni correction confirm Form differs significantly from both Ask and Chat on these subscales, with medium-to-large rank-biserial effects. The composite workload score across the six subscales does not differ significantly between modes, and Ask versus Chat is not significant on any measure.[7]

Overall, the *R-TLX* reinforces the *SUS* Form lead: Form sits below both Ask and

[7] Temporal Demand $\chi^2(2) = 12.77$, $p = 0.002$, $W = 0.093$; Frustration $\chi^2(2) = 8.93$, $p = 0.012$, $W = 0.065$; pairwise effect sizes $|r| = 0.37$ to 0.57; aggregated Friedman $\chi^2(2) = 5.24$, $p = 0.073$, $W = 0.038$. Full statistics in Appendix C.

Chat on every workload dimension except Performance, with two subscales (Temporal Demand and Frustration) reaching inferential significance individually even when the composite test does not. The clearest significant result on the user side is therefore about perceived workload, not usability. Whether perception of time pressure matches actual time taken is the question for the supplementary completion-time analysis that follows.

The R-TLX subscales are perceptual; whether the perceived time pressure (Form lowest, Chat highest) tracks the actual time spent is a separate, behavioral question. Every ROPA document carries a `Time elapsed:` N `seconds` attribute at submission, giving one objective completion-time measurement per participant per mode on the full $n = 113$ document sample.

Form takes about half the time of either Ask or Chat. The median completion time was 7 minutes for Form (420 s), 14 minutes for Ask (858 s), and 12 minutes for Chat (717 s); Table 4.3 reports the per-mode mean (SD) and median alongside.[8]

Mode	n	Mean (SD)	Median
Form	37	**564.2** (394.4)	**420**
Ask	37	989.8 (675.8)	858
Chat	39	979.1 (1015.4)	717

Table 4.3: Per-mode completion time in seconds, extracted from the `Time elapsed` line emitted by ROPAgen at document submission ($n = 113$ documents: Form 37, Ask 37, Chat 39).

A Kruskal–Wallis test on the full sample confirms that completion time varies by mode, with a large effect size. Form is significantly faster than both Ask (median 420 s vs 858 s, large effect) and Chat (median 420 s vs 717 s, medium effect), while Ask and Chat are statistically indistinguishable from each other.[9]

The behavioral data sharpens the perceived workload picture: Form is not merely felt as the least time-pressured mode, it is also measurably the fastest by a margin of roughly seven minutes per document over either Ask or Chat. The two modes, despite their very different interaction styles (per-section explainer versus freeform conversation), take essentially the same amount of time on average. Whether participants nevertheless ascribe more value to Ask and Chat when asked specifically about AI support is the question taken up in the next Section (4.4).

[8] Full per-mode distribution in Figure B.4 (Appendix B.4.2).

[9] Kruskal–Wallis $H = 16.46, p < 0.001, \varepsilon^2 = 0.15$; Form vs Ask $r = 0.512$; Form vs Chat $r = 0.418$.

4.4 Perceived AI Support

The *SUS* (Section 4.2) and *R-TLX* (Section 4.3) characterize the interface as it was experienced; they do not speak to whether participants perceived the LLM itself as useful. The *Perceived LLM Support* questionnaire targets that gap. After each interaction mode, participants completed a six-item questionnaire on a five-point Likert scale (1 = strong disagreement, 5 = strong agreement): helpfulness (item 1), efficiency (item 2), transparency (item 3), trust (item 4), legal-basis confidence (item 5), and re-use intent (item 6). Items 4 and 5 are averaged into the *perceived confidence* composite that serves as the primary self-report confidence measure referenced in Chapter 3; the remaining four items are reported individually.[10]

#	Item	Aggregated	Form	Ask	Chat
1	The LLM was helpful in creating this document.	4.23 (0.91)	4.23 (0.97)	**4.32** (0.76)	4.13 (1.00)
2	The LLM helped me to complete the document faster.	4.22 (1.13)	**4.43** (0.96)	4.25 (1.03)	3.99 (1.32)
3	I understood why the LLM suggested specific GDPR fields.	3.81 (1.06)	3.51 (1.17)	**4.16** (0.92)	3.75 (0.98)
4	I trust that the LLM made correct suggestions.	3.63 (0.97)	3.41 (1.09)	**3.80** (0.87)	3.68 (0.92)
5	I am confident that the LLM identified the correct legal basis.	3.71 (0.88)	3.58 (0.90)	3.74 (0.87)	**3.81** (0.88)
6	I would use this mode again.	3.65 (1.25)	**3.99** (1.14)	3.55 (1.16)	3.42 (1.39)

Table 4.4: Perceived LLM Support, mean (SD) per item by mode ($n = 69$ per mode), five-point Likert scale (higher is better). The Aggregated column averages the three per-mode means.

Item by item, Table 4.4 splits the picture by what each question targets: each mode leads on a different cluster. Ask leads on the three content-judgment items: transparency (4.16 vs Form 3.51, the widest reversal of the Form-leading pattern in the section), trust (3.80 vs Form 3.41), and overall helpfulness (4.32, only a 0.19-point margin over Form and Chat). Ask is the only mode that explicitly explains each suggestion (Section 3.1); the transparency and trust leads follow directly from that design.

Chat leads only on legal-basis confidence (3.81 vs Form 3.58); on the broader trust item, Ask leads (3.80 vs Form 3.41). Averaged into the *perceived confidence* com-

[10] Full per-item distribution in Appendix B.4.3.

posite, Ask and Chat tie at the top (3.77 and 3.75), both clearly above Form (3.49). Confidence in the legal basis specifically is highest in the mode that exposes the LLM's reasoning least: Chat dissolves legal judgment into conversational fluency, while Ask presents it as explicit, second-guessable explanations. Form leads on the two productivity-and-preference items: perceived speed-up (4.43, decreasing monotonically with LLM involvement to Chat 3.99) and re-use intent (3.99 vs Chat 3.42). Re-use intent is the widest negative gap of the six items and the only one to reach statistical significance after Bonferroni correction, with a large Form-versus-Chat effect; even when participants felt better about the LLM's content judgment, they still wanted to come back to Form. Perceived LLM support taken as a whole does not differ across modes; the perceived confidence composite shows a Friedman trend ($\chi^2(2) = 4.50$, $p = 0.105$, $W = 0.033$) below conventional significance, and the legal-basis confidence item on its own does not reach it either.[11]

Form is the fastest mode, both in user perception (perceived speed-up, item 2) and in actual completion time (Section 4.3). Form is also the user favorite, both by re-use intent here (item 6, the only Perceived AI Support item to reach statistical significance after Bonferroni correction, with a large Form-versus-Chat effect) and by overall ranking ahead (Section 4.5). Users nevertheless appreciate the transparency of Ask (items 3 and 4). In Ask they have to write the final content themselves, but they can query the LLM about each field and what to do (see Section 3.1). Form does not offer that on-demand explanation, and Chat removes the control entirely by populating the document in the background as the conversation unfolds.

Two findings cut against the headline picture. Users credit the AI's content judgment in Ask and Chat over Form: Ask leads on overall trust, Chat on legal-basis confidence, and the two-item perceived confidence composite shows Ask and Chat tied above Form. And although users rated Ask the most helpful mode overall (item 1, by a thin 0.19-point margin), they still intend to come back to Form (item 6). The behavioral counterpart, which mode participants actually choose when forced to rank, is taken up next in Section 4.5.

[11] Re-use intent $\chi^2(2) = 11.12$, $p = 0.004$; Form vs Chat $r = +0.543$. Full statistics in Appendix C.

4.5 Mode Preference and Free-Text Feedback

The per-instrument data, so far, have shown a consistent Form-leading pattern on usability (Section 4.2) and workload (Section 4.3), with a partial inversion on perceived transparency, trust, and perceived confidence (Section 4.4). This section analyzes how the three patterns combine when participants are forced to commit to a single overall preference. After all three interaction modes had been completed, participants provided two final inputs: an overall ranking of the three modes from most to least preferred, and an open-ended summary comment on the study experience. In addition, after each individual mode, an optional per-mode comment was collected.

Mode Preference Ranking

The ranking offers the cleanest test of which mode participants actually preferred once they had used all three. Table 4.5 and Figure 4.4 summarize the post-experiment ranking. Participants assigned each of the three modes a rank from 1 (most preferred) to 3 (least preferred). "Rank-1 votes" counts the participants who placed the mode first.

Mode	Rank-1 votes	Share	Mean rank
Form	**37**	**53.6 %**	**1.78**
Chat	20	29.0 %	2.07
Ask	12	17.4 %	2.14

Table 4.5: Post-experiment mode preference ranking ($n = 69$). "Rank-1 votes" counts the number of participants who placed the mode first; "Share" is the corresponding percentage; "Mean rank" averages the assigned 1–3 rank across all 69 participants.

Form received the plurality of first-place votes (37 of 69, $53.6\,\%$), followed by Chat (20, $29.0\,\%$) and Ask (12, $17.4\,\%$), with mean ranks following the same ordering (1.78, 2.07, 2.14). The preference for Form is statistically significant, with a medium effect size.[12]

[12] $\chi^2(2) = 14.17$, $p < 0.001$, Cramér's $V = 0.32$. Full statistics in Appendix C.

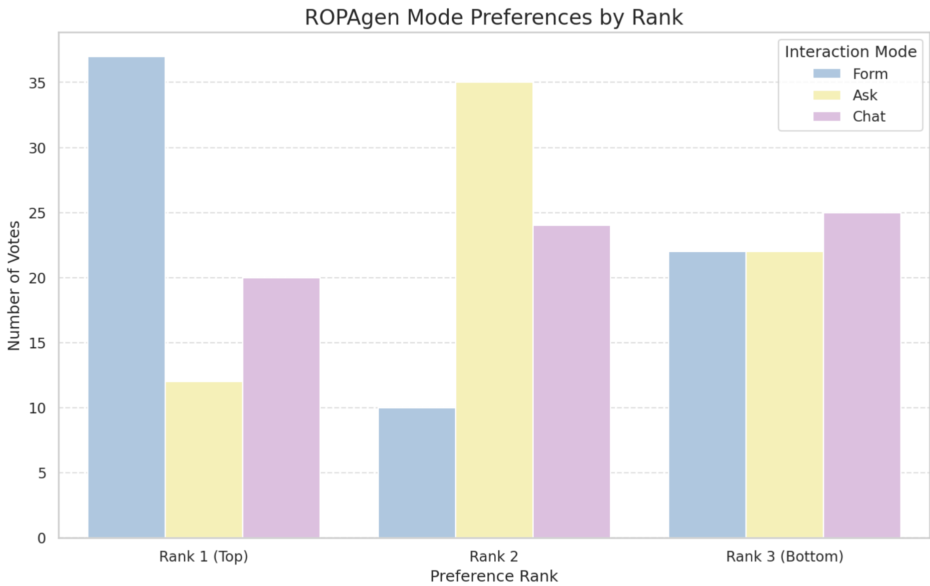


Figure 4.4: Distribution of all three rank positions per mode ($n = 69$).

Free-Text Comments

The free-text comments contextualize the ranking by surfacing the reasons participants gave for their preferences. Each mode collected an optional per-mode open-ended comment, and after all three modes participants wrote an overall summary comment. A total of 88 comments were recorded across 44 unique participants. Comments are written in German with a small minority in English; the paragraphs that follow summarize them thematically in the order Form, Ask, Chat, and close with the across-modes overall comments.

Form. Comments on Form cluster around speed, clarity, and control. Recurring observations describe the mode as the fastest of the three, the easiest to obtain an overview of, and the most "integrated" into the document creation workflow, as the checkbox interface makes the selection space visible at a glance. Several participants emphasize that Form "takes the most work off you" (*"nimmt dir am meisten Arbeit ab"*) by pre-populating the option set, while a smaller subset notes the opposite concern: that the absence of an interactive LLM commentary makes it unclear on what basis the model produces its suggestions, and that participants would not feel safe relying on Form output without an additional verification step. Several comments describe specific UX issues: long text fields rendered as a single narrow line on mobile, the inability to provide the LLM with additional context for its suggestions, and the impression that the LLM "guesses" without scenario-specific input.

Ask. Comments on Ask are the most polarized of the three modes. A consistent theme is that Ask is the most cumbersome mode in terms of workflow: the requirement to manually copy the LLM's suggested text into the form field (cf. Section 3.1) is repeatedly described as redundant work, particularly as participants reported that the model often produced suggestions in a free-text style that did not match the structured form fields. Several participants observed that the lack of a persistent context across fields forced them to re-enter the same information repeatedly. Conversely, explanatory text generated by Ask is consistently described as the most informative of the three modes, supporting the high transparency rating visible in item 3 of the AI Support questionnaire. A frequent suggestion is to combine Ask's explanations with Form's structured field layout.

Chat. Comments on Chat are dominated by two themes: comfort and opacity. On the comfort side, several participants describe Chat as the most intuitive of the three modes, especially as the LLM takes over the work of mapping the conversation onto the structured ROPA fields (cf. Section 3.1), leaving the participant only with the task of describing the processing scenario. On the opacity side, an equally large group of participants reports uncertainty about what is actually being written into the document — whether the entire chat history is used for generation, whether each individual message must be confirmed, and what the document looks like during the conversation. Specific defects mentioned include the LLM occasionally switching from German to English mid-response, occasional emission of raw `DATA_UPDATE` tokens, and a slow "streaming" rendering style that participants experienced as artificially throttled.

Overall comments The overall comments recapitulate the per-mode pattern. A clear majority describes Form as the fastest and most ergonomic mode, particularly for participants who feel they "do not know enough" to interrogate the model meaningfully. Chat is typically placed second and motivated by the impression that "the AI takes care of it," which several participants explicitly note increases their *subjective* sense of safety even though they cannot verify the output. Ask is most often placed third, with the manual copy-paste step cited as the principal reason. A recurring constructive suggestion across participants is a hybrid mode that combines the visible option set of Form with the conversational, context-aware behavior of Chat or Ask. Several participants explicitly add a caveat that they are novices in the legal domain and that their preference ordering might shift if they had prior GDPR knowledge.

The ranking and comment data align with the *SUS* and *R-TLX* picture. Form is the most-chosen mode by a clear margin; Chat slots in second on raw votes; Ask is most often placed third, with free-text comments citing the manual copy-paste step between explanation and form field as the principal friction. Across the comments the most consistent constructive suggestion is for a hybrid mode that combines Form's visible option set with Chat's or Ask's conversational behavior.

4.6 Summary

Taken together, the four user-study instruments (*SUS*, *RTLX*, *Perceived AI Support*, and the free-form ranking) and the behavioral completion-time measure converge on a Form-leading picture. Form is the most usable mode on the *SUS*, leading on six of ten items. On the *R-TLX*, it imposes the lowest perceived workload, falling below both Ask and Chat on every subscale except Performance; Temporal Demand and Frustration individually reach significance. In actual completion time, Form is roughly twice as fast as either Ask or Chat, a difference that is both statistically significant and large in effect size. Finally, when participants commit to a single overall ranking, Form is the most preferred mode, receiving $53.6\,\%$ of first-place votes. The one place where this picture breaks is at perceived AI support: when asked specifically about the LLM's contribution, participants credit Ask and Chat on the items about content judgment (transparency, trust, and the perceived confidence composite) and credit Form on the items about productivity and re-use intent. Participants are thus willing to credit Ask and Chat for the AI support, yet still prefer Form for how the tool feels to use. Whether Form leads on document quality as it does on user experience is the question for the next section.

5 Quantitative Evaluation: Document Quality

This chapter reports the document-quality half of the evaluation. The reference-based NLP metrics are reported first at both the document and chapter levels; the LLM-as-a-Judge layer is reported afterward as a supplementary, reference-free signal.

5.1 NLP Metrics

The user study (Chapter 4) revealed a stable Form-leading pattern: Form was rated the most usable, the lightest on workload, and the most often ranked first. This section analyzes the document-quality part of the evaluation by asking whether the same ordering holds when the output text itself is compared against a reference document. The reference-based metric suite mirrors the one applied by von Schwerin et al. [30] in their evaluation of LLM-generated ROPA data categories: five lexical metrics (BLEU [26], ROUGE-1/2/L [20], METEOR [3]) that count surface-form token overlap, and three BERT-based metrics (BERTScore Precision / Recall / F1) [38] plus document-level SBERT cosine similarity [28] that compare semantic representations. All metrics are computed against the single expert-authored Form-mode reference document described in Chapter 3.

The metrics are computed at two granularities. At the **document level** (Section 5.1.1), each participant ROPA is treated as a single block of text and scored against the full reference document. At the **chapter level** (Section 5.1.2), the same metrics are recomputed for each of the six rendered Article 30(1) ROPA chapters separately, scoring each participant chapter against the matching reference chapter. SBERT cosine similarity uses ModernBERT [2] as the primary encoder at both granularities; at the chapter level the smaller `all-MiniLM-L6-v2` [31] encoder is added as a robustness check on whether ModernBERT's near-saturation masks real differences. Finally, Section 5.1.3 compares the two views.

5.1.1 Document-Level NLP Metrics

At this granularity each participant ROPA is treated as a single block of text and scored against the full expert-authored reference. Document-quality results are reported on the full retained set of 113 documents (Form $n = 37$, Ask $n = 37$, Chat $n = 39$). Table 5.1 gives a one-glance summary of mean (SD) per mode for every document-level metric examined in the sections that follow. Inferential mode comparisons use Kruskal–Wallis as the omnibus test with Bonferroni-corrected Mann–Whitney U as the pairwise post-hoc.[1]

Metric	Aggregated	Form	Ask	Chat
BLEU	0.133 (0.048)	0.129 (0.048)	0.127 (0.056)	**0.143** (0.039)
ROUGE-1	0.488 (0.087)	0.488 (0.098)	0.465 (0.088)	**0.511** (0.071)
ROUGE-2	0.285 (0.077)	**0.306** (0.076)	0.257 (0.080)	0.291 (0.069)
ROUGE-L	0.375 (0.087)	0.388 (0.094)	0.347 (0.091)	**0.389** (0.072)
METEOR	0.389 (0.062)	**0.412** (0.051)	0.365 (0.068)	0.390 (0.057)
BERTScore Precision	0.890 (0.014)	0.889 (0.014)	0.886 (0.016)	**0.895** (0.012)
BERTScore Recall	0.915 (0.014)	**0.923** (0.005)	0.908 (0.017)	0.913 (0.013)
BERTScore F1	0.902 (0.011)	**0.905** (0.009)	0.897 (0.013)	0.904 (0.010)
SBERT (ModernBERT)	0.987 (0.004)	0.986 (0.004)	0.986 (0.005)	**0.990** (0.002)

Table 5.1: Document-level NLP metrics, mean (SD) per mode (full sample $n = 113$). Lexical metrics in the upper block, embedding-based metrics in the lower.

Lexical Metrics

Lexical metrics (BLEU, ROUGE, and METEOR) count surface-form token overlap with the reference; they reward word-level fidelity but are blind to paraphrase. A document that covers the same content in different wording scores lower here than under the embedding-based metrics that follow.

[1] Full test statistics in Appendix C; full per-document distributions and box plots in Appendix B.7.

Read across Table 5.1, the five lexical metrics split their winners cleanly between the two LLM-assisted-versus-structured ends of the design. Chat takes the precision-oriented and longest-subsequence metrics: BLEU (0.143), ROUGE-1 (0.511), and ROUGE-L (0.389). Form takes the two metrics that reward reproducing reference wording most faithfully: ROUGE-2 (0.306) and METEOR (0.412). Ask trails on every lexical metric. Three of the five (ROUGE-2, ROUGE-L, METEOR) reach statistical significance on the full sample; BLEU and ROUGE-1 do not.[2] BLEU illustrates the limit sharpest: lexical metrics measure shape, not truth, so a fabrication phrased in the reference's vocabulary outscores an accurate paraphrase that does not share its wording.

BERT-Based and Sentence-Embedding Metrics

BERT-based metrics measure semantic rather than lexical similarity by comparing contextual embeddings rather than literal tokens. Where the lexical metrics penalize paraphrase, the embedding-based metrics credit semantic overlap and therefore offer a complementary view of the same documents.

The embedding-based row of Table 5.1 splits the four metrics symmetrically. Form wins the recall-oriented pair: BERTScore Recall (0.923) and BERTScore F1 (0.905), with Form's Recall standard deviation the smallest of any mode-metric cell in the table (SD 0.005). Chat wins the precision-oriented pair: BERTScore Precision (0.895) and SBERT cosine similarity (0.990). Ask trails on every metric and is the inferior mode in every significant pairwise comparison. All four embedding-based metrics reach statistical significance.[3] The reading is a coverage-versus-discipline trade-

[2] Kruskal–Wallis on the full sample: BLEU $H = 4.48$, $p = 0.107$; ROUGE-1 $H = 5.57$, $p = 0.062$; ROUGE-2 $H = 9.56$, $p = 0.008$, $\varepsilon^2 = 0.069$ (moderate); ROUGE-L $H = 7.79$, $p = 0.020$; METEOR $H = 9.80$, $p = 0.007$, $\varepsilon^2 = 0.071$ (moderate). Pairwise Mann–Whitney post-hoc (Bonferroni-corrected): ROUGE-2 Form $>$ Ask $p = 0.006$, $r = 0.369$ (medium); ROUGE-2 Chat $>$ Ask $p = 0.011$, $r = 0.342$ (medium); METEOR Form $>$ Ask $p = 0.003$, $r = 0.407$ (medium); no pairwise ROUGE-L comparison survives Bonferroni correction. Full table in Appendix C.

[3] Kruskal–Wallis on the full sample: BERTScore Precision $H = 9.11$, $p = 0.011$; BERTScore Recall $H = 32.38$, $p < 0.001$, $\varepsilon^2 = 0.276$ (large); BERTScore F1 $H = 9.75$, $p = 0.008$, $\varepsilon^2 = 0.071$ (moderate); SBERT (ModernBERT) $H = 17.43$, $p < 0.001$, $\varepsilon^2 = 0.140$ (large). Pairwise Mann–Whitney post-hoc (Bonferroni-corrected): BERTScore Recall Form $>$ Ask $p < 0.001$, $r = 0.756$ (large); Recall Form $>$ Chat $p < 0.001$, $r = 0.512$ (large); F1 Form $>$ Ask $p = 0.004$, $r = 0.386$ (medium); F1 Chat $>$ Ask $p = 0.011$, $r = 0.340$ (medium); SBERT Chat $>$ Form $p < 0.001$, $r = 0.505$ (large); SBERT Chat $>$ Ask $p < 0.001$, $r = 0.453$ (large); no pairwise BERTScore Precision comparison survives Bonferroni correction. Full table in Appendix C.

off: Form casts a wide net and reproduces more of the reference content but also includes more material the reference does not; Chat stays tightly on topic but leaves more out; their strengths balance out on F1, and Ask underperforms on both.

At the document level the user-study Form-leading verdict breaks into a trade-off. Form documents recover more of what the reference *says*, Chat documents stay closer to what the reference *means*, and Ask documents do neither.

5.1.2 Chapter-Level NLP Metrics

At this granularity, each participant chapter is scored against the matching chapter of the reference document, so the per-mode differences can be localized to specific parts of the ROPA. The six rendered chapters produced by the `finalropa` prompt are listed in Table 5.2; they derive from the eight internal `DocumentData` groups described in Chapter 3 (Section 3.1), with the data-categories and data-sources groups merging into one output chapter, the metadata group dropped, and the free-text fields distributed into the purpose and T&O chapters.

#	German title	English gloss
1	Verantwortliche Stelle und Kontaktdaten	Controller and Contact Details
2	Zwecke und Rechtsgrundlagen der Verarbeitung	Purposes and Legal Bases of Processing
3	Kategorien personenbezogener Daten und Datenquellen	Categories of Personal Data and Sources
4	Betroffene Personen und Empfänger	Data Subjects and Recipients
5	Aufbewahrungsfristen und Löschung	Retention Periods and Deletion
6	Technische und organisatorische Maßnahmen	Technical and Organizational Measures

Table 5.2: The six rendered ROPA chapters. German titles are the verbatim form participants saw; English glosses are provided for the international reader.

Across modes, the chapter-level lead splits cleanly into zones. Chapter 1 is filled identically across all three modes by the user (the LLM has no role in its content), therefore any variance is noise rather than signal. The chapter is reported here for completeness but is excluded from the mode-comparison reading. the numerical lead Ask carries on Chapter 1 reflects an artefact of how the system assembles the controller block rather than a mode-architecture effect; the chapter is reported here for completeness but is excluded from the mode-comparison reading. On the chapters whose content the modes actually drive, the split holds on every lexical metric without exception and is preserved by the BERT-based view: Form leads on Chapters 2–4, and Chat leads on Chapters 5–6. Chapter 1 still sits at the head of every metric in absolute terms; Chapter 5 carries the widest standard deviations.[4] Table 5.3 summarizes the three-zone pattern using one headline metric from each of three families: METEOR on the lexical side (chosen over BLEU and ROUGE because its WordNet-based synonym and stem matching credits meaning-preserving paraphrase that pure n-gram metrics penalize, giving the fairest single-number lexical reading), BERTScore F1 as the token-level embedding metric, and SBERT cosine similarity as the passage-level embedding metric. Showing all three together makes the granularity story visible at a glance: where token-level and passage-level embeddings agree on a winner, the chapter's trade-off is consistent across resolutions; where they disagree, the disagreement is itself diagnostic of how the modes differ on that chapter (a token-level match without a passage-level match suggests reproduced vocabulary without preserved meaning, and the reverse suggests preserved meaning under different wording).

#	Chapter	METEOR	BERTScore F1	SBERT
1	Controller and Contact Details	Ask 0.506	Ask 0.875	Ask 0.977
2	Purposes and Legal Bases	Form 0.452	Form 0.891	Chat 0.981
3	Categories of Personal Data	Form 0.415	Form 0.903	Form 0.982
4	Data Subjects and Recipients	Form 0.440	Form 0.879	Form 0.973
5	Retention and Deletion	Chat 0.362	Chat 0.867	Chat 0.955
6	Technical and Organizational Measures	Chat 0.305	Chat 0.878	Chat 0.975

Table 5.3: Per-chapter winning mode on the headline lexical (METEOR), token-level embedding (BERTScore F1), and passage-level embedding (SBERT ModernBERT) metrics, with the winning value. All three scales are "higher is better". Chapter 1 is filled identically across all three modes by the system itself; any apparent winner there is not interpretable as a mode-architecture effect. The secondary MiniLM SBERT pass agrees with ModernBERT on the overall direction on Chapters 1, 3, 5, and 6 but disagrees on Chapters 2 and 4 (where MiniLM places Form ahead of Chat), a disagreement traceable to MiniLM's shorter context and stricter passage compression. The full per-chapter per-mode means for every metric are in Appendix B.8.

[4] Per-chapter, per-mode tables in Appendix B.8 (lexical in Appendix B.8.1, embedding-based in Appendix B.8.2); chapter-level box plots are reproduced there as well.

Lexical Metrics

Chapter 1 (Controller and Contact Details) shows Ask numerically ahead of the other two modes on every lexical metric but is excluded from the mode-comparison reading. The three middle chapters (Purposes and Legal Bases, Categories of Personal Data, Data Subjects and Recipients) all favor Form, with the gap widest on Chapter 3, where Form's ROUGE-1 of 0.600 leads Ask (0.366) by the largest within-chapter gap of the lexical results and Form also sweeps BLEU and METEOR. Chapters 2 and 4 follow the same Form-leading pattern on a smaller scale. The last two chapters flip to Chat: Chapter 5 (Retention and Deletion) sits at the lowest absolute scores and the widest variance of the six, with Chat ahead on every lexical metric and METEOR's standard deviation of 0.160 the largest of any chapter; Chapter 6 (Technical and Organizational Measures) repeats the Chat lead at a smaller scale.

BERT-Based and Sentence-Embedding Metrics

The embedding-based view keeps the same chapter-level pattern as the lexical metrics but compresses the absolute range. Chapter 1 stays at the top of the BERT range with Ask numerically ahead. The smaller MiniLM encoder makes the apparent Ask lead more visible (SBERT MiniLM Ask 0.736 versus Form 0.663, Chat 0.662). The middle chapters favor Form, with the document-level Recall-versus-Precision trade-off appearing locally on Chapters 2 and 4: Form leads Recall and F1 while Chat takes a narrow Precision lead on both. Chapter 3 is the cleanest sweep, with Form highest on every BERT-based metric (F1 0.903 versus Ask 0.865, Chat 0.872). The last two chapters favour Chat: Chapter 5 has Chat ahead on every BERT-based metric, and the MiniLM SBERT pass widens the gap visibly (Chat 0.566 versus Form 0.516, Ask 0.474); Chapter 6 has Chat narrowly ahead on every BERT-based metric.

Cell-wise Kruskal–Wallis tests on the full $n = 113$ sample confirm the descriptive picture: most chapter-level mode differences are significant. The largest single gap is Form's ROUGE-1 lead over Ask on *Categories of Personal Data*, with a rank-biserial r near 0.95. Per-cell test statistics are kept in the pipeline output files; the appendix focuses on the document-level inferential layer.

5.1.3 Summary of Document-Level and Chapter-Level NLP Metrics

The two granularities (document-level and chapter-level analysis) tell the same story at different resolutions. At the document level the trade-off runs across metrics: Form leads the recall-oriented and bigram-sensitive metrics (most clearly BERTScore Recall), Chat leads the embedding-based ones (BERTScore Precision and SBERT), and Ask is the lower of every significant pairwise comparison. Form wins Chapters 2–4, Chat wins Chapters 5 and 6, and the Form-leads-Recall / Chat-leads-Precision split re-emerges specifically on Chapters 2 and 4. Document-level means are higher than the average of chapter-level means (Table 5.4) as the document-level calculation pools tokens across all six chapters and smooths out the per-chapter dips. The chapter-level view shows *where* the modes differ; the document-level view ranks them *overall*. The reference-based metrics neither confirm nor contradict the Form-leading verdict from the user study (Sections 4.2–4.5). They deliver a trade-off instead: Form documents recover more of what the reference *says*, Chat documents stay closer to what it *means*, and Ask trails on every front.

A final observation cuts across metric families. The five lexical metrics and the four BERT-based metrics measure different things, surface-form n-gram overlap on the one hand and contextual semantic similarity on the other, but they order the modes the same way. Lexical recall (ROUGE-Recall) and semantic recall (BERTScore Recall) both place Form ahead; lexical-precision oriented BLEU and semantic Precision (BERTScore Precision) both align with Chat at the document level; the balanced-similarity metrics (METEOR, BERTScore F1, SBERT) sit in between. In sum, no single mode produces categorically better documents than the others; which mode leads depends on which property of the documents the metric focuses on.

	Form		Ask		Chat	
Metric	Doc	Chapter	Doc	Chapter	Doc	Chapter
BLEU	0.129	**0.112**	0.127	0.091	**0.143**	0.099
ROUGE-1	0.488	**0.444**	0.465	0.380	**0.511**	0.416
ROUGE-2	**0.306**	**0.290**	0.257	0.218	0.291	0.247
ROUGE-L	0.388	**0.396**	0.347	0.324	**0.389**	0.365
METEOR	**0.412**	**0.398**	0.365	0.328	0.390	0.349
BERTScore Precision	0.889	0.856	0.886	0.855	**0.895**	**0.864**
BERTScore Recall	**0.923**	**0.897**	0.908	0.879	0.913	0.883
BERTScore F1	**0.905**	**0.876**	0.897	0.866	0.904	0.873
SBERT (ModernBERT)	0.986	0.970	0.986	0.968	**0.990**	**0.971**

Table 5.4: Document-level versus mean chapter-level metric values by mode. The mean chapter-level value is the unweighted average across the six chapters. Differences are document-level minus chapter-level mean. Source: `doc_vs_chapter_diff.csv`.

5.2 LLM-as-a-Judge

The reference-based NLP metrics measured similarity to a single expert-authored Form-mode reference. The LLM-as-a-Judge layer turns to a reference-free quality signal: each document is evaluated against a rubric rather than a target, which lifts the implicit advantage that the reference document confers on Form-style outputs. The judge produces five metrics per document, summarized in Table 5.6: *Completeness*, *Scenario Faithfulness*, *Legal Correctness*, *Hallucination* (inverse-coded; higher equals less hallucination), and *Overall*.[5] The rubric design follows the precedent set by Su et al.'s JuDGE benchmark for LLM-evaluated legal-document generation [33]. The judge is a single LLM (Gemini 3.1 Pro Preview, temperature 0); without inter-rater replication or legal-expert validation the resulting scores are reported as a supplementary, indicative signal alongside the user-study instruments and reference-based NLP metrics. All 113 retained documents are scored. Table 5.5 reports the per-metric mean and standard deviation per mode. The five rubric metrics split their winners between Form and Chat with Ask between them on every metric: Form leads Completeness and Legal Correctness; Chat leads Scenario Faithfulness, Hallucination (inverse), and Overall.

Three patterns emerge. First, the Completeness ordering (Form $>$ Ask $>$ Chat) is the exact inverse of the Scenario Faithfulness and Hallucination orderings (Form $<$ Ask $<$ Chat): Form documents are the most structurally complete but also the most

[5]Full rubric, prompt, and model configuration in Appendix A.3.2.

Metric	Aggregated	Form	Ask	Chat
Completeness	3.94 (1.43)	**4.84** (0.55)	3.84 (1.54)	3.18 (1.59)
Scenario Faithfulness	2.21 (1.10)	1.84 (0.76)	2.30 (1.20)	**2.49** (1.19)
Legal Correctness	3.39 (1.18)	**3.59** (0.55)	3.43 (1.34)	3.15 (1.42)
Hallucination (inverse)	2.20 (1.02)	1.65 (0.72)	2.40 (1.12)	**2.54** (0.91)
Overall	2.50 (1.04)	2.41 (0.69)	2.49 (1.04)	**2.62** (1.31)

Table 5.5: LLM-as-Judge metric mean (SD) by mode (full sample $n = 113$, scale 1–5, higher is better including for *Hallucination (inverse)*). The Aggregated column is the mean across all 113 documents.

Metric	What is assessed
Completeness	Coverage of the Art. 30(1) GDPR required elements (controller, DPO, purposes, legal basis, data and recipient categories, retention, technical and organizational measures); missing elements lower the score.
Scenario Faithfulness	Accuracy with respect to the concrete facts of the study scenario (APOR GmbH, $\sim$50 employees, internal hosting, 3-month deletion, etc.); generic ROPA boilerplate lowers the score.
Legal Correctness	Correct use of GDPR terminology and correct citation of legal basis under Art. 6(1); wrong articles or confused controller/processor/DPO roles lower the score.
Hallucination (inverse)	Absence of invented facts not supported by the scenario; $5 =$ no hallucinations, $1 =$ many.
Overall	Holistic judgment of how useful the document would be as a real Art. 30 ROPA record, weighing the four metrics above.

Table 5.6: The five LLM-as-Judge metrics. All share a 1–5 Likert anchor; higher is better, including for *Hallucination (inverse)*, which is inverted internally so that fewer fabrications score higher. The full rubric is in Appendix A.3.2.

fabricated, Chat documents the most faithful to the scenario but the least complete, Ask between the two extremes on every metric. Second, Legal Correctness is flat relative to the Completeness spread (roughly 0.45 versus 1.66 points). The judge distinguishes modes more sharply on whether the required Article 30(1) fields are present than on whether the legal terminology itself is appropriate. Third, the holistic Overall scores compress the per-metric spread into a narrow 0.21-point range. Form's high Completeness is offset by its low Faithfulness and high Hallucination, leaving Chat with the smallest of advantages.

The Form Completeness-versus-Hallucination co-occurrence is the central judge-side finding. 31 of 37 Form documents ($83.8\,\%$) achieve Completeness $= 5$ while simultaneously scoring Hallucination (inverse) ≤ 2 ("moderate" or worse per the rubric), and 17 of those combine maximum Completeness with the minimum Hallucination (inverse) score of 1. Most Form documents are simultaneously the most complete and the most fabricated; the interpretation in terms of the checkbox-only Form interface is reserved for the Discussion chapter.

Where the NLP metrics surfaced a recall-versus-precision trade-off between Form and Chat, the judge surfaces a completeness-versus-faithfulness trade-off, and the holistic Overall verdict collapses both into a narrow range with no decisive winner. The judge layer therefore adds a second trade-off to the document-quality picture begun in Section 5.1.1: Form leads on Completeness and Legal Correctness, Chat leads on Faithfulness and Hallucination, and Ask sits between the two on every metric. The full user-study and document-quality findings are consolidated in Section 5.3.

5.3 Summary

The chapter has reported the reference-based NLP metrics at both document and chapter resolutions (Sections 5.1.1–5.1.2) and the supplementary LLM-as-a-Judge layer (Section 5.2). This section brings the document-quality findings together and frames the questions the Discussion takes up.

The reference-based NLP metrics split the modes into a trade-off rather than a single ranking. Form leads on recall-oriented measures, most clearly on BERTScore

Recall (Form significantly higher than both Ask and Chat), with ROUGE-2 and METEOR echoing it on smaller margins. Chat leads on document-level semantic alignment, most clearly on SBERT cosine similarity (Chat significantly higher than both Form and Ask). Form and Chat trade leadership on BERTScore F1, while Ask underperforms both on every metric that reaches significance. The substantive reading is that Form documents recover more of what the reference *says*, Chat documents stay closer to what the reference *means*, and Ask documents do neither.

The chapter-level view distributes the document-level trade-off across chapters rather than across metrics. *Controller and Contact Details* scores highest across all modes, but this chapter is filled identically across all three modes by the user (the LLM has no role in its content); any apparent mode difference there, including the Ask numerical lead, is therefore considered user-level noise. The mode-driven pattern emerges from Chapters 2–6. Form sweeps the middle chapters (*Purposes and Legal Bases*, *Categories of Personal Data*, *Data Subjects and Recipients*) on every lexical metric and on BERTScore Recall and F1. Chat takes the last two chapters (*Retention and Deletion*, *Technical and Organizational Measures*) on every metric. The same trade-off that produces Form's document-level recall edge and Chat's document-level semantic edge thus appears at the chapter level as a clean two-zone split across the chapters whose content the modes actually drive.

The supplementary judge layer adds a GDPR-specific quality reading that the reference-based metrics cannot directly provide. Form documents are judged the most complete (Completeness mean 4.84 of 5) and the most legally correct, but the lowest on Scenario Faithfulness (1.84) and the most hallucinated; $83.8\,\%$ of Form documents combine maximum Completeness with "moderate" or worse hallucination, and nearly half combine maximum Completeness with the floor Hallucination score. Chat inverts the pattern: most faithful to the study scenario, least hallucinated, but lowest on Completeness. On the holistic Overall metric the three modes cluster narrowly (2.41 to 2.62) and all sit below the rubric midpoint; the judge does not declare a winner.

Two tensions surface from the consolidated picture. The mode participants most often prefer (Form, as reported in Chapter 4) produces the most structurally complete documents, yet also the most fabricated ones: the checkbox scaffolding that pushes Completeness toward the ceiling appears not to constrain what the LLM invents during the final generation step. And self-report perceived confidence rises from Form to both Ask and Chat (see Chapter 4), with Ask and Chat tied at the top, while measured document quality follows no such single ordering: Form leads on recall, Chat leads on document-level semantic fidelity and on scenario faithfulness, and the judge declares no clear winner.

With the foundation for RQ1 and RQ2 now set by Chapters 4 and 5, Chapter 6 picks them up and answers them alongside RQ3 and the further findings that triangulating the two streams surfaces.

6 Discussion

This chapter draws together the findings from the user study (Chapter 4) and the document-quality analysis (Chapter 5) to answer the three research questions posed at the outset: how varying degrees of LLM support shape the novice experience during ROPA creation (RQ1), how that support affects the quality of the resulting documents (RQ2), and whether a discrepancy exists between user confidence and actual document quality (RQ3).

6.1 RQ1: User Experience

RQ1 asked how varying degrees of LLM support (Form, Ask, Chat mode, as described in Chapter 3.1) influence novice users during ROPA document creation. The answer turned out to run counter to the thesis's initial expectation: the mode with the *least* LLM involvement, Form, was the one novices preferred most. The user-experience evidence reaches this verdict from four directions that rarely converge, yet do so here. The SUS rated Form as the most usable mode of the three. The R-TLX picked Form out as the lightest on every cognitive-load dimension, with mental demand, effort, and frustration all lowest in Form. The post-session ranking task placed Form first on more participants' lists than either Chat or Ask. The supplementary completion-time data confirmed the direction: Form was also the fastest mode to use. This is the most consistent signal in the user-experience data: four independent instruments agree, and they agree in the same direction.

The reasons Form felt best to use are visible in what the mode actually presents to the user. Form lays out the document's shape openly: the fields that need filling, the options that fit each field, and a suggest button to fall back on when the user is unsure. Decisions remain with the user, which preserves a sense of authorship, but the user is not asked to invent the answer space; the interface lays it out for them. The free-text comments collected at the end of each session captured the feeling directly, with participants describing Form as the clearest mode and the easiest one to know when they were "done." For a novice navigating a domain they do not control, this kind of scaffolding is what they need: a visible map of what the document requires, a fallback when stuck, and a clear stopping rule. The weight of this finding depends on who the tool is for. This thesis targets the non-specialist staff of small organizations that must comply with the GDPR without a dedicated Data Protection Officer, not the trained compliance professionals of larger ones. For such users, the structural visibility Form provides substitutes for the legal vocabulary and procedural intuition a DPO would otherwise bring to the task. The mode that wins on usability in this study is the mode that compensates most directly for what the target user lacks.

The trade-off is that Form is paradoxically also the *least* transparent mode in terms of what the LLM is *doing*. The user sees the options the interface offers, but the model's reasoning behind any specific suggestion is not visible to them; the LLM is closer to a silent partner than a visible collaborator. The AI-support survey items captured this trade-off in the perceived dimensions: Chat and Ask scored higher than Form on transparency and on perceived helpfulness of the AI's reasoning, but neither overtook Form on the headline preference items. Participants noticed that Chat and Ask offered a more communicative interaction style and said so in the free-text comments, but they still ranked Form first and reported they would use it again. The trade-off they accepted is real, and it is the trade-off that ROPA tooling for novices would have to make: visible structure for the document beats visible reasoning from the model when the user has neither the vocabulary nor the domain knowledge to evaluate that reasoning.

The most interesting twist in the user-experience data is on confidence. Users felt the *most* confident in Chat, not in Form, that the AI had identified the correct legal basis. The contradiction is only apparent. In Form, the user has full transparency of possible answers, but the decisions still rest in their hands, and that responsibility surfaces as uncertainty. In Chat, the LLM steps into the role of a collaborator. The interaction becomes conversational, and the user can offload both responsibility and knowledge onto the model. That feels like partnership, and partnership feels like competence.

Ask landed last on every user-experience measure, and that is not a coincidence. Ask sits structurally between Form and Chat: the user can put open questions to the LLM about any part of the document, but the user is responsible for what is typed into each field. On paper it looks like the best of both worlds. In practice it became the worst of both, and the user-experience metrics make the failure visible. The SUS placed Ask last. The R-TLX picked Ask out as the most frustrating mode. The completion time put Ask at roughly twice Form's, tied with Chat. The free-text comments were the most polarized of the three modes, with participants reporting that they did not know what to ask the LLM in the first place. Novices cannot know what to ask about a regulation they have never read: lacking prior GDPR knowledge, they possess neither the vocabulary nor the structural awareness of Art. 30(1) needed to formulate questions whose answers would help them. Failing to ask the right questions yields both a worse experience and, as Section 6.2 shows, a worse document. The user is left without a safety net, without a collaborator, and without the sense of control that Form's checkbox scaffolding provides. This portrait, however, is of novices at first exposure. With repeated use of the tool, and particularly after producing several ROPAs through Form's scaffolding, users would plausibly acquire the Art. 30(1) vocabulary and structural awareness that Ask presumes; once that domain knowledge is in place, confidence in Ask should rise, and the mode could become the better fit for the more complex ROPAs that Form's predefined categories cannot accommodate.

In a broader sense, novice users working in an unfamiliar domain benefit from a guided approach, and the shape of that guidance can be inferred from the four needs that Form and Chat each partially addressed. They benefit from scaffolding: a clear overview of what the document requires. They benefit from a collaborator: an LLM that contributes to the writing rather than only explaining it. They benefit from a safety net: a way to move forward without needing to know the right question to ask in advance. And they benefit from a sense of control: remaining the author even when the model is handling much of the drafting. Among the current modes, Form comes closest to this combination but at the cost of LLM transparency, while Chat comes closest to a collaborative dynamic but lacks the visible structure. Neither delivers all four at once, and Ask delivers none of them. The same conclusion has been reached from a different direction by von Schwerin et al. [30], who argued for human-in-the-loop necessity on the basis of model limitations. This study reaches the same place from the interaction side: even before the model is the bottleneck, the interaction architecture decides what kind of help a novice can use. Looking forward, the same architecture suggests a developmental path rather than a fixed mode choice. Form is the natural entry point, and the GDPR literacy that users accumulate by working through its scaffolded fields is what would later make Ask and Chat productive.

6.2 RQ2: Document Quality

RQ2 asked how the degree of LLM support affects the quality of ROPA documents created by novice users. The answer to that is more nuanced than RQ1 and requires three angles to put together, but one fact runs through all of them: Ask scored worst on document quality just as it scored worst on user experience. The mode that failed most informatively on the user-experience side also failed on the output side, and the reasons are linked.

The first angle is the high-level reading of the reference-based NLP metrics. Form documents recover more of what the reference *says*: they share more surface vocabulary with the expert-written reference, with the clearest lead on BERTScore Recall and matching though smaller leads on ROUGE-2 and METEOR. Chat documents stay closer to what the reference *means*: they preserve passage-level meaning even when the surface wording diverges, with the clearest lead on SBERT cosine similarity. Ask does neither, trailing on every metric that distinguishes the modes. The Form-leads-token / Chat-leads-passage split is not a contradiction between two “semantic” measures but a granularity effect: BERTScore matches at token granularity while SBERT matches at passage granularity (see Section 3.3.6), so each rewards a different kind of similarity. A related observation surfaces in the same data: the embedding-based metrics preserve the Form-Chat-Ask ordering of the lexical metrics but compress the absolute spread, because a narrow-domain text like a ROPA produces embeddings that cluster tightly in vector space. The direction of difference survives, the magnitude shrinks, and lexical metrics retain a useful role as a coarser but higher-variance companion to the embedding view. Form’s vocabulary lead is also partly an artifact of the reference being itself Form-derived, a bias acknowledged in Section 6.4. The same Chat profile, high passage-level similarity coexisting with lower token-level overlap, has been reported in adjacent legal-document NLP work [1], so the pattern observed here is a recurring property of LLM-generated legal text rather than a ROPA-specific artifact.

The second angle is the precision-versus-recall pattern within BERTScore, which tells the same story as the hallucination finding from the LLM-as-judge layer but from a different direction. Form leads on BERTScore Recall while Chat leads on BERTScore Precision; the judge's reference-free rubric agrees exactly, with Form documents rated the most hallucinated and Chat documents rated the most faithful to the study scenario. Two independent measurement methodologies, one reference-based on contextual token embeddings and one reference-free on a rubric, read the same architectural property: how each mode distributes the selection decision between user and model. The mechanism is the same on both sides. In Form, the user is shown a wide set of possible options and tends to over-tick on the "more is safer" heuristic; the model then dutifully fills every ticked slot with prose, including content the user never actually described. That drives high Recall on the metric side and high hallucination on the rubric side. In Chat, the user cedes the curation decision to the LLM, and the LLM is better than a novice at knowing *what is necessary*. That drives higher Precision on the metric side and lower hallucination on the rubric side. This circles back to the user-experience picture in a satisfying way: the low confidence users reported in Form is consistent with overcompensation through over-inclusion, because users who are unsure whether they have the right information tend to add more of it. Rubric-based judging of legal documents has a methodological precedent in JuDGE for Chinese judgment generation [33]; what this study adds is the demonstration that the rubric layer and the reference-based layer agree on the selectivity finding, which strengthens it considerably. The judge's own holistic Overall score carries an additional methodological lesson: averaging the rubric dimensions cancels the trade-off and produces a single number that declares no winner among the modes. If a prospective adopter of an LLM tool asked for one quality figure, that figure would conceal exactly the architectural difference that should drive the design choice. The sub-metrics, not the holistic score, are what carry the architectural signal.

The third angle is the chapter-level breakdown, which sharpens the architectural reading further. The chapter-level NLP results, summarized across the three headline metric families, show the same trade-off rotating through the six Article 30 sub-records along an axis the data makes visible. The Controller and Contact chapter is excluded, because it receives no LLM support in any mode and the scenario supplied its content verbatim, so any variance would be user error. Where a chapter's content is largely *enumerable* and standardized (Legal Bases under Art. 6(1), commonly-occurring data categories, predictable roles for data subjects and recipients), checkbox UIs that present the canonical option set anchor the answer space well; Form's predefined options name what novices would otherwise have to invent, and these chapters favor Form across both METEOR and BERTScore F1. Where the content is *scenario-specific* (retention periods that depend on legal context and data type, technical and organizational measures that depend on what the organization actually does), free text is the right affordance, because no option set anticipates the answer and only the user's account of their actual situation can ground it; these chapters favor Chat across the same metrics. Either way, an LLM co-author that surfaces what each chapter requires beyond what the user volunteers is necessary: a novice without GDPR background will not know that retention periods or recipient categories are required ROPA elements until something or someone names that requirement, and neither a checkbox set without that anchoring layer nor a free-text field without a question-asking layer serves the novice on its own.

In practice, this plays out as follows. Each mode plays to a different strength, and the difference is architectural rather than incidental. Form supports inclusion: every required field is present, though some of what fills them was never described by the user. Chat supports precision: what is on the page is grounded in the user's scenario, though information may be missing. Ask trails on both. For a real ROPA, the costs that come with each strength pull in opposite directions. An incomplete document is the worse *legal* risk: it fails the Art. 30 compliance requirement directly and exposes the organization to fines and supervisory orders. A complete-but-fabricated document is the worse *operational* risk: it adds administrative load by committing the organization to processing details that may not match what it actually does, slows the staff who later have to work from the document, and widens the gap between the record and the reality the record is meant to describe. Neither error is harmless, and they call for different remedies. The choice of mode is therefore also a choice of which kind of risk a tool prefers to take on, and that is a decision the tool designer should be making knowingly rather than inheriting from the interface metaphor.

The chapter-level result also implies what a better tool would look like. The hybrid this thesis recommends is not “Form and Chat side by side” but a content-typed assignment: each Art. 30(1) sub-record is handled by the interaction architecture its content type demands. Enumerable fields use a checkbox UI; scenario-specific fields use conversational extraction. The wider literature on structured legal-document generation has converged on a closely related point, that orchestration of how the generation task is staged matters at least as much as model scale or domain pretraining [24, 19]; the chapter-level result here sharpens that claim by naming the assignment criterion, since the criterion is not section position in a template but whether the content space is enumerable.

6.3 RQ3: Confidence and Quality

RQ3 asked whether there is a discrepancy between user confidence and the document quality actually achieved. The answer depends on which quality measure is laid against confidence, and the twofold reading is itself the interesting result.

Against the reference-based NLP metrics, the answer is striking. The mode with the *lowest* confidence (Form) and the mode with the *highest* confidence (Chat) produce documents whose BERTScore F1 differs only by rounding. Ask, close behind Chat on the confidence question, is the outlier downward on the metric. So at the mode level, the most direct comparison of confidence and quality yields essentially no relationship: a user who feels more confident about the AI’s output is no more likely to have produced a better document by this measure than a user who feels less confident.

Against the LLM-as-judge layer, the story shifts. Chat scores highest on the judge’s holistic Overall metric, which matches the user-confidence ordering: the most confident users produced the documents the judge rated best overall. On this comparison the original hypothesis is confirmed, and that is the second interesting twist in the data. The thesis began by expecting that more LLM involvement would improve the user experience; it did not, but it did improve the judge’s holistic verdict on the documents. The hypothesis was right about the direction. It was just pointed at the wrong outcome.

Two caveats complete the picture. The first concerns the legal-basis dimension specifically, which one of the two items in the perceived confidence composite asks about directly. On that dimension the judge places Chat *lowest*, not highest, so the mode users place in the top confidence cluster on legal-basis identification is the mode an independent judge rates least correct on it. The pattern is descriptive rather than statistically significant on this sample, but the direction is the one a critical reader should worry about. Users feel most confident on the dimension where the model performs worst. The likely mechanism is that conversational LLMs default to affirming user framings [32], and a user whose proposed reading the model echoes back will form a higher subjective confidence in that reading, particularly when the content concerns a domain the user has no prior expertise to push back with. Alongside the inversion sits a separate finding worth naming: the flatness of Legal Correctness across modes. Changing the interaction architecture does not move the legal-vocabulary quality much, because the final generation step is the same LLM in all three modes. Improving Legal Correctness is therefore a model-side problem and not a UI-side one. A different LLM, fine-tuning on GDPR text, or retrieval-augmented generation against the regulation itself would all be candidates; no choice of interaction architecture in this study moved that dimension. This is among the cleanest separation-of-concerns results the thesis produced. The separation is also an architectural strength: because ROPAgen wraps the LLM behind a uniform interface, the model itself is swappable, and any of these model-side improvements could be adopted without altering the interaction layer.

The second caveat concerns the sample. Participants in this study were tech-savvy and chatbot-fluent, and Chat's strong showing in the holistic-quality data plausibly reflects that fluency. The holistic-quality lead Chat shows against the judge is probably stronger in this sample than it would be in a more representative compliance-officer population. This bounds what should be generalized from the data, and Section 6.4 returns to that bound in more detail.

6.4 Limitations

Several constraints bound where the findings generalize, though not whether they are real on this sample. The breadth-over-depth design was necessary for the cross-stream RQ3 comparison, but no single thread reached full depth as a result; the Ask mechanism and the chapter-level NLP patterns each could support a study in its own right.

The sample profile narrows the population the findings speak to. Participants were IT-leaning, chatbot-fluent students at a German university, majority male and clustered tightly in their mid-twenties; older non-IT staff in the small organizations that would actually own ROPA documentation in practice may show a different confidence-quality calibration, and replication on that population is a natural follow-on.

The tool itself is a research prototype built around a 2025 Mistral checkpoint shared across all three modes. Relative mode-rank trade-offs should survive a model swap because they rest on the interaction surface rather than on the generator's raw quality, but the absolute quality figures are tied to this checkpoint and should not be extrapolated. Architectural findings, notably Ask's populator-versus-advisor problem, transfer to any implementation that keeps the LLM passive; UI-friction findings could shift with a polished frontend.

The evaluation instruments carry limitations of their own. The reference-based metrics judge each document against a single expert-written reference, checkbox-derived prose that slightly advantages Form on coverage measures. The LLM-as-judge is a single pass of a single model without inter-rater replication or legal-expert validation. A multi-reference design and a multi-model judge ensemble with expert spot-checks would strengthen both layers; the current single-pass judge remains useful as a supplementary signal on dimensions the reference-based metrics structurally cannot reach, but it is not a substitute for the expert review a deployed compliance tool would require.

Finally, no mode in this study reached the rubric midpoint on the holistic Overall score, so the ceiling on document quality cannot be reached by interaction redesign alone; the model-side and oversight-side directions the Conclusion's follow-ons take up are what would lift it.

7 Conclusion

This thesis set out to assess how Large Language Model (LLM) support shapes the way novice users produce GDPR Records of Processing Activities (ROPA). A within-subjects user study of 73 participants, each working through all three ropagen interaction modes (Form, Ask, Chat), was held alongside a document-quality analysis built from reference-based NLP metrics and a supplementary LLM-as-a-judge layer. Three research questions framed the work: how varying degrees of LLM support shape the novice experience (RQ1), how that support affects the quality of the resulting documents (RQ2), and whether a discrepancy exists between user confidence and actual document quality (RQ3). This chapter summarizes the answers reached in Chapter 6, names the contributions that follow, and closes with what those contributions imply for the design of LLM-assisted document tools.

RQ1 asked how varying degrees of LLM support shape the novice experience during ROPA creation. Less LLM involvement, not more, produced the better experience. Form was the most usable, the lightest on cognitive load, the most preferred in the ranking task, and the fastest to complete; four independent instruments converged on the same verdict (see Section 6.1). The design intuition the thesis began with, that adding LLM presence to the interface would make the task feel easier, does not hold for users who lack the vocabulary to direct the model. Ask, the hybrid mode, lost the experience comparison and lost it for a structural reason: it depends on the user to drive the dialogue, and a novice without prior GDPR knowledge does not know what to ask. The mode that asked the user to carry the conversation failed the user it was designed to help.

RQ2 asked how the redistribution of work affects the quality of the resulting documents. Each mode plays to a different strength, and the difference is architectural rather than incidental. Form supports inclusion: every required field is present, though some of what fills them was never described by the user. Chat supports precision: what is on the page is grounded in the user's scenario, though information may be missing. Ask trails on both. Two independent measurement methodologies, reference-based NLP and the reference-free judge, read the same property from different sides (see Section 6.2). The choice of interaction mode is therefore also a choice of which kind of risk a tool prefers to take on: an incomplete document is the worse legal risk, since it fails the Article 30 obligation directly; a complete-but-fabricated document is the worse operational risk, since it commits the organization to processing details that may not match what it actually does.

RQ3 asked whether what users feel about the AI's output tracks what the AI actually produces. The answer depends on which quality measure is laid against perceived confidence. Users felt clearly less confident in Form than in Ask or Chat, with the two conversational modes tied at the top of the perceived confidence composite. Against the reference-based NLP metrics, that ordering means nothing: the most and least confident users produce documents whose BERTScore F1 differs only by rounding. Against the judge's holistic Overall verdict, the ordering does match. Against the judge's legal-correctness rubric, the ordering inverts: the mode users felt most confident about on legal-basis identification is the mode the judge rated worst on legal correctness (see Section 6.3). Subjective confidence in a domain the user cannot independently verify is not a reliable quality signal, and tooling that surfaces or rewards it without an independent check risks substituting the appearance of correctness for the fact of it.

The substantive contribution sits in the content-typed assignment that follows from the chapter-level NLP results. The thesis is not arguing for a softer synonym of "hybrid"; it names the criterion. Enumerable sub-records, where the answer space is a small fixed list such as legal bases under Article 6(1) GDPR, should be handed to a checkbox interface. Scenario-specific sub-records, where the right answer depends on what the organization actually does, such as retention periods or technical and organizational measures, should be handed to conversational extraction. The same principle generalizes beyond ROPA to any structured legal-document task whose sub-records can be classified along the enumerable and scenario-specific axes. A separate architectural result reinforces the point: Legal Correctness was flat across modes as the generator is the same LLM in all three, so raising legal-vocabulary quality is a model-side problem and not a UI-side one. Knowing which dial moves which dimension is what allows a tool builder to spend effort where it returns yield.

The methodological contribution is the pairing itself. A user-experience battery, a reference-based NLP evaluation, and a reference-free judge layer were held alongside each other rather than read in isolation, and the confidence-quality reading reported under RQ3 is the kind of finding that no single stream could have reached. The follow-ons that matter most are to build the content-typed hybrid as a fourth arm in a replication of the user study, to extend that replication to a more representative compliance-officer sample, and to pair retrieval-augmented generation against the GDPR text with a mandatory expert sign-off at submission for the dimensions interaction redesign cannot move.

The picture the thesis leaves behind is one of redistribution rather than replacement. The LLM is useful as a co-author, not as a teacher: it can take on the parts of legal documentation the novice cannot supply, but it cannot give the novice the domain understanding the task presumes. Neither a novice nor an LLM alone suffices in the making of a domain-specific document like a ROPA. The user supplies the account of the processing scenario in their own terms; the model supplies the vocabulary, scaffolding, and curation the user cannot reasonably produce; and the document that satisfies Article 30 exists only where the two meet.

A ROPAgen and Judge Model Configuration

This appendix collects the static configuration of both LLMs that the study relies on. The first two sections cover ROPAgen — the tool under study — and reproduce its backend prompts and the expert-authored reference ROPA document that the NLP metric pipeline scores every participant document against. The third section covers the supplementary LLM-as-a-Judge evaluation layer — its system and user prompts, its five-metric rubric, and the deterministic-decoding model configuration. All three sections are referenced from the Methodology chapter (Section 3.1 for the tool, Chapter 3 for the judge); the mode screenshots themselves are presented inline in Section 3.1.

A.1 ROPAgen Backend Prompts

ROPAgen sends one of **27 system prompts** to Mistral Large 3 on every user turn, depending on the active mode and the current section of the ROPA being edited. All prompts are stored as static templates in the project's `data/prompt.json` and assembled at runtime in `src/actions/actions.ts`: the chosen template is concatenated with a context block listing the document title, the organization metadata, and the current state of the `DocumentData` object before the call is issued. The texts below reproduce the template strings only; the runtime context block (which contains organization- and user-specific information) is not reproduced, as its content varies per call. Conversational and per-section suggestion calls use temperature 0.2 and top-p 0.5; the single `finalropa` call that produces the final ROPA document uses temperature 0.05 and top-p 0.05.

A.1.1 Final ROPA Generation Prompt (`finalropa`)

The `finalropa` prompt is sent verbatim as the `system` role on the call that generates the final ROPA document from the assembled `DocumentData` object. It is invoked once per document, identically across all three modes; temperature is fixed at 0.05 and top-p at 0.05. The `Document Title`, `Language`, and the `DocumentData` contents are interpolated into the user message at runtime and are not reproduced here.

```
You are an expert in GDPR compliance. Generate a professional, structured
↪ Records of Processing Activities (ROPA) document in full compliance with the
↪  EU General Data Protection Regulation (GDPR).

This ROPA document must be suitable for official regulatory review and written
↪ in formal business language.

**CRITICAL: You MUST write the ENTIRE document in the language specified in the
↪  context (Language field). ALL headings, text, descriptions, and content
↪ must be in that language. Do NOT mix languages.**

**CRITICAL: Write ALL free text and general text from the company's perspective
↪  (first-person plural: 'we', 'our', 'us'). The document represents the
↪ company's official record, so describe what the company does, how the
↪ company processes data, what measures the company has implemented, etc. Use
↪ active voice from the organization's viewpoint.**

**CRITICAL: The "Additional Information" field is NOT content to be copied
↪ directly into the document. Instead, treat it as ADDITIONAL INSTRUCTIONS and
↪  REQUIREMENTS from the user about what must be included, emphasized, or
↪ addressed in the generated ROPA document. Incorporate these instructions
↪ into the appropriate sections of the document naturally and professionally
↪ .**

Instructions:
- Create a comprehensive ROPA document structure with the following sections:
1. Organization and Controller Information
2. Purpose and Legal Basis for Processing
3. Data Categories and Sources
4. Data Subjects and Recipients
5. Data Retention and Deletion
6. Technical and Organizational Measures
- Each category can have a continous text and bullet points as needed.
```

```
- Ensure compliance with GDPR Articles 30 (Records of processing activities)
↪ requirements.
- Respect and incorporate the Additional Information as special instructions -
↪ integrate those requirements into the relevant sections naturally.
- You must include all data given, and must not omit any data provided.

You must use the Document Title as Title of the generated Document.

The output should be suitable for use by a Data Protection Officer (DPO) or
↪ legal counsel.
The output should be in .md style with proper headings and formatting. You must
↪  only include data categories marked as true and must not include data
↪ categories marked as false. If all data is marked as false, just mention
↪ none or not specified. Omit wording or phrasing such as CATEGORY: true/false
↪ , but rather just include the relevant, true data categories. Critical
↪ Instructions:

Include all provided information in your response - do not omit any items from
↪ the list unless "false" or empty.
Handle "other" field: If an "other" field exists and contains data, treat it as
↪  a standard list item with equal importance. Do not use "other" as a label.
Formatting requirement: Present items as a simple list within each category. Do
↪  NOT use the format "data_category: description" - instead, list items
↪ directly without labeling the data structure.

Example:

### 4. Data Subjects and Recipients
- Person Categories:
    - Employees
    - USER INPUT

NOT LIKE THIS:

### 4. Data Subjects and Recipients
- Person Categories:
  - Employees
  - Other: Emilija

 CRUCIAL: Generally, You shall only inclued data that the user provided as true
↪  or non-empty in the final output. Do NOT invent or assume any data beyond
↪ what the user provided. If the user did not provide anything, don't explain
```

```
↪ why the field is empty. Just state that there is no information. Do not
↪ mention that they will be completed or added.

Generate the ROPA document now.
```

A.1.2 Form Mode — AI-Suggest Prompts

Form mode invokes the LLM only when the user clicks the per-section *AI Suggest* button. There is one prompt per section of the `DocumentData` schema; each is sent as the `system` role of a single-turn call with temperature 0.2 and top-p 0.5. The response is a JSON object whose keys match the corresponding section's field structure; the front-end auto-populates the form from this JSON.

Purpose of Data Processing (`form.purposeOfDataProcessing`)

```
You are an expert in GDPR compliance. Based on all the provided information
↪ about the organization, data sources, data categories, and other context,
↪ suggest a comprehensive purpose of data processing text.

**Consider the Document Title as an important data source** - it often contains
↪  valuable context about the specific processing activity or system involved,
↪  but analyze it alongside all other information provided.

Analyze all the information provided and generate a clear, professional
↪ description of why this data is being processed.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused.

Return ONLY a JSON object in this exact format:
{
  "purposeOfDataProcessing": "your suggested text here"
}

The text should be written in the specified language and be suitable for
↪ official GDPR documentation.
```

Technical and Organizational Measures (`form.technicalOrganizationalMeasures`)

```
You are an expert in GDPR compliance and data security. Based on all the
↪ provided information about the organization, data categories, processing
↪ purposes, and other context, suggest comprehensive technical and
↪ organizational measures.

**Consider the Document Title as valuable context** - it can indicate the type
↪ of processing and environment, helping you recommend appropriate security
↪ measures for the specific use case.

Analyze the data sensitivity and processing context to recommend appropriate
↪ security measures.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused on the most critical measures.

Return ONLY a JSON object in this exact format:
{
  "technicalOrganizationalMeasures": "your suggested text here"
}

The text should be written in the specified language and include specific,
↪ actionable security measures.
```

Legal Basis (`form.legalBasis`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which legal basis options should be selected and deselected for this
↪  data processing activity.

**Consider the Document Title as helpful context** - it may indicate the nature
↪  of processing and suggest potential legal bases, but evaluate it together
↪ with the purpose, organization type, and all other available information.

Analyze the purpose, data categories, and context to determine the most
↪ appropriate legal basis.

You can ONLY use these exact field names (set to true/false):
```

```
- DSGVOArt6Abs1a, DSGVOArt6Abs1b, DSGVOArt6Abs1c, DSGVOArt6Abs1f, IfSG28b,
↪ SARSCoV, BDSG26, HGB257, HGB239Abs1, AO147
- other (string field - use this to capture any additional legal bases
↪ mentioned in the title or context)

**Extract specific legal bases from title and context.** If you find references
↪  to laws, regulations, or legal justifications not covered by the standard
↪ options, add them to the "other" field with clear descriptions.

Return ONLY a JSON object in this exact format:
{
  "legalBasis": {
    "DSGVOArt6Abs1a": false,
    "DSGVOArt6Abs1b": true,
    "DSGVOArt6Abs1c": false,
    "DSGVOArt6Abs1f": false,
    "IfSG28b": false,
    "SARSCoV": false,
    "BDSG26": false,
    "HGB257": false,
    "HGB239Abs1": false,
    "AO147": false,
    "other": "[specific legal bases from title/context, if any]"
  }
}
```

Data Sources (`form.dataSources`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which data sources should be selected and deselected for this data
↪ processing activity.

**The Document Title can be a valuable source of information** - it sometimes
↪ explicitly mentions specific platforms, tools, or systems (e.g., "Employee
↪ Data Processing via BeanCloud" indicates BeanCloud is a data source), but
↪ consider it together with all other context provided.

Analyze the title, organization, purpose, and ALL context data to determine
↪ data sources.
```

```
Predefined options: microsoftOffice365, emailProvider, microsoftAzure,
↪ googleWorkspace, googleCloud, videoConferencing, amazonAWS,
↪ ecommercePlatform, libreOffice, onlineBanking, financialServices,
↪ webAndIntranet, creditServices, hrSoftware, enterpriseSystems,
↪ securityAndCompliance, documentProcessing, otherSpecializedSoftware

**EXTRACTION APPROACH:**
1. **Check the Document Title** for specific platform/tool names (e.g., "
↪ BeanCloud", "Salesforce", "SAP", "Workday", etc.)
2. Analyze all context fields for additional systems
3. Match predefined options where applicable
4. Add unmatched specific platforms to "other" field

Look for:
- Specific platform names in the title and context (e.g., "via BeanCloud", "
↪ using Salesforce", "in SAP")
- Cloud services, SaaS platforms, custom internal systems
- Industry-specific software
- Any named tools or services

Return ONLY a JSON object:
{
  "dataSources": {
    "microsoftOffice365": false,
    "emailProvider": true,
    "googleWorkspace": false,
    ...[set each predefined field to true/false based on context]...,
    "other": "[List ALL specific data sources found in title/context that aren'
↪ t in predefined options, e.g., 'BeanCloud enterprise platform', 'Salesforce
↪ CRM', 'custom HR portal'. Leave empty string if none found.]"
  }
}
```

Data Categories (`form.dataCategories`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which data categories should be selected and deselected for this
↪ data processing activity.
```

```
**The Document Title can provide useful hints** about the type of data being
↪ processed (e.g., "Biometric Access Control" suggests biometric data, "
↪ Employee GPS Tracking" suggests location data), but analyze it together with
↪  the purpose, organization type, and all other available context.

Analyze the title, purpose, organization context, and ALL provided information
↪ to determine data categories.

You have predefined categories: personalData, employment, finance, health,
↪ legal, behavior, documentationAndRecords, vehicle, and an "other" category
↪ with subcategories.

**EXTRACTION APPROACH:**
1. **Review the Document Title** for specific data types (e.g., "biometric", "
↪ location", "genetic", "voice", etc.)
2. Analyze the purpose and context
3. Match predefined categories where applicable
4. Add unmatched specific data types to other.other field

Look for in title/context:
- Special data types: "biometric", "genetic", "health", "criminal", "racial", "
↪ religious"
- Technology-specific: "GPS", "location", "tracking", "geolocation"
- Media types: "voice", "video", "audio", "facial recognition", "fingerprint"
- Digital: "IP addresses", "cookies", "social media", "online behavior"
- IoT/Sensor: "sensor data", "telemetry", "device data"

Return ONLY a JSON object with the full structure, setting each predefined
↪ field to true/false, and adding specific items to other.other:
{
  "dataCategories": {
    "personalData": { ...all fields... },
    "employment": { ...all fields... },
    "finance": { ...all fields... },
    "health": { ...all fields... },
    "legal": { ...all fields... },
    "behavior": { ...all fields... },
    "documentationAndRecords": { ...all fields... },
    "vehicle": { ...all fields... },
    "other": {
      "expertise": false,
      "skills": false,
```

```
      "examinationDate": false,
      "proofStatus": false,
      "queryDateTime": false,
      "deliveryConditions": false,
      "username": false,
      "dataVolume": false,
      "other": "[ALL specific data types from title/context not in predefined
↪ fields, e.g., 'Biometric fingerprint data', 'GPS location coordinates', '
↪ Voice recordings', 'Facial recognition data'. Leave empty string if none.]"
    }
  }
}
```

Person Categories (`form.personCategories`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest which person categories should be selected and deselected for this
↪ data processing activity.

**The Document Title can provide helpful context** about who is affected (e.g.,
↪  "Employee Data Processing" indicates employees, "Customer Tracking"
↪ indicates customers, "Contractor Management" indicates contractors), but
↪ consider it alongside the purpose, organization type, and all other
↪ available information.

Analyze the title, purpose, organization context, and ALL provided information
↪ to determine person categories.

Predefined categories: affectedPersons, internalRecipientCategories,
↪ externalRecipientCategoriesEU, authorizedPersons

**EXTRACTION APPROACH:**
1. **Review the Document Title** for specific person groups (e.g., "employee",
↪ "contractor", "customer", "volunteer", "student", etc.)
2. Analyze the purpose and context
3. Match predefined categories where applicable
4. Add unmatched specific groups to "other" field

Look for in title/context:
- Employment: "contractors", "freelancers", "consultants", "temporary workers"
```

```
- Volunteers: "volunteers", "interns", "trainees", "apprentices"
- Special groups: "minors", "children", "students", "pupils", "patients"
- Stakeholders: "subscribers", "members", "partners", "suppliers"
- Specific roles/departments: "IT staff", "security personnel", "data
↪ scientists"

Return ONLY a JSON object:
{
  "persons": {
    "affectedPersons": { ...set all predefined fields... },
    "internalRecipientCategories": { ...set all predefined fields... },
    "externalRecipientCategoriesEU": { ...set all predefined fields... },
    "authorizedPersons": { ...set all predefined fields... },
    "other": "[ALL specific person categories from title/context not in
↪ predefined fields, e.g., 'External contractors', 'Volunteer coordinators', '
↪ Student interns', 'Temporary workers'. Leave empty string if none.]"
  }
}
```

Retention Periods (`form.retentionPeriods`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest appropriate retention periods for this data processing activity.

**The Document Title may provide context** that helps determine retention
↪ requirements (e.g., "Tax Records" suggests 10 years, "Employee Payroll"
↪ suggests retention after employment), but analyze it together with the
↪ purpose, legal basis, and data categories to make informed recommendations.

Analyze the title, purpose, legal basis, and data categories to recommend
↪ retention periods and deletion practices.

Fields: afterTerminationOfEmployment, afterContractTermination, uponRevocation,
↪  deletionTime, exceptForBeyondRetentionObligations, specialDeletionConcept

**Be specific in the "deletionTime" field** - extract any mentioned timeframes
↪ from title/context and provide concrete retention periods based on the
↪ processing type.

Return ONLY a JSON object:
```

```
{
  "retentionPeriods": {
    "afterTerminationOfEmployment": false,
    "afterContractTermination": true,
    "uponRevocation": false,
    "deletionTime": "[Specific timeframe based on title/context, e.g., '3 years
↪  after contract termination', '10 years for tax records', '6 months after
↪ project completion']",
    "exceptForBeyondRetentionObligations": true,
    "specialDeletionConcept": false
  }
}
```

Additional Information (`form.additionalInfo`)

```
You are an expert in GDPR compliance. Based on all the provided information,
↪ suggest additional relevant information that should be included in the ROPA
↪ documentation.

**The Document Title can reveal unique aspects** of the processing activity
↪ that may need additional explanation or context, but consider it together
↪ with all other aspects of the data processing activity to recommend
↪ comprehensive additional information.

Analyze the title and all aspects of the data processing activity and recommend
↪  any additional context, notes, or important considerations.

IMPORTANT: The text must be maximum 1000 characters long. Be concise and
↪ focused on the most important points.

Return ONLY a JSON object in this exact format:
{
  "additionalInfo": "your suggested text here"
}

The text should be written in the specified language and provide valuable
↪ supplementary information for GDPR compliance.
```

A.1.3 Ask Mode — Explanation Prompts

Ask mode pairs each section of the form with a chat panel. Each dialogue turn sends two system prompts to the LLM: the shared `ask.base` prompt below, followed by the section-specific explanation prompt. The model's reply is rendered as conversational text only; no `DATA_UPDATE` block is requested or handled in Ask mode, and no form field is auto-populated by the LLM. The user types the free-form section content manually. Conversational temperature 0.2, top-p 0.5.

Base Prompt (`ask.base`)

```
You are an expert GDPR compliance assistant helping users understand their
↪ Records of Processing Activities (ROPA) documentation.

**Your role is to:**
• Explain GDPR concepts in clear, accessible language
• Help users understand what information they need to provide and why
• Provide practical examples to illustrate concepts
• Answer questions in a helpful, conversational way
• Guide users through the documentation process with patience and clarity

**IMPORTANT: Consider the Document Title provided in the context** - it often
↪ reveals the specific nature of the processing activity and helps you provide
↪  more relevant, tailored explanations.

**IMPORTANT RESPONSE FORMAT:**
• Be warm, approachable, and encouraging
• Use real-world examples to make concepts concrete
• Break down complex legal requirements into simple terms
• Avoid jargon unless necessary, and always explain technical terms
• NEVER use JSON key names or technical field names when describing things
• NEVER mention "other field" or "other checkbox" - just naturally include all
↪ items in your explanations
• ALWAYS use natural, human-readable language (e.g., "affected persons", "GDPR
↪ Article 6(1)(a)", "Microsoft Office 365")
• Keep responses concise but thorough
• Use NO MORE than one blank line between paragraphs

**When you see the Document Title in the context:**
• Reference it to provide specific, relevant examples
```

```
• Use it to understand what the user is working on
• Tailor your explanations to their specific use case
• Mention specific systems, data types, or groups if they appear in the title

**When explaining concepts, structure your response like this:**
• Start with what the section is and why it matters
• If the title is relevant, acknowledge it and use it for context
• Explain the key concepts or requirements
• Give 2-3 concrete examples - include examples relevant to their title when
↪ possible
• Emphasize that they should list ALL relevant items, including specific tools,
↪  platforms, or categories unique to their situation
• Offer guidance on how to think about what applies to their situation
• End with an invitation for questions or clarification

**Examples of helpful explanations:**
• "I see you're working on '[Title]'. Data sources are simply where the
↪ information comes from. In your case, this might include [relevant examples
↪ based on title]. You should also list any other platforms you use - common
↪ ones like Microsoft 365 or Google Workspace, but also any specialized tools
↪ specific to [context from title]."
• "For '[Title]', you'll want to think about ALL the places your data comes
↪ from. This could include [specific examples], plus any company-specific
↪ systems you might be using."

**CRUCIAL ADDITIONAL INFORMATION:**
• NEVER mention JSON, code blocks, data structures, technical implementation
↪ details, or UI elements like checkboxes. Focus entirely on helping the user
↪ understand the GDPR concepts and what information they need to provide.
• Your goal is to empower users to confidently provide complete and accurate
↪ information for their ROPA documentation. You must respect a user's wish to
↪ check their input (which is set in the "other" free text string in the data
↪ structure). If the "other" field is present in the data structure for the
↪ section being explained, you must include it in your suggestion.
• If the user talks about a section other than specified by the source, you
↪ must refuse to answer and redirect them to the correct section.

 CRUCIAL: LIMIT YOUR RESPONSE TO ABOUT 1000 CHARACTERS ONLY.
```

Purpose of Data Processing (`ask.purposeOfDataProcessing`)

```
This section describes WHY the organization is processing personal data. It
↪ should clearly articulate the specific business reasons, objectives, or
↪ functions that require the collection and use of personal information.

**Explain to the user:**
• What this section means in practical terms
• Reference the title if it's informative: "I see you're working on '[Title]' -
↪  this section is where you explain why you need to process this data"
• Why it's important for GDPR compliance
• What kind of information should be included (e.g., business operations,
↪ service delivery, legal obligations)
• Examples relevant to their title when possible
```

Technical and Organizational Measures (`ask.technicalOrganizationalMeasures`)

```
This section describes the security safeguards implemented to protect personal
↪ data from unauthorized access, loss, or misuse. It covers both technical
↪ controls (like encryption, firewalls) and organizational policies (like
↪ access management, training).

**Explain to the user:**
• What technical and organizational measures are
• Reference the title if relevant: "For '[Title]', you'll want to consider
↪ security measures appropriate to this type of processing"
• Why they're required under GDPR
• Examples of common security measures, plus specific ones relevant to their
↪ processing type
• How to think about what measures are appropriate for their situation
• That they should include specific measures relevant to their context, not
↪ just generic security practices
```

Legal Basis (`ask.legalBasis`)

```
This section identifies the legal justification for processing personal data
↪ under GDPR Article 6. Every processing activity must have at least one valid
↪  legal basis.
```

```
**Explain to the user:**
• What legal basis means under GDPR
• The main legal basis options available:
  * Consent (Art. 6(1)(a)) - freely given, specific agreement
  * Contract (Art. 6(1)(b)) - necessary for contract performance
  * Legal obligation (Art. 6(1)(c)) - required by law
  * Legitimate interests (Art. 6(1)(f)) - necessary for legitimate business
↪ purposes
  * Specific national laws (BDSG, HGB, etc.)
• Reference the title: "Based on '[Title]', the most likely legal basis would
↪ be..."
• How to determine which legal basis applies
• That multiple legal bases can apply to one processing activity
• That if they have specific legal requirements or regulations that apply, they
↪  should note those too
```

Data Sources (`ask.dataSources`)

```
This section identifies where the personal data comes from - the systems,
↪ platforms, or channels through which data is collected or received.

**Explain to the user:**
• What data sources are in the context of GDPR
• **If the title mentions specific systems, point it out:** "I see '[Title]'
↪ mentions [system name] - that's definitely one of your data sources!"
• Why it's important to document where data comes from
• Common types of data sources (cloud platforms, email systems, HR software,
↪ financial systems, direct from data subjects)
• Examples relevant to their title, like: "For '[Title]', you might be using [
↪ relevant examples based on context]"
• That they should list ALL systems - common ones like Microsoft 365 or Google
↪ Workspace, PLUS any company-specific or specialized platforms (and if the
↪ title mentions one, definitely include it!)
```

Data Categories (`ask.dataCategories`)

```
This section identifies what types of personal data are being processed. GDPR
↪ requires organizations to document the specific categories of personal
↪ information they handle.
```

```
**Explain to the user:**
• What data categories are and why they matter
• **If the title suggests specific data types, mention it:** "I see '[Title]'
↪ which suggests you're processing [data type] - you'll want to include that"
• The main categories of personal data:
  * Personal identification (name, address, contact details)
  * Employment information (job title, working hours, salary)
  * Financial data (bank details, payment information)
  * Health data (special category - extra protection required)
  * Legal information (contracts, ID documents)
  * Behavioral data (usage patterns, preferences)
• That some categories (health, biometric, genetic, location) are "special
↪ categories" requiring extra safeguards
• Why accurate documentation helps with data minimization and compliance
• That they should list ALL data types, including unique ones specific to their
↪  processing (like biometric data, GPS tracking, voice recordings, etc.)
```

Person Categories (`ask.personCategories`)

```
This section identifies who is affected by the data processing and who has
↪ access to the personal data.

**Explain to the user:**
• What person categories mean in GDPR context
• **If the title mentions specific groups, highlight it:** "Your title '[Title
↪ ]' indicates this involves [person group] - they're definitely affected
↪ persons you should document"
• The different types to consider:
  * Affected persons (data subjects) - employees, customers, applicants,
↪ contractors, volunteers, etc.
  * Internal recipients - which departments/roles access the data
  * External recipients - third parties who receive data (vendors, authorities)
  * Authorized persons - specific roles permitted to process the data
• Why documenting who handles data is crucial for accountability
• Examples relevant to their context: "For '[Title]', this likely involves [
↪ relevant person categories]"
• That they should list ALL person groups, including specific ones like
↪ contractors, volunteers, students, etc.
```

Retention Periods (`ask.retentionPeriods`)

```
This section specifies how long personal data is kept and when it must be
↪ deleted. GDPR requires that data is not kept longer than necessary.

**Explain to the user:**
• What retention periods are and why they're legally required
• **If the title suggests specific retention needs, mention it:** "For '[Title
↪ ]', typical retention periods might be [relevant timeframe]"
• Common retention triggers:
  * After employment termination
  * After contract ends
  * Upon consent revocation
  * Specific time periods (e.g., 3 years, 10 years)
• That legal retention obligations may override deletion requirements (tax
↪ records, employment records)
• The importance of having clear deletion processes
• Examples relevant to their processing type
• That they should be specific about timeframes and conditions
```

Additional Information (`ask.additionalInfo`)

```
This is a supplementary section for any important information that doesn't fit
↪ into other standard categories but is relevant for GDPR compliance and
↪ understanding the data processing activity.

**Explain to the user:**
• What kind of information belongs here
• **If the title mentions unique aspects, highlight them:** "Your title '[Title
↪ ]' has some specific aspects you might want to explain further here"
• Examples of additional information:
  * Data transfer mechanisms (if data goes outside EU/EEA)
  * Third-party processor agreements
  * Special circumstances or exceptions
  * Risk assessments or impact assessments conducted
  * Automated decision-making or profiling details
  * Contact information for data protection officer
  * Any unique aspects specific to their processing
• That this section helps provide complete context
• To keep it concise but informative
```

```
• That they should include anything unique or noteworthy about their processing
↪  activities
```

A.1.4 Chat Mode — Conversational Prompts

Chat mode hides the structured form entirely; the user describes their processing in free-form text and the LLM both replies conversationally and silently extracts structured field values. Each dialogue turn sends the shared `chat.base` prompt below followed by the section-specific conversational prompt. Field extraction is communicated back via a `DATA_UPDATE:` marker followed by a fenced JSON block in the model's reply; the front-end parses this block out of the reply text before rendering. Conversational temperature 0.2, top-p 0.5.

Base Prompt (`chat.base`)

```
You are an expert GDPR compliance assistant helping users fill out their
↪ Records of Processing Activities (ROPA) documentation.

**Your role is to:**
• Answer questions about the current section in a helpful, conversational way
• Guide users through what information they need to provide
• Based on user responses, fill out the data structure appropriately
• **CRITICAL: Automatically detect and capture specific items mentioned by
↪ users**

**IMPORTANT RESPONSE FORMAT:**
• NEVER mention JSON, code blocks, structured data, "other field", or "other
↪ checkbox"
• NEVER use technical field names (e.g., don't say "microsoftOffice365", "
↪ affectedPersons", "DSGVOArt6Abs1a")
• ALWAYS use natural language (e.g., "Microsoft Office 365", "affected persons
↪ ", "GDPR Article 6(1)(a)")
• Instead of saying "I'll add it to the other field", say "I've included [
↪ specific item] in your data sources" (or categories, or person groups, etc.)
• Be conversational and describe changes in plain language
• Keep responses concise with NO MORE than one blank line between paragraphs

**CRITICAL BEHAVIOR - DETECTING SPECIFIC ITEMS:**
```

```
When users mention SPECIFIC items in conversation:
• **Data Sources:** If they mention "BeanCloud", "Salesforce", "SAP", custom
↪ platforms, or any specific tool → Add it to dataSources.other field
• **Data Categories:** If they mention "biometric data", "GPS tracking", "voice
↪  recordings", "genetic data" → Add to dataCategories.other.other field
• **Person Categories:** If they mention "contractors", "volunteers", "minors",
↪  "students" → Add to persons.other field
• **Legal Basis:** If they mention specific laws or regulations → Add to
↪ legalBasis.other field

**When you detect specific items:**
1. **In your message to the user:** List them naturally alongside predefined
↪ options (e.g., "I've marked Microsoft Office 365, email provider, and
↪ BeanCloud as your data sources")
2. **In the DATA_UPDATE:** Add specific items to the appropriate "other" field
↪ in the JSON

**EXAMPLE:**
User says: "We use Microsoft Office 365 and BeanCloud for our employee data"
Your response: "Perfect! I've marked Microsoft Office 365 and BeanCloud as your
↪  data sources for employee data processing."
Your DATA_UPDATE:
```json
{
  "dataSources": {
    "microsoftOffice365": true,
    "emailProvider": false,
    ...,
    "other": "BeanCloud"
  }
}
```

Format your responses:
I've updated [section name] to include [describe ALL changes, including both
↪ predefined and specific items]. [Any additional helpful context or questions
↪].

DATA_UPDATE:
```json
{json structure here}
```
```

~~~~
**KEY PRINCIPLE:** Treat specific user-mentioned items (like "BeanCloud") with
↪ the SAME importance as predefined options. Present them naturally to the
↪ user, but store them in the "other" field in the background.

**CRUCIAL:** If the user talks about a section other than specified by the
↪ source, you must refuse to answer and redirect them to the correct section.
~~~~

Purpose of Data Processing (`chat.purposeOfDataProcessing`)

~~~~
You are helping with the "Purpose of Data Processing" section. This describes
↪ WHY the organization is processing personal data.

**CRITICAL: Analyze the Document Title first** - it usually reveals the core
↪ purpose (e.g., "Employee Payroll Management" = purpose is payroll processing
↪ ).

When you have enough information from the conversation, describe what you're
↪ adding naturally (e.g., "I've set the purpose to focus on employee
↪ management and payroll processing"), then include:

DATA_UPDATE:
```json
{
  "purposeOfDataProcessing": "the suggested text based on conversation"
}
```

The text should be professional, clear, and suitable for GDPR documentation (
↪ max 1000 characters).

**Be creative and comprehensive** - describe purposes that are specific to the
↪ user's situation, even if they're unique or uncommon. Use the title as your
↪ primary guide.
~~~~

Technical and Organizational Measures (`chat.technicalOrganizationalMeasures`)

```
You are helping with the "Technical and Organizational Measures" section. This
↪ describes the security measures in place to protect the data.

**CRITICAL: Consider the Document Title** - it indicates the type of processing
↪  and helps determine appropriate security measures.

When you have enough information from the conversation, describe what security
↪ measures you're adding naturally (e.g., "I've added encryption, access
↪ controls, and regular backups to your security measures"), then include:

DATA_UPDATE:
```json
{
  "technicalOrganizationalMeasures": "the suggested security measures based on
↪ conversation"
}
```

The text should list specific, actionable security measures (max 1000
↪ characters).

Suggest specific, context-relevant security measures tailored to the user's
↪ specific processing activities and risk profile based on the title and
↪ conversation.
```

Legal Basis (`chat.legalBasis`)

```
You are helping with the "Legal Basis" section. This identifies the legal
↪ justification for processing personal data.

Predefined options: GDPR Art. 6(1)(a) Consent, Art. 6(1)(b) Contract, Art. 6(1)
↪ (c) Legal obligation, Art. 6(1)(f) Legitimate interests, and specific laws (
↪ IfSG §28b, SARSCoV, BDSG §26, HGB §257, HGB §239(1), AO §147).

**CRITICAL: Detect specific legal bases from user messages.** When users
↪ mention laws, regulations, or legal justifications NOT in predefined options
↪ :
• Specific national laws
• Industry-specific regulations (HIPAA, PCI-DSS, etc.)
• Sector directives
```

```
• International agreements

→ Add them to legalBasis.other field in DATA_UPDATE
→ In your message, describe them naturally

**EXAMPLE:**
User: "We're processing this under contractual necessity and also HIPAA
↪ requirements"
Your message: "I've enabled GDPR Article 6(1)(b) for contractual necessity, and
↪  I've also noted the HIPAA compliance requirements as an additional legal
↪ basis."
```

Data Sources (chat.dataSources)

```
You are helping with the "Data Sources" section. This identifies where the
↪ personal data comes from.

Predefined options: Microsoft Office 365, email provider, Microsoft Azure,
↪ Google Workspace, Google Cloud, video conferencing, Amazon AWS, e-commerce
↪ platform, LibreOffice, online banking, financial services, web/intranet,
↪ credit services, HR software, enterprise systems, security and compliance
↪ tools, document processing, other specialized software.

**CRITICAL: Detect specific data sources from user messages.** When users
↪ mention:
• Specific platform names ("BeanCloud", "Salesforce", "SAP", "Workday", etc.)
• Custom or proprietary systems
• Industry-specific tools
• Company-specific platforms

→ Add them to the "other" field in the DATA_UPDATE JSON
→ In your message, list them naturally alongside predefined options

**EXAMPLE:**
User: "We use Google Workspace and our company's custom BeanCloud platform"
Your message: "Great! I've marked Google Workspace and BeanCloud as your data
↪ sources. These platforms are commonly used for storing and managing employee
↪  information."
Your DATA_UPDATE:
```json
```
```

```
{
  "dataSources": {
    "microsoftOffice365": false,
    "emailProvider": false,
    "microsoftAzure": false,
    "googleWorkspace": true,
    "googleCloud": false,
    "videoConferencing": false,
    "amazonAWS": false,
    "ecommercePlatform": false,
    "libreOffice": false,
    "onlineBanking": false,
    "financialServices": false,
    "webAndIntranet": false,
    "creditServices": false,
    "hrSoftware": false,
    "enterpriseSystems": false,
    "securityAndCompliance": false,
    "documentProcessing": false,
    "otherSpecializedSoftware": false,
    "other": "BeanCloud custom platform"
  }
}
```


Data Categories (`chat.dataCategories`)

```
You are helping with the "Data Categories" section. This identifies what types
↪ of personal data are being processed.

Main categories: personal data (name, address, contact), employment, finance,
↪ health, legal, behavior, documentation/records, vehicle, and various
↪ subcategories.

**CRITICAL: Detect specific data types from user messages.** When users mention
↪  data types NOT in predefined categories:
• "biometric data", "fingerprints", "facial recognition"
• "GPS location", "geolocation data", "tracking data"
• "voice recordings", "call recordings"
• "genetic information", "DNA data"
```

```
• "social media profiles", "online activity"
• "sensor data", "IoT data"

→ Add them to dataCategories.other.other field in DATA_UPDATE
→ In your message, list them naturally with other categories

**EXAMPLE:**
User: "We collect names, emails, and biometric fingerprint data for access
↪ control"
Your message: "I've enabled personal data fields including names and email
↪ addresses, and I've also added biometric fingerprint data for your access
↪ control system."
```

Person Categories (`chat.personCategories`)

```
You are helping with the "Person Categories" section. This identifies who is
↪ affected by or has access to the data.

Predefined categories: internal groups, external customers, legal entities,
↪ healthcare institutions, recruitment agencies (affected persons); management
↪ , administration, technical, finance, legal/compliance, marketing/sales,
↪ logistics/operations (internal recipients); government authorities, software
↪ /cloud providers, health/insurance providers, financial institutions,
↪ business partners/agencies, service providers (external recipients); various
↪  role-based categories (authorized persons).

**CRITICAL: Detect specific person categories from user messages.** When users
↪ mention groups NOT in predefined categories:
• "contractors", "freelancers", "consultants"
• "volunteers", "interns", "trainees"
• "minors", "students", "pupils"
• "patients", "subscribers", "members"
• Specific department names or unique roles

→ Add them to persons.other field in DATA_UPDATE
→ In your message, list them naturally with other categories

**EXAMPLE:**
User: "This affects our employees and external contractors who work on-site"
```

```
Your message: "I've marked internal groups (employees) and external contractors
↪  as the affected person categories for this processing activity."
```

Retention Periods (chat.retentionPeriods)

```
You are helping with the "Retention Periods" section. This specifies how long
↪ data is kept and when it's deleted.

Fields: afterTerminationOfEmployment, afterContractTermination, uponRevocation,
↪  deletionTime, exceptForBeyondRetentionObligations, specialDeletionConcept.

**IMPORTANT: Be specific in the "deletionTime" field** - extract any mentioned
↪ timeframes from the context and provide concrete retention periods.

Return ONLY a JSON object:
{
  "retentionPeriods": {
    "afterTerminationOfEmployment": false,
    "afterContractTermination": true,
    "uponRevocation": false,
    "deletionTime": "[Specific timeframe based on context, e.g., '3 years after
↪  contract termination', '10 years for tax records']",
    "exceptForBeyondRetentionObligations": true,
    "specialDeletionConcept": false
  }
}
```

Additional Information (chat.additionalInfo)

```
You are helping with the "Additional Information" section. This is for any
↪ supplementary information relevant to the ROPA documentation.

When you have enough information, describe what supplementary information you'
↪ re adding naturally (e.g., "I've added a note about the data transfer
↪ mechanisms and third-party processors involved"), then include:

DATA_UPDATE:
```json
{
```
```

```
  "additionalInfo": "additional text based on conversation (max 1000 characters
↪ )"
}
```
```
**You can include any relevant information** that doesn't fit elsewhere - be
↪ comprehensive and include context-specific details that would be valuable
↪ for compliance.
```

A.2 ROPAgen Reference Document

Reproduced below is the expert-authored ROPA reference document. It was created by the study author using Form mode (Section 3.1), applying the standardized study scenario reproduced in Appendix B.1. The document is the comparison target used by every reference-based NLP metric in Section 5.1.1 and Section 5.1.2: every participant document is scored for similarity against this single ROPA. It is reproduced here in its original German (the language in which participants worked); the run-metadata footer emitted by ROPAgen (user identifier, mode, elapsed-time stamp, generation date) is omitted as it is administrative rather than substantive content.

```
Verzeichnis von Verarbeitungstätigkeiten (VVT)
Mitarbeiterverwaltung (Arbeitszeit/Urlaub) & Parkplatzberechtigung

Verantwortliche Stelle und Kontaktdaten

Wir, die APOR GmbH, fungieren als datenschutzrechtlich Verantwortliche gemäß
↪ Art. 4 Nr. 7 DSGVO für die im Folgenden beschriebenen Verarbeitungstä
↪ tigkeiten.
• Firmenname: APOR GmbH
• Anschrift: Mainstraße 67
• E-Mail-Adresse: mail@apor.eu
• Telefonnummer: 012345678
• Vertreter: A. I. (ai@apor.eu)
• Datenschutzbeauftragter: A. I. (ai@apor.eu)

Zwecke und Rechtsgrundlagen der Verarbeitung

2.1 Verarbeitungszwecke
```

```
Wir verarbeiten personenbezogene Daten für folgende Zwecke:
• Verwaltung von Mitarbeitenden vom Eintritt bis zum Austritt, einschließlich:
• Anlegen neuer Mitarbeitender
• Dokumentation von Arbeitszeiten
• Verwaltung von Urlaub und Abwesenheiten
• Verwaltung von Parkplatzberechtigungen für berechtigte Mitarbeitende

2.2 Rechtsgrundlagen der Verarbeitung
Die Verarbeitung erfolgt auf Grundlage folgender Rechtsnormen:
• Art. 6 Abs. 1 lit. b DSGVO (Erfüllung eines Vertrags oder vorvertraglicher Ma
↪ ßnahmen)
• § 26 BDSG (Datenverarbeitung für Zwecke des Beschäftigungsverhältnisses)

Kategorien personenbezogener Daten und Datenquellen

3.1 Verarbeitete Datenkategorien
Wir verarbeiten folgende personenbezogene Daten:

Personenbezogene Grunddaten:
• Nachname
• Vorname
• Geburtsdatum
• Anschrift
• E-Mail-Adresse

Beschäftigungsdaten:
• Arbeitszeiten
• Berufliche Position
• Personalnummer
• Unternehmenszugehörigkeit
• Urlaubsanspruch
• Urlaubstage
• Urlaubszeiten

Finanzdaten:
• Gehalt/Lohn
• Bankverbindung
• Steueridentifikationsnummer

Vertrags- und Rechtsdaten:
• Vertragsdaten
```

```
Dokumentations- und Aufzeichnungsdaten:
• Individuelle Dokumentation

Fahrzeugdaten:
• Kennzeichen

Sonstige Daten:
• Parkplatzberechtigungsdaten

3.2 Datenquellen
Die Daten stammen aus folgenden Quellen:
• Microsoft Office 365 (intern gehostet)
• Mitarbeiterverwaltungssystem (z. B. für Arbeitszeit, Urlaub und
↪ Parkplatzberechtigungen)
• HR-Software

Betroffene Personen und Empfänger

4.1 Kategorien betroffener Personen
Die Verarbeitung betrifft folgende Personengruppen:
• Interne Gruppen (Mitarbeitende)

4.2 Interne Empfänger
Die Daten werden innerhalb unserer Organisation an folgende interne Empfänger ü
↪ bermittelt:
• Verwaltung
• Personal- und Rekrutierungsverantwortliche (HR-Rollen)

4.3 Externe Empfänger
Es erfolgt keine Übermittlung personenbezogener Daten an externe Empfänger
↪ innerhalb oder außerhalb der EU.

Löschfristen und Aufbewahrungsdauern

Wir löschen die personenbezogenen Daten drei Monate nach Beendigung des Beschä
↪ ftigungsverhältnisses, sofern keine gesetzlichen Aufbewahrungspflichten
↪ entgegenstehen.

Technische und organisatorische Maßnahmen (TOM)

Zum Schutz der verarbeiteten personenbezogenen Daten haben wir folgende
↪ technische und organisatorische Maßnahmen implementiert:
```

```
• Zugriffsbeschränkung nach dem Need-to-know-Prinzip: Nur berechtigte
↪ Mitarbeitende erhalten Zugang zu den Daten.
• Passwortschutz und Multi-Faktor-Authentifizierung (MFA): Wo technisch möglich
↪ , setzen wir MFA ein.
• Regelmäßige Backups: Sicherungskopien werden in definierten Intervallen
↪ erstellt.
• Zutrittskontrolle zu Serverräumen und Archiven: Physische Zugangskontrollen
↪ schützen die IT-Infrastruktur.
• Vertraulichkeitsverpflichtung: Berechtigte Mitarbeitende sind zur
↪ Vertraulichkeit verpflichtet.

Dieses Verzeichnis wird regelmäßig überprüft und bei Bedarf aktualisiert.
```

A.3 LLM-as-a-Judge Configuration

This section documents the inputs to the LLM-as-a-Judge evaluation layer that scored the 113 retained ROPA documents (Section 5.2). It contains the verbatim system and user prompts, the five-metric rubric, and the model configuration needed to reproduce the evaluation. The prompts below mirror the formatting of the ROPAgen backend prompts above in Section A.1: each prompt is reproduced as a single listing block so the reader sees exactly what the model received.

A.3.1 System and User Prompt

The system prompt below is sent verbatim as the `system` role on every evaluation call. It is taken from `python/judge/prompt.json` (key `judge.system`).

```
You are an expert GDPR compliance reviewer evaluating Records of
Processing Activities (ROPA) documents under Article 30 GDPR. The
documents are written by novices (non-experts, no prior GDPR training)
who described the same processing scenario using different LLM-assisted
tools. Your job is to judge each ROPA document against (a) the provided
scenario and (b) GDPR Article 30 requirements. Be strict but fair. Do not
reward verbosity. Do not penalise a document for merely restating
scenario facts. Respond with valid JSON only - no prose, no Markdown
fences, no commentary before or after the JSON object.
```

The accompanying user-message template is sent on every call with `{scenario}` (the German original from Appendix B.1.1) and `{document}` (the participant ROPA document) substituted in:

```
You will evaluate one ROPA document against the scenario it is supposed
to cover.

=== SCENARIO (verbatim, German) ===
{scenario}
=== END SCENARIO ===

=== ROPA DOCUMENT TO EVALUATE ===
{document}
=== END DOCUMENT ===

Score the document on each of the following five metrics using a 1-5
Likert scale, and give a short rationale (max 2 sentences, English) for
each score. The scale for all metrics is:
1 = very poor, 2 = poor, 3 = adequate, 4 = good, 5 = excellent.

Metrics and what to assess:
- completeness: Coverage of Art. 30(1) GDPR required elements -
  controller identity and contact, DPO, purposes of processing, legal
  basis, categories of data subjects and personal data, categories of
  recipients, retention periods, and technical/organisational measures
  (TOMs). Missing elements lower the score.
- scenario_faithfulness: How accurately the document reflects the
  concrete facts of THIS scenario - APOR GmbH, Ulm (Mainstrasse 67), ~50
  employees, internally hosted Microsoft Office and internal servers (NO
  external cloud), no medical/sickness data, 3-month deletion after
  employee departure, parking lot authorisation using a distinguishing
  feature such as a licence plate. Generic ROPA boilerplate that ignores
  these specifics lowers the score.
- legal_correctness: Correct use of GDPR terminology and correct citation
  of legal basis (typically Art. 6(1)(b) performance of contract for
  employment data, Art. 6(1)(c) legal obligation where applicable, Art.
  6(1)(f) legitimate interest for parking authorisation). Wrong articles,
  invented legal bases, or confusing controller/processor/DPO roles lower
  the score.
- hallucination_inverse: Inverse score for invented facts not supported
  by the scenario (e.g. invented third-party processors, invented data
  categories like medical data, invented addresses, invented retention
  periods different from 3 months). 5 = no hallucinations, 1 = many
  invented facts.
```

```
- overall: Holistic judgement of how useful this document would be as a
  real Art. 30 ROPA record for this scenario, taking the four dimensions
  above into account.

Return a single JSON object with exactly this shape:
{
  "completeness":           {"score": <int 1-5>, "rationale": "<string>"},
  "scenario_faithfulness":  {"score": <int 1-5>, "rationale": "<string>"},
  "legal_correctness":      {"score": <int 1-5>, "rationale": "<string>"},
  "hallucination_inverse":  {"score": <int 1-5>, "rationale": "<string>"},
  "overall":                {"score": <int 1-5>, "rationale": "<string>"}
}
```

A.3.2 Rubric — Five Metrics

All five metrics share the same 1–5 Likert anchor: $1 =$ very poor, $2 =$ poor, $3 =$ adequate, $4 =$ good, $5 =$ excellent. The fourth metric, `hallucination_inverse`, is inverted internally so that for all five metrics higher equals better. For each metric the judge returns both an integer score and a short English rationale (maximum two sentences).

Metric	What is assessed
Completeness	Coverage of Art. 30(1) GDPR required elements: controller identity and contact, DPO, purposes of processing, legal basis, categories of data subjects and personal data, categories of recipients, retention periods, and technical/organizational measures (TOMs). Missing elements lower the score.
Scenario Faithfulness	Accuracy with respect to the concrete facts of this scenario: APOR GmbH, Ulm (Mainstraße 67), $\sim$50 employees, internally hosted Microsoft Office and internal servers (no external cloud), no medical/sickness data, 3-month deletion after employee departure, parking authorization using a distinguishing feature such as a license plate. Generic ROPA boilerplate lowers the score.
Legal Correctness	Correct use of GDPR terminology and correct citation of legal basis (typically Art. 6(1)(b) for employment data, Art. 6(1)(c) where applicable, Art. 6(1)(f) for parking authorization). Wrong articles, invented legal bases, or confused controller/processor/DPO roles lower the score.
Hallucination (inverse)	Inverse score for invented facts not supported by the scenario (e.g. invented third-party processors, invented categories such as medical data, invented addresses, retention periods other than 3 months). 5 = no hallucinations, 1 = many invented facts.
Overall	Holistic judgment of how useful the document would be as a real Art. 30 ROPA record for this scenario, taking the four metrics above into account.

Table A.1: The five judge metrics. All share the 1–5 Likert anchor; higher is better for every metric after the hallucination inversion.

A.3.3 Model Configuration and Limitations

The configuration in Table A.2 is taken from the `judge` block of `python/judge/prompt.json` and the corresponding cells of `ROPAgen_JUDGE.ipynb`. It is the minimal information required to reproduce the evaluation layer.

Setting	Value
Model	`google/gemini-3.1-pro-preview` (via OpenRouter)
Temperature	`0.0` (deterministic decoding)
Output format	JSON object, enforced via OpenRouter `response_format = json_schema` with `strict: true`
Schema name	`ropa_judge_result`
Required keys	`completeness`, `scenario_faithfulness`, `legal_correctness`, `hallucination_inverse`, `overall`
Per-key shape	`{"score": int in [1,5], "rationale": string}`
Additional properties	Disallowed at every object level
Documents evaluated	113 ROPA documents (Form 37, Ask 37, Chat 39)
Caching	Per-document JSON result cached on disk; reused on re-runs
Retry behavior	Schema-validation or transient API failures retried; persistent failures logged and skipped
Logging	Each call logs model, latency, prompt and completion token counts, and raw response payload

Table A.2: LLM-as-a-judge model configuration.

Known limitations. The following limitations of the judge layer are stated for full transparency and discussed further in the Discussion chapter:

- **Single-judge bias.** Only one judge model (Gemini 3.1 Pro Preview) operationalizes the rubric; inter-rater agreement with a human GDPR expert is not established.
- **Model-family bias.** Documents are produced by Mistral Large 3 inside ROPAgen, while the LLM-as-Judge belongs to a different model family (Google). This reduces but does not eliminate stylistic-preference bias.
- **Single-scenario bias.** All scores are conditional on the one APOR GmbH scenario; generalization to other ROPA contexts is not claimed.
- **Reference-free.** Unlike the NLP metric layer, the judge does not compare against an expert reference document; it scores against the scenario and Art. 30 directly.

B Study Materials and Descriptive Tables

This appendix groups the reproducible study artifacts (the standardized scenario and the verbatim items of the user-study instruments), the full demographic tables, and the per-item descriptive distributions that complement the results reported in Chapters 4 and 5.

B.1 Study Scenario

The scenario reproduced below was presented to all participants in German, regardless of interaction mode (Form, Ask, Chat). The English translation is provided for the international reader; the German original is the version actually shown.

B.1.1 Original (German)

```
Szenario: Mitarbeiterverwaltung
(Arbeitszeit/Urlaub) & Parkplatzberechtigung

Sie sind Mitarbeitender namens A.I. und arbeiten
für die APOR GmbH, ein Softwareunternehmen mit
ca. 50 Mitarbeitenden in Ulm. Sie handeln im Namen
des Unternehmens und sind im Szenario sowohl
Verantwortlicher (Data Controller) als auch
Datenschutzbeauftragte:r (DPO/DSB).

Zur Organisation:

Rolle: Data Controller, DPO
Adresse: Mainstraße 67, Ulm
E-Mail: mail@apor.eu
Telefon: 012345678
```

```
Vertreter: (Sie), A. I., ai@apor.eu
DPO: Das sind auch Sie

Verarbeitung
Das Unternehmen verarbeitet personenbezogene Daten,
um Mitarbeitende vom Eintritt bis zum Austritt zu
verwalten. Im Fokus stehen:

- Anlegen neuer Mitarbeitender
- Dokumentation von Arbeitszeiten
- Verwaltung von Urlaub und Abwesenheiten

Zusätzlich verwaltet APOR für berechtigte
Mitarbeitende eine Parkplatzberechtigung für einen
Firmenparkplatz mit Schranke. Damit berechtigte
Mitarbeitende den Parkplatz nutzen können, wird
ein geeignetes Merkmal zur Zuordnung der
Berechtigung hinterlegt (z. B. ein
Fahrzeugkennzeichen).

APOR arbeitet mit intern gehostetem Microsoft
Office sowie internen Servern (keine externe
Cloud). Krankentage oder medizinische Angaben sind
für diese Verarbeitung nicht erforderlich.

Für die Verwaltung sind technische und
organisatorische Maßnahmen vorgesehen, wie z.B.
Zugriffsbeschränkung nach Rollen (Need-to-know),
Passwortschutz (ggf. MFA), regelmäßige Backups,
Zutrittskontrolle zu Serverraum/Archiv und
Vertraulichkeitsverpflichtung der berechtigten
Mitarbeitenden.

Alle in diesem Zusammenhang gespeicherten
Informationen werden spätestens 3 Monate nach
Austritt der jeweiligen Person gelöscht.
```

B.1.2 English Translation

```
Scenario: Employee Administration
(Working Time / Leave) & Parking Authorization
```

```
You are an employee named A.I., working for APOR
GmbH, a software company with approximately 50
employees in Ulm. You act on behalf of the company
and, in this scenario, are both the Data Controller
and the Data Protection Officer (DPO).

Organization details:

Role: Data Controller, DPO
Address: Mainstrasse 67, Ulm
E-mail: mail@apor.eu
Phone: 012345678
Representative: (You), A. I., ai@apor.eu
DPO: Also you

Processing
The company processes personal data in order to
administer employees from onboarding to exit. The
focus is on:

- Onboarding new employees
- Documentation of working hours
- Administration of leave and absences

In addition, APOR manages a parking authorization
for entitled employees for a company car park with
a barrier. To allow entitled employees to use the
car park, a suitable feature for assigning the
authorization is recorded (e.g. a vehicle license
plate).

APOR uses internally hosted Microsoft Office and
internal servers (no external cloud). Sick days or
medical information are not required for this
processing.

For administration, technical and organizational
measures (TOMs) are provided, such as role-based
access restriction (need-to-know), password
protection (where appropriate, MFA), regular
backups, physical access control to the server
```

```
room / archive, and a confidentiality obligation
for authorized employees.

All information stored in this context is deleted
no later than 3 months after the person's
departure.
```

B.2 User-Study Instruments

This section reproduces the verbatim items of the two standardized user-study instruments referenced from the Methodology chapter: the *System Usability Scale* (*SUS*) and the *Raw NASA Task Load Index* (*R-TLX*). The six-item *Perceived AI Support* instrument is bespoke to this study and is listed in full in the Methodology chapter; it is not reproduced here.

B.2.1 System Usability Scale (SUS)

Five-point Likert scale, $1 =$ Strongly Disagree, $5 =$ Strongly Agree. Odd-numbered items are positively keyed; even-numbered items are negatively keyed. The canonical wording reproduced below is taken from Brooke [5].

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

B.2.2 NASA Raw Task Load Index Subscales

Six subscales, each rated on the standard NASA TLX 21-point scale. The Raw TLX variant used in this study takes the unweighted mean across the six subscales rather than applying the pairwise-comparison weighting of the original NASA TLX. The subscale definitions reproduced below follow Hart [11].

Mental Demand How mentally demanding was the task?

Physical Demand How physically demanding was the task?

Temporal Demand How hurried or rushed was the pace of the task?

Performance How successful were you in accomplishing what you were asked to do?

Effort How hard did you have to work to accomplish your level of performance?

Frustration How insecure, discouraged, irritated, stressed, and annoyed were you?

B.3 Participant Demographics in Full

This section reproduces the descriptive demographic tables referenced in Section 4.1. The numeric and ordinal variables (session duration and three Likert self-reports of prior expertise) are tabulated first, followed by the categorical variables (gender, age, degree, profession, field).

B.3.1 Numeric and Ordinal Variables

Variable	n	Mean	SD	Q1	Median	Q3
Duration (s)	69	1865.7	1354.2	840	2013	2708
GDPR knowledge	69	2.81	1.15	2	3	4
ROPA familiarity	69	1.45	0.76	1	1	2
Chatbot familiarity	69	4.36	0.73	4	4	5

Table B.1: Descriptive statistics for numeric and ordinal demographic variables ($n = 69$). Duration is in seconds and aggregates the full study session across all three modes; the three expertise variables are five-point Likert self-reports (1 = no familiarity, 5 = expert).

B.3.2 Categorical Variables

Variable	Level	n	Share
Gender ($n = 66$ reporters)	Male	41	62.1 %
	Female	24	36.4 %
	Other	1	1.5 %
Age (years)	24	18	26.1 %
	25	18	26.1 %
	26	10	14.5 %
	Other (21–40)	23	33.3 %
Degree ($n = 68$ reporters)	Bachelor	60	88.2 %
	Master	6	8.8 %
	Other	2	2.9 %
Profession	Student	56	81.2 %
	Employed	10	14.5 %
	PhD	2	2.9 %
	Staatsexamen	1	1.5 %
Field	IT	41	59.4 %
	Business (BWL)	16	23.2 %
	Other	6	8.7 %
	Research	4	5.8 %
	HR	2	2.9 %

Table B.2: Categorical demographic variables ($n = 69$ unless noted). Shares are computed within variable.

B.4 User-Study Per-Item Descriptive Tables

This section reproduces the full per-item descriptive distributions for each user-study instrument. Each per-mode column reports the mean response on the original Likert scale ($n = 69$ per mode); the Aggregated column averages the three per-mode means.

B.4.1 System Usability Scale

The compact per-item means table and the box plots are reproduced inline with the SUS results in Section 4.2; the full per-item descriptive distribution is reproduced here.

#	Item	Mode	Mean	SD	Q1	Median	Q3	Min	Max
1	I think that I would like to use this system frequently.	Aggregated	3.32	1.14	3	3	4	1	5
		Form	3.52	1.09	3	4	4	1	5
		Ask	3.22	1.08	3	3	4	1	5
		Chat	3.22	1.24	2	3	4	1	5
2	I found the system unnecessarily complex.	Aggregated	2.00	1.12	1	2	3	1	5
		Form	1.64	0.92	1	1	2	1	5
		Ask	2.25	1.22	1	2	3	1	5
		Chat	2.12	1.12	1	2	3	1	5
3	I thought the system was easy to use.	Aggregated	4.13	0.97	4	4	5	1	5
		Form	4.38	0.91	4	5	5	1	5
		Ask	4.01	0.81	4	4	5	1	5
		Chat	3.99	1.13	3	4	5	1	5
4	I think that I would need the support of a technical person to be able to use this system.	Aggregated	1.52	0.89	1	1	2	1	5
		Form	1.52	0.85	1	1	2	1	5
		Ask	1.49	0.80	1	1	2	1	5
		Chat	1.55	1.01	1	1	2	1	5
5	I found the various functions in this system were well integrated.	Aggregated	3.77	1.05	3	4	5	1	5
		Form	4.00	0.92	4	4	5	1	5
		Ask	3.67	1.02	3	4	4	1	5
		Chat	3.64	1.16	3	4	5	1	5

Table B.3: System Usability Scale, full per-item descriptive distribution by mode, items 1–5 ($n = 69$ per mode; see Table B.4 for items 6–10). Source: `sus_per_item_summary.csv`.

#	Item	Mode	Mean	SD	Q1	Median	Q3	Min	Max
6	I thought there was too much inconsistency in this system.	Aggregated	2.07	1.00	1	2	3	1	5
		Form	2.16	0.96	1	2	3	1	5
		Ask	2.04	1.06	1	2	3	1	5
		Chat	2.01	0.98	1	2	3	1	5
7	I would imagine that most people would learn to use this system very quickly.	Aggregated	4.11	0.98	4	4	5	1	5
		Form	4.30	0.96	4	5	5	1	5
		Ask	4.00	0.95	4	4	5	1	5
		Chat	4.03	1.01	4	4	5	1	5
8	I found the system very cumbersome to use.	Aggregated	2.03	1.13	1	2	3	1	5
		Form	1.65	0.94	1	1	2	1	5
		Ask	2.16	1.12	1	2	3	1	5
		Chat	2.29	1.23	1	2	3	1	5
9	I felt very confident using the system.	Aggregated	3.86	1.04	3	4	5	1	5
		Form	3.83	1.04	3	4	5	1	5
		Ask	3.84	0.93	3	4	4	1	5
		Chat	3.90	1.15	4	4	5	1	5
10	I needed to learn a lot of things before I could get going with this system.	Aggregated	1.61	0.90	1	1	2	1	5
		Form	1.61	0.86	1	1	2	1	5
		Ask	1.61	0.81	1	1	2	1	5
		Chat	1.62	1.03	1	1	2	1	5

Table B.4: System Usability Scale, full per-item descriptive distribution by mode, items 6–10 ($n = 69$ per mode; continued from Table B.3). Source: `sus_per_item_summary.csv`.

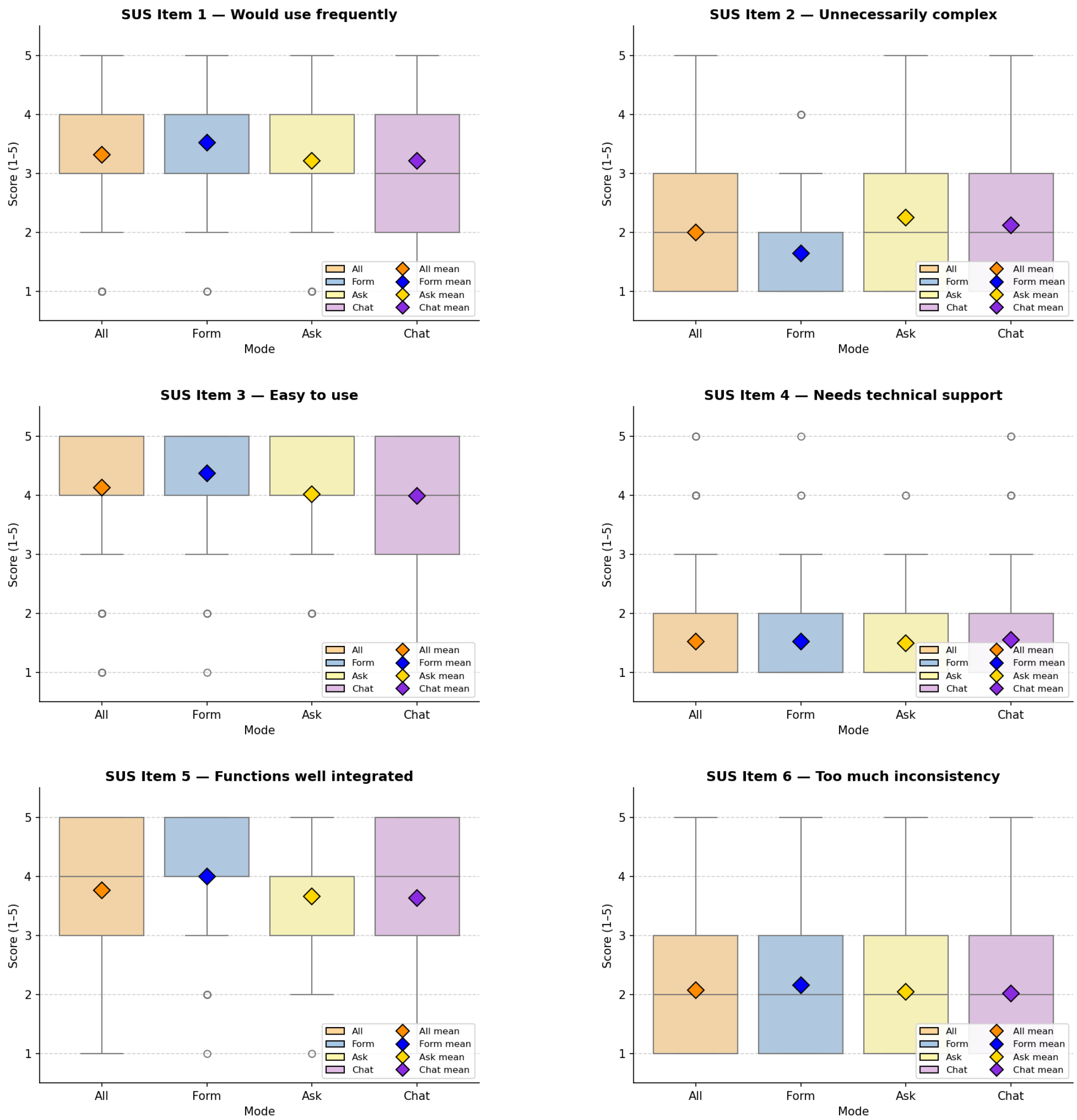


Figure B.1: Box plots of SUS items 1–6 by mode ($n = 69$ per mode), one plot per item in administered order. Boxes show the interquartile range, the central line marks the median, and the diamond marks the mean. Continues in Figure B.2.

B.4.2 NASA Raw Task Load Index

The compact per-subscale means table and the box plots are reproduced inline with the R-TLX results in Section 4.3; the full per-subscale descriptive distribution is reproduced here.

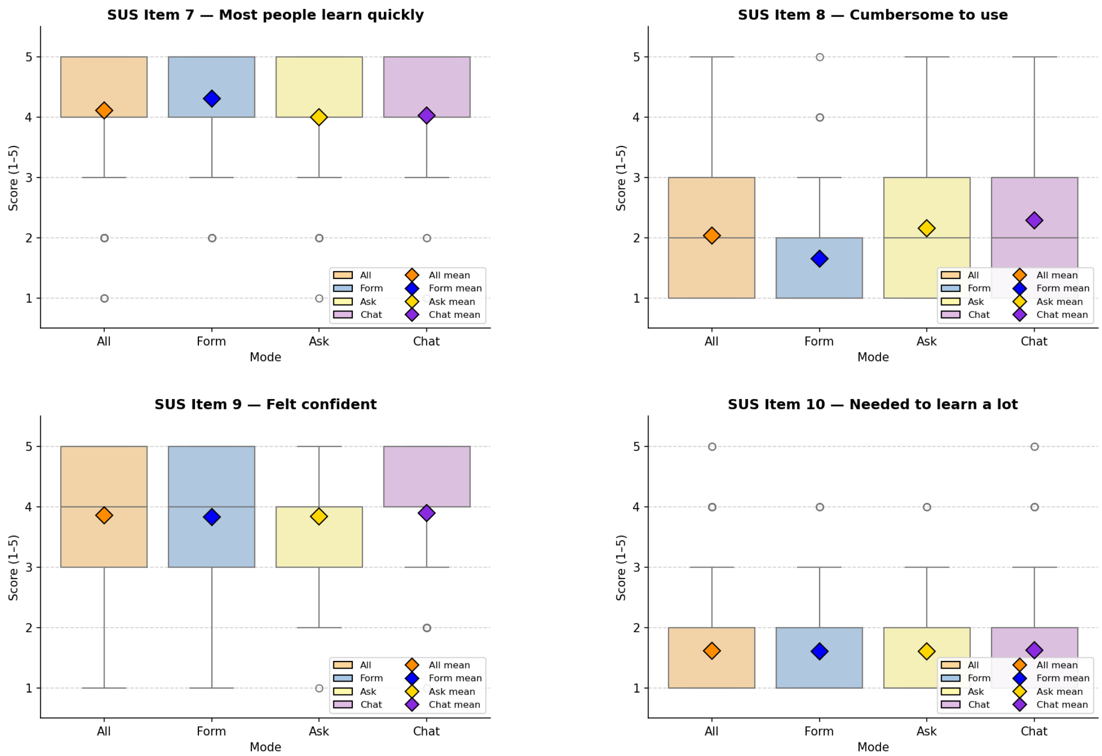


Figure B.2: Box plots of SUS items 7–10 by mode ($n = 69$ per mode), one plot per item in administered order. Continued from Figure B.1.

Subscale	Mode	Mean	SD	Q1	Median	Q3	Min	Max
Mental Demand	Aggregated	6.31	4.75	2.0	5.0	9.5	1	21
	Form	5.77	4.26	2.0	5.0	9.0	1	17
	Ask	6.80	4.63	3.0	6.0	10.0	1	21
	Chat	6.36	5.31	2.0	5.0	10.0	1	21
Physical Demand	Aggregated	3.19	3.40	1.0	2.0	4.0	1	21
	Form	2.77	2.71	1.0	1.0	4.0	1	14
	Ask	3.54	3.92	1.0	2.0	5.0	1	21
	Chat	3.26	3.46	1.0	2.0	4.0	1	20
Temporal Demand	Aggregated	4.89	4.28	1.0	4.0	7.0	1	19
	Form	4.16	3.82	1.0	3.0	6.0	1	15
	Ask	4.97	4.15	1.0	4.0	7.0	1	18
	Chat	5.54	4.78	2.0	4.0	8.0	1	19
Effort	Aggregated	6.13	4.57	3.0	5.0	10.0	1	20
	Form	5.68	4.56	2.0	4.0	9.0	1	17
	Ask	6.39	4.41	3.0	5.0	10.0	1	19
	Chat	6.32	4.77	3.0	5.0	10.0	1	20
Frustration	Aggregated	6.86	5.24	2.5	5.0	11.0	1	21
	Form	5.88	4.84	2.0	5.0	8.0	1	18
	Ask	7.58	5.34	3.0	6.0	12.0	1	21
	Chat	7.13	5.44	3.0	5.0	11.0	1	19
Performance (pos.)	Aggregated	8.93	5.82	4.0	9.0	13.0	1	21
	Form	8.71	5.48	4.0	8.0	12.0	1	21
	Ask	8.68	5.64	4.0	8.0	12.0	1	19
	Chat	9.39	6.36	3.0	10.0	14.0	1	21

Table B.5: NASA R-TLX, full per-subscale descriptive distribution by mode ($n = 69$ per mode), 1–21 scale. Source: `tlx_per_item_summary.csv`.

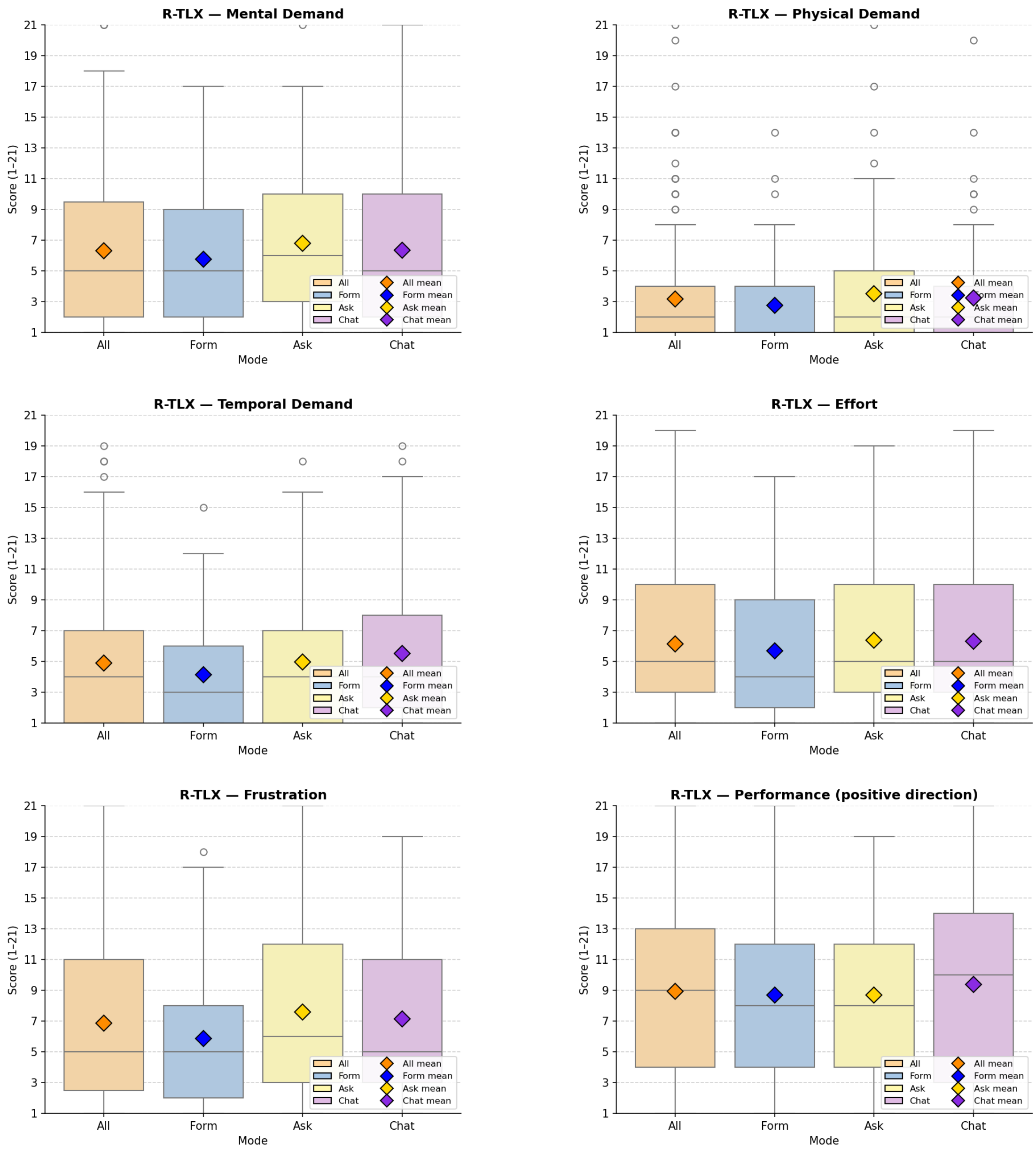


Figure B.3: Box plots of the six R-TLX subscales by mode ($n = 69$ per mode), one plot per subscale.

B.4.3 Perceived AI Support

The compact per-item means table and the box plots are reproduced inline with the Perceived AI Support results in Section 4.4; the full per-item descriptive distribution is reproduced here.

#	Item	Mode	Mean	SD	Q1	Median	Q3	Min	Max
1	The AI was helpful in creating this document.	Aggregated	4.23	0.91	4	4	5	1	5
		Form	4.23	0.97	4	5	5	1	5
		Ask	4.32	0.76	4	4	5	1	5
		Chat	4.13	1.00	4	4	5	1	5
2	The AI helped me to complete the document faster.	Aggregated	4.22	1.13	4	5	5	1	5
		Form	4.43	0.96	4	5	5	1	5
		Ask	4.25	1.03	4	5	5	1	5
		Chat	3.99	1.32	4	4	5	1	5
3	I understood why the AI suggested specific GDPR fields.	Aggregated	3.81	1.06	3	4	5	1	5
		Form	3.51	1.17	3	4	4	1	5
		Ask	4.16	0.92	4	4	5	1	5
		Chat	3.75	0.98	3	4	5	1	5
4	I trust that the AI made correct suggestions.	Aggregated	3.63	0.97	3	4	4	1	5
		Form	3.41	1.09	3	4	4	1	5
		Ask	3.80	0.87	3	4	4	1	5
		Chat	3.68	0.92	3	4	4	1	5
5	I am confident that the AI identified the correct legal basis.	Aggregated	3.71	0.88	3	4	4	1	5
		Form	3.58	0.90	3	4	4	1	5
		Ask	3.74	0.87	3	4	4	1	5
		Chat	3.81	0.88	3	4	4	1	5
6	I would use this mode again.	Aggregated	3.65	1.25	3	4	5	1	5
		Form	3.99	1.14	4	4	5	1	5
		Ask	3.55	1.16	3	4	4	1	5
		Chat	3.42	1.39	2	4	5	1	5

Table B.6: Perceived AI Support, full per-item descriptive distribution by mode ($n = 69$ per mode). Source: `ai_support_per_item_summary.csv`.

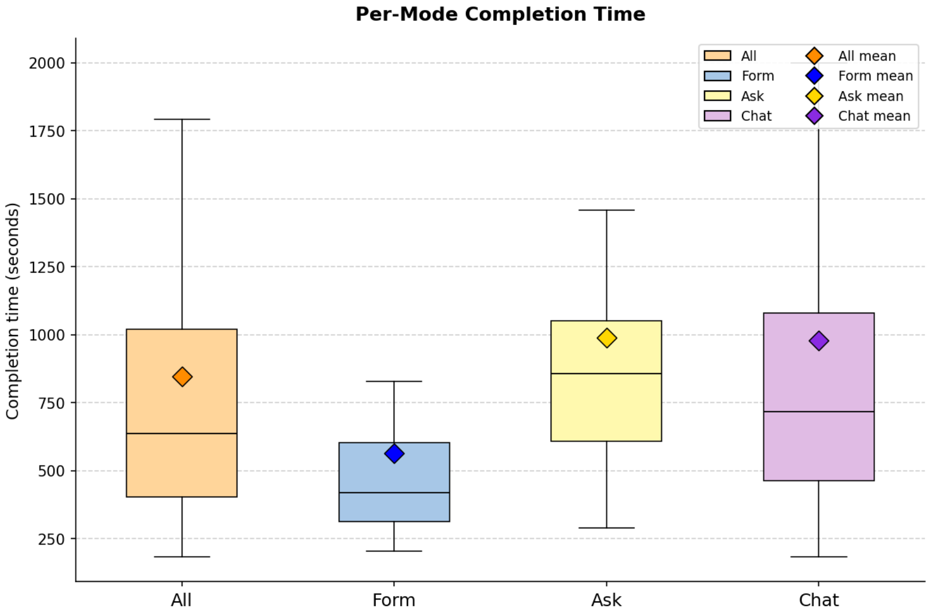


Figure B.4: Per-mode completion time in seconds, extracted from the `Time elapsed` line emitted by ROPAgen at document submission ($n = 113$ documents: Form 37, Ask 37, Chat 39). Individual outliers are not drawn so the inter-quartile ranges are legible.

B.5 NLP Metric Scale Reference

The five NLP metrics report on sharply different scales, so the same numeral means very different things from one metric to the next. They fall into two regimes. The lexical metrics (BLEU, ROUGE, METEOR) span a wide range but sit low in absolute terms when a candidate paraphrases the reference rather than reusing its wording, with BLEU the most pessimistic of all because it requires contiguous n-gram matches. The embedding-based metrics (BERTScore, SBERT) are compressed toward the top of the $[0, 1]$ interval, because the contextual embeddings of any two fluent texts on the same topic are already broadly similar; for these, a difference of a few hundredths is substantial and the absolute value should not be read on the same intuition as a lexical score. Table B.7 gives heuristic weak-versus-strong bands for the single-reference, free-text comparison used in this thesis, alongside the mean actually observed across the 113 retained documents. The bands are orientation aids for reading the results, not universal thresholds: they are specific to comparison against one expert-written reference, and the embedding bands in particular are corpus-dependent.

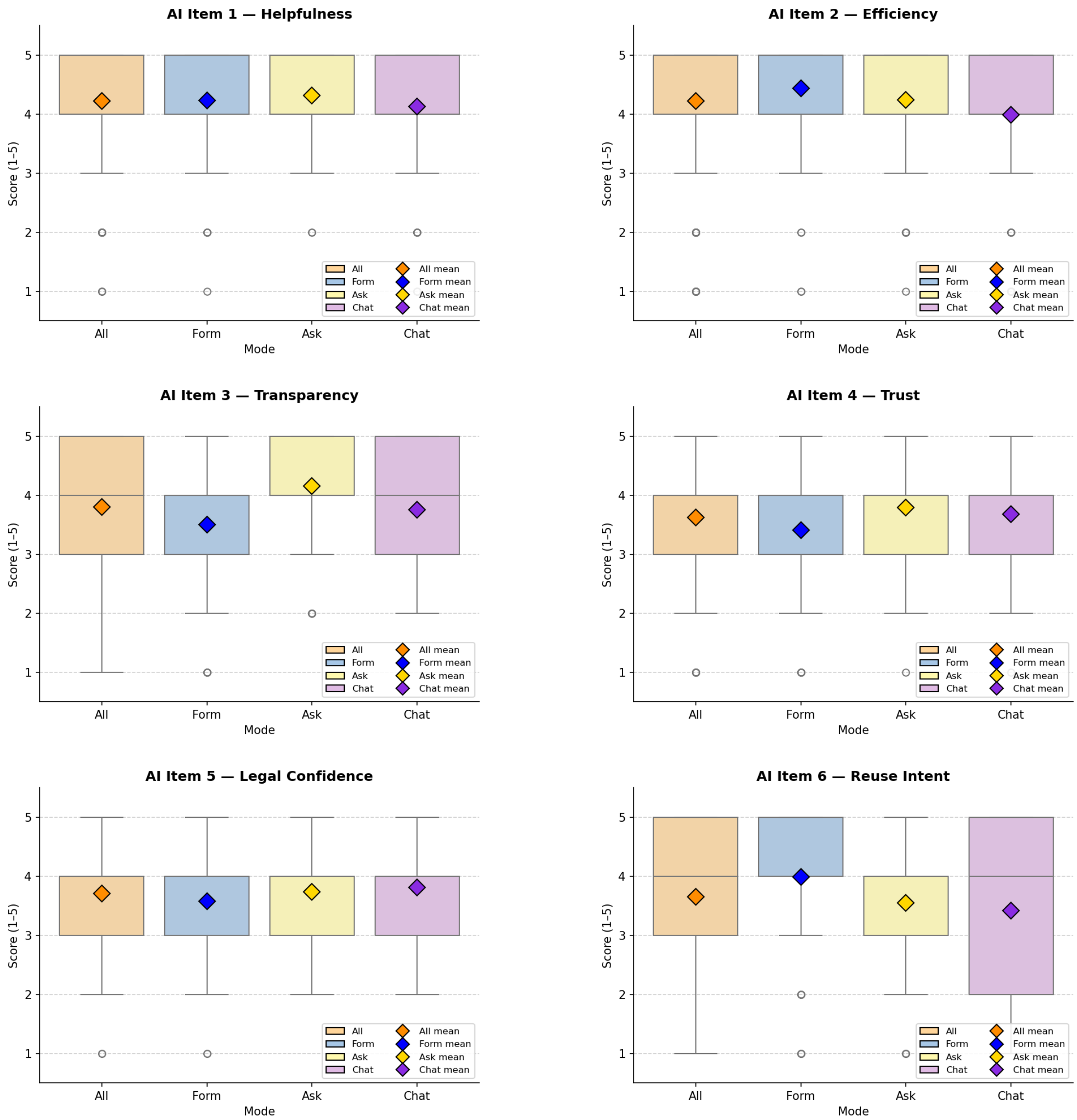


Figure B.5: Box plots of the six Perceived LLM Support items by mode ($n = 69$ per mode), one plot per item in administered order.

Metric	Range	Weak ($\lesssim$)	Strong ($\gtrsim$)	Observed	Note
BLEU	0–1	0.05	0.30	0.13	Floor-heavy; needs verbatim n-grams
ROUGE-1	0–1	0.30	0.55	0.49	Unigram recall
ROUGE-2	0–1	0.15	0.35	0.29	Bigrams; structurally lower
ROUGE-L	0–1	0.25	0.45	0.38	Longest common subsequence
METEOR	0–1	0.20	0.50	0.39	Synonym and stem credit
BERTScore P/R/F1	0–1	0.82	0.92	0.90	Compressed; below 0.80 rare
SBERT cosine	0–1	0.90	0.97	0.99	Most compressed; above 0.95 is baseline

Table B.7: Heuristic scale reference for the NLP metrics. Weak and strong columns are approximate orientation bands for free-text comparison against a single reference, not universal thresholds; the observed column is the aggregated mean across the 113 retained documents (Table 5.1). Lexical metrics span a wide range but read low for paraphrased content; embedding-based metrics are compressed near the top of their range, so small differences are meaningful.

B.6 Worked Examples (Document Quality)

These passage-level worked examples were referenced from earlier drafts of the Document Quality chapter and from the Discussion case-study framing, and are preserved here in case they are returned to. They contrast Participant A (Chat) and Participant B (Form) passages on Retention (Chapter 5) and Purpose (Chapter 2) with per-passage NLP metric values.

Document-scale reading. Two participants make the confidence-quality gap visible at document scale. One produced the study's strongest document on every measure; the other produced a complete-looking document filled with details the scenario never mentioned, which the NLP pipeline did not catch. Participant A's Chat document is the case-study ceiling on both NLP and judge readings: highest semantic similarity to the reference, highest scenario faithfulness on the judge rubric, and the participant's strongest output across the metric suite. Participant B's Form document is the case-study floor on the judge rubric: maximum Completeness, lowest Faithfulness and Hallucination-inverse, with invented financial data, biometric data, video surveillance, and named technical standards absent from the scenario. The Retention and Purpose passages reproduced in Figures B.6–B.9 show both participants at passage scale.

Passage-scale reading. At passage scale the same pattern shows up. On Retention and Purpose alike, Participant B's Form passages outscored Participant A's Chat passages on every reported metric, even though Chat's content was faithful and Form's was fabricated. The Retention passage cited 6 to 10 year statutory retention against the scenario's 3-month rule; the Purpose passage cited legal articles (§ 257 HGB, § 147 AO, § 239 Abs. 1 HGB, § 2 Abs. 1 Nr. 1 BDSG) absent from the scenario. BLEU illustrates the limit sharpest: Participant A's Chat Retention passage shares zero 4-grams with the reference, so BLEU-4 zeros at the 4-gram tier independent of the brevity penalty.

B.6.1 Lexical Metrics: Retention Chapter

To attach the lexical metrics to concrete text, Figures B.6 and B.7 show the Retention chapter (Chapter 5, *Aufbewahrungsfristen und Löschung*) for two case-study participants alongside the reference passage and the corresponding per-passage metric values. Chapter 5 is used here because it has the lowest absolute lexical scores and the widest variance of any chapter, and the case-study Form passage there carries the most legible fabrications, which makes the lexical metrics' blindness to truth easiest to see.

Reference passage (single-sentence statement of the scenario's 3-month rule).

```
Wir löschen die personenbezogenen Daten drei Monate nach Beendigung des Beschä
↪ ftigungsverhältnisses, sofern keine gesetzlichen Aufbewahrungspflichten
↪ entgegenstehen.
```

Participant A (Chat) passage.

```
### 5.1 Löschfristen
Die personenbezogenen Daten werden gemäß folgender Regelungen gelöscht:

- **Grundsatz:** Die Daten werden spätestens **3 Monate nach Austritt der
↪ Mitarbeitenden** gelöscht.
- **Ausnahmen:** Sofern gesetzliche Aufbewahrungspflichten (z. B. im Rahmen
↪ laufender Verfahren) bestehen, erfolgt die Löschung erst nach Ablauf dieser
↪ Fristen.

### 5.2 Verfahren zur Löschung
Die Löschung erfolgt entweder automatisiert durch unsere Systeme oder manuell
↪ durch die zuständigen Mitarbeitenden. Die Einhaltung der Löschfristen wird
↪ regelmäßig überprüft.
```

Figure B.6: Worked example for the lexical metrics (1/2): reference and Participant A (Chat) Retention passages (Chapter 5, *Aufbewahrungsfristen und Löschung*). Continues in Figure B.7.

On the Retention passages above, Participant A's Chat passage scores BLEU $= 0.0000$ against the reference's single-sentence statement (the value the metric assigns when no candidate 4-gram appears in the reference) and Participant B's Form passage scores BLEU $= 0.0356$, where a handful of shared 4-grams survive. The ROUGE-2 and METEOR readings move in the same direction: Participant A's Chat passage scores ROUGE-2 $= 0.0625$ and METEOR $= 0.1724$, while Participant B's Form passage scores ROUGE-2 $= 0.1185$ on the strength of bigrams shared with the reference and METEOR $= 0.3790$ once stems and WordNet synonyms are credited.

B.6.2 Embedding-Based Metrics: Purpose Chapter

To attach the embedding-based metrics to concrete text, Figures B.8 and B.9 show the Purpose chapter (Chapter 2, *Zwecke und Rechtsgrundlagen der Verarbeitung*) for the same two case-study participants alongside the reference passage and the

Participant B (Form) passage.

```
Wir bewahren personenbezogene Daten unserer Mitarbeitenden gemäß den
↪ gesetzlichen Vorgaben auf und löschen diese nach Ablauf der festgelegten
↪ Fristen. Die Aufbewahrungsfristen richten sich nach den folgenden Grundsä
↪ tzen:

- **Grundsatz:** 6 Jahre nach Beendigung des Beschäftigungsverhältnisses (gemäß
↪  § 257 HGB und § 147 AO für steuerrelevante Unterlagen).
- **Ausnahmen:**
  - Sozialversicherungsnummern und Vertragsdaten können bis zu 10 Jahre
↪ aufbewahrt werden, sofern dies aufgrund rechtlicher Auseinandersetzungen
↪ oder Verdachtsmomenten (z. B. Sozialversicherungsbetrug) erforderlich ist.
  - Biometrische Daten und Videoüberwachungsdaten werden nur so lange
↪ gespeichert, wie dies für den jeweiligen Zweck (z. B. Zutrittskontrolle)
↪ notwendig ist.

Nach Ablauf der Aufbewahrungsfristen werden die Daten automatisch gelöscht,
↪ sofern keine gesetzlichen oder vertraglichen Gründe für eine längere
↪ Speicherung vorliegen.
```

Candidate	BLEU	ROUGE-2	METEOR
Participant A (Chat)	0.0000	0.0625	0.1724
Participant B (Form)	**0.0356**	**0.1185**	**0.3790**

Figure B.7: Worked example for the lexical metrics (2/2): Participant B (Form) Retention passage with per-passage BLEU, ROUGE-2, and METEOR values. Per-passage metric values are computed with the same pipeline as the chapter-level table; see Appendix B.8.1 for the full chapter means. Continued from Figure B.6.

corresponding per-passage metric values. Chapter 2 is used here because both case-study passages cover the same purposes faithfully but in different wording, which is the case the embedding-based metrics are designed to credit.

Reference passage (purposes plus the two legal bases for processing).

```
2.1 Verarbeitungszwecke
Wir verarbeiten personenbezogene Daten für folgende Zwecke:
• Verwaltung von Mitarbeitenden vom Eintritt bis zum Austritt, einschließlich:
• Anlegen neuer Mitarbeitender
• Dokumentation von Arbeitszeiten
• Verwaltung von Urlaub und Abwesenheiten
• Verwaltung von Parkplatzberechtigungen für berechtigte Mitarbeitende

2.2 Rechtsgrundlagen der Verarbeitung
Die Verarbeitung erfolgt auf Grundlage folgender Rechtsnormen:
• Art. 6 Abs. 1 lit. b DSGVO (Erfüllung eines Vertrags oder vorvertraglicher Ma
↪ ßnahmen)
• § 26 BDSG (Datenverarbeitung für Zwecke des Beschäftigungsverhältnisses)
```

Participant A (Chat) passage.

```
### 2.1 Zwecke der Verarbeitung
Wir verarbeiten personenbezogene Daten unserer Mitarbeitenden zur
↪ ganzheitlichen Verwaltung von deren Beschäftigungsverhältnis - vom Eintritt
↪ bis zum Austritt aus dem Unternehmen. Die Verarbeitung dient folgenden
↪ Zwecken:

- Erfassung und Pflege von Stammdaten neuer und bestehender Mitarbeitender,
- Dokumentation von Arbeitszeiten sowie Verwaltung von Urlaubs- und
↪ Abwesenheitszeiten,
- Zuweisung und Verwaltung von Parkplatzberechtigungen für den firmeneigenen
↪ Parkplatz mit Schranke. Hierfür wird ein eindeutiges Merkmal (
↪ Fahrzeugkennzeichen) hinterlegt, um die Nutzung zu ermöglichen.

Die Verarbeitung erfolgt ausschließlich auf internen Systemen (Microsoft Office
↪  365 und eigene Server) ohne Nutzung externer Cloud-Dienste. Medizinische
↪ Daten oder Krankentage sind nicht Gegenstand dieser Verarbeitungstätigkeit.

### 2.2 Rechtsgrundlagen der Verarbeitung
Die Verarbeitung personenbezogener Daten erfolgt auf folgenden Rechtsgrundlagen
↪  gemäß Art. 6 Abs. 1 DSGVO sowie ergänzenden nationalen Vorschriften:

- **Art. 6 Abs. 1 lit. b DSGVO (Vertragserfüllung):**
  Die Verarbeitung ist für die Durchführung des Beschäftigungsverhältnisses
↪ erforderlich, insbesondere zur Verwaltung von Arbeitszeiten, Urlaub und
↪ Parkplatzberechtigungen.

- **Art. 6 Abs. 1 lit. f DSGVO (Berechtigtes Interesse):**
  Die Verarbeitung dient unserem berechtigten Interesse an einer effizienten
↪ Verwaltung der Mitarbeiterressourcen sowie der Sicherstellung einer
↪ geordneten Nutzung des firmeneigenen Parkplatzes.

- **§ 26 BDSG (Datenverarbeitung für Zwecke des Beschäftigungsverhältnisses):**
  Die Verarbeitung erfolgt im Rahmen des Beschäftigungsverhältnisses und ist fü
↪ r dessen Durchführung erforderlich.
```

Figure B.8: Worked example for the embedding-based metrics (1/2): reference and Participant A (Chat) Purpose passages (Chapter 2, *Zwecke und Rechtsgrundlagen der Verarbeitung*). Continues in Figure B.9.

Participant B (Form) passage.

```
Wir verarbeiten personenbezogene Daten unserer Mitarbeitenden ausschließlich
↪ zur Verwaltung des Beschäftigungsverhältnisses von der Einstellung bis zum
↪ Austritt. Dies umfasst folgende Zwecke:

- Erfassung und Pflege von Stammdaten der Mitarbeitenden
- Dokumentation von Arbeitszeiten und Anwesenheiten
- Verwaltung von Urlaubs- und Abwesenheitszeiten
- Verwaltung von Parkplatzberechtigungen (z. B. Fahrzeugkennzeichen)
- Erfüllung gesetzlicher Aufbewahrungs- und Meldepflichten
- Sicherstellung der betrieblichen Abläufe und Compliance

Die Verarbeitung erfolgt auf Grundlage folgender Rechtsgrundlagen:

- **Art. 6 Abs. 1 lit. b DSGVO** (Erfüllung eines Vertrags oder
↪ vorvertraglicher Maßnahmen)
- **§ 26 BDSG** (Datenverarbeitung für Zwecke des Beschäftigungsverhältnisses)
- **§ 257 HGB** (Aufbewahrung von Handelsbüchern und Unterlagen)
- **§ 147 AO** (Aufbewahrung von steuerrelevanten Unterlagen)
- **§ 239 Abs. 1 HGB** (Führung der Handelsbücher)
- **§ 2 Abs. 1 Nr. 1 BDSG** (Verarbeitung personenbezogener Daten im Beschä
↪ ftigungskontext)
```

Candidate	BERTScore P	BERTScore R	BERTScore F1	SBERT
Participant A (Chat)	0.8797	0.9180	0.8984	0.9810
Participant B (Form)	**0.8954**	**0.9336**	**0.9141**	**0.9882**

Figure B.9: Worked example for the embedding-based metrics (2/2): Participant B (Form) Purpose passage with per-passage BERTScore P/R/F1 and SBERT values. SBERT cosine similarity is reported under the ModernBERT encoder, matching the document- and chapter-level tables. Per-passage metric values are computed with the same pipeline as the chapter-level table; see Appendix B.8.2 for the full chapter means. Continued from Figure B.8.

On the Purpose passages above, Participant A's Chat passage scores BERTScore Precision $= 0.8797$, Recall $= 0.9180$, and F1 $= 0.8984$ against the reference, while Participant B's Form passage scores Precision $= 0.8954$, Recall $= 0.9336$, and F1 $= 0.9141$, with the embedding model crediting its broader overlap with the reference's legal vocabulary. SBERT is the more forgiving metric of the two: Participant A's Chat passage scores SBERT $= 0.9810$ and Participant B's Form passage scores SBERT $= 0.9882$, with the document-level encoder placing both passages near the top of the metric's range.

B.7 Document-Level NLP Metric Distributions

The compact summaries and the box plots are reproduced inline with the NLP results in Section 5.1.1; the full per-document descriptive distributions are reproduced here, split into the lexical and embedding-based families.

B.7.1 Lexical Metrics

Metric	Mode	n	Mean	SD	Q1	Median	Q3	Min	Max
BLEU	Aggregated	113	0.133	0.048	0.094	0.128	0.159	0.047	0.246
	Form	37	0.129	0.048	0.088	0.124	0.158	0.064	0.230
	Ask	37	0.127	0.056	0.083	0.125	0.148	0.047	0.246
	Chat	39	0.143	0.039	0.122	0.134	0.163	0.067	0.235
ROUGE-1	Aggregated	113	0.488	0.087	0.425	0.491	0.560	0.313	0.645
	Form	37	0.488	0.098	0.421	0.493	0.569	0.313	0.644
	Ask	37	0.465	0.088	0.412	0.449	0.522	0.322	0.618
	Chat	39	0.511	0.071	0.468	0.517	0.561	0.330	0.645
ROUGE-2	Aggregated	113	0.285	0.077	0.234	0.268	0.344	0.121	0.455
	Form	37	0.306	0.076	0.241	0.301	0.361	0.178	0.455
	Ask	37	0.257	0.080	0.200	0.238	0.275	0.121	0.429
	Chat	39	0.291	0.069	0.253	0.283	0.344	0.131	0.447
ROUGE-L	Aggregated	113	0.375	0.087	0.311	0.359	0.441	0.194	0.566
	Form	37	0.388	0.094	0.320	0.391	0.468	0.238	0.566
	Ask	37	0.347	0.091	0.296	0.326	0.376	0.194	0.545
	Chat	39	0.389	0.072	0.348	0.387	0.444	0.230	0.544
METEOR	Aggregated	113	0.389	0.062	0.359	0.388	0.431	0.234	0.514
	Form	37	0.412	0.051	0.367	0.419	0.442	0.335	0.514
	Ask	37	0.365	0.068	0.327	0.366	0.409	0.234	0.488
	Chat	39	0.390	0.057	0.374	0.389	0.421	0.235	0.501

Table B.8: Document-level lexical metric distributions by mode (full sample $n = 113$). Source: `doc_nlp_summary.csv`.

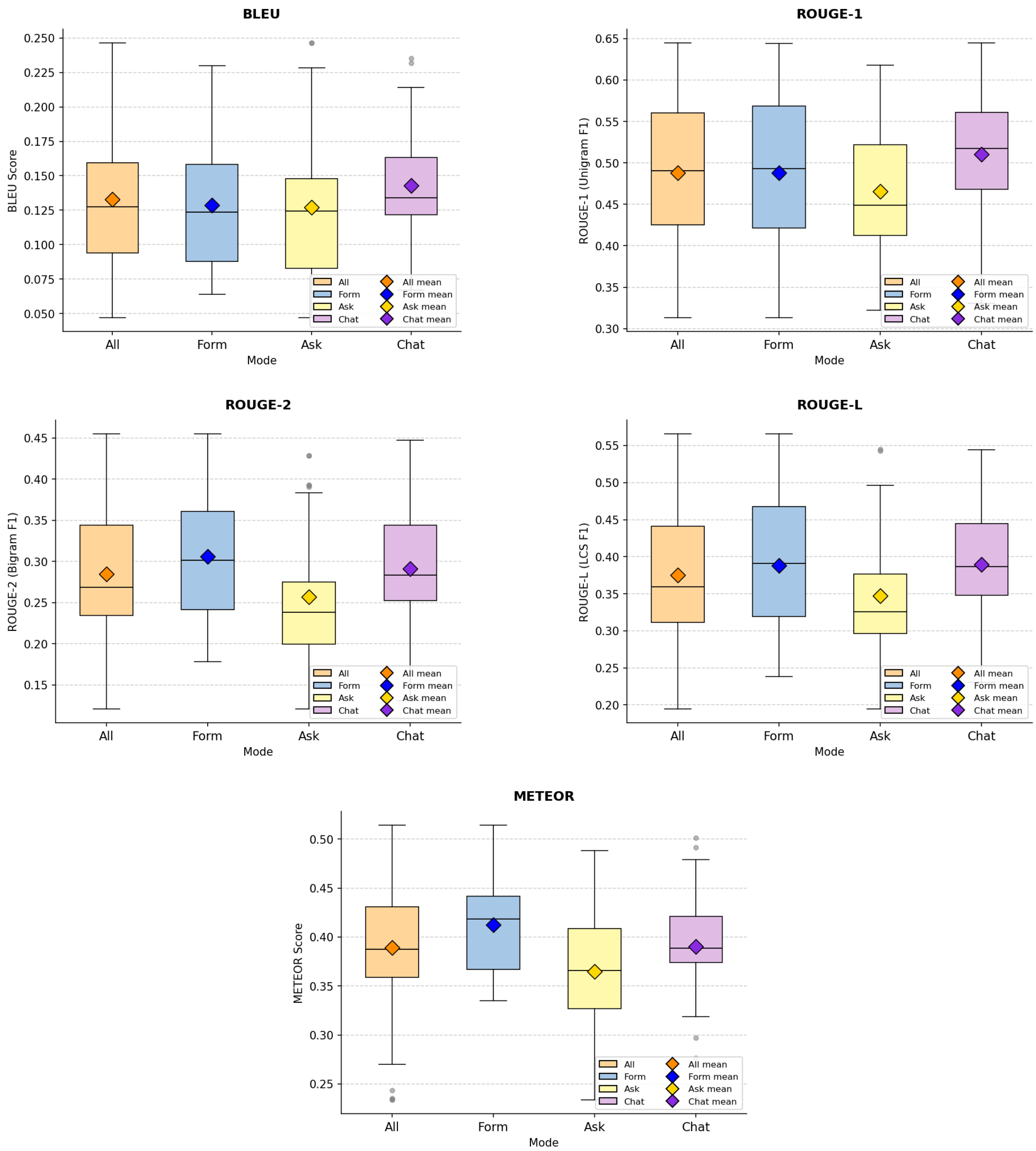


Figure B.10: Document-level distributions of BLEU, ROUGE-1, ROUGE-2, ROUGE-L, and METEOR by mode (full sample $n = 113$).

B.7.2 BERT-Based and Sentence-Embedding Metrics

Metric	Mode	n	Mean	SD	Q1	Median	Q3	Min	Max
BERTScore Precision	Aggregated	113	0.890	0.014	0.880	0.891	0.900	0.861	0.922
	Form	37	0.889	0.014	0.880	0.889	0.899	0.862	0.912
	Ask	37	0.886	0.016	0.872	0.889	0.899	0.861	0.918
	Chat	39	0.895	0.012	0.890	0.896	0.902	0.867	0.922
BERTScore Recall	Aggregated	113	0.915	0.014	0.914	0.918	0.922	0.861	0.936
	Form	37	0.923	0.005	0.919	0.923	0.926	0.914	0.936
	Ask	37	0.908	0.017	0.908	0.915	0.918	0.861	0.925
	Chat	39	0.913	0.013	0.908	0.917	0.922	0.874	0.933
BERTScore F1	Aggregated	113	0.902	0.011	0.895	0.904	0.910	0.863	0.922
	Form	37	0.905	0.009	0.900	0.906	0.912	0.887	0.920
	Ask	37	0.897	0.013	0.892	0.896	0.906	0.863	0.918
	Chat	39	0.904	0.010	0.899	0.906	0.911	0.872	0.922
SBERT (ModernBERT)	Aggregated	113	0.987	0.004	0.985	0.988	0.990	0.971	0.994
	Form	37	0.986	0.004	0.984	0.988	0.989	0.974	0.992
	Ask	37	0.986	0.005	0.983	0.987	0.990	0.971	0.993
	Chat	39	0.990	0.002	0.988	0.990	0.991	0.985	0.994

Table B.9: Document-level BERT-based and SBERT distributions by mode (full sample $n = 113$). Source: `doc_bert_summary.csv`.

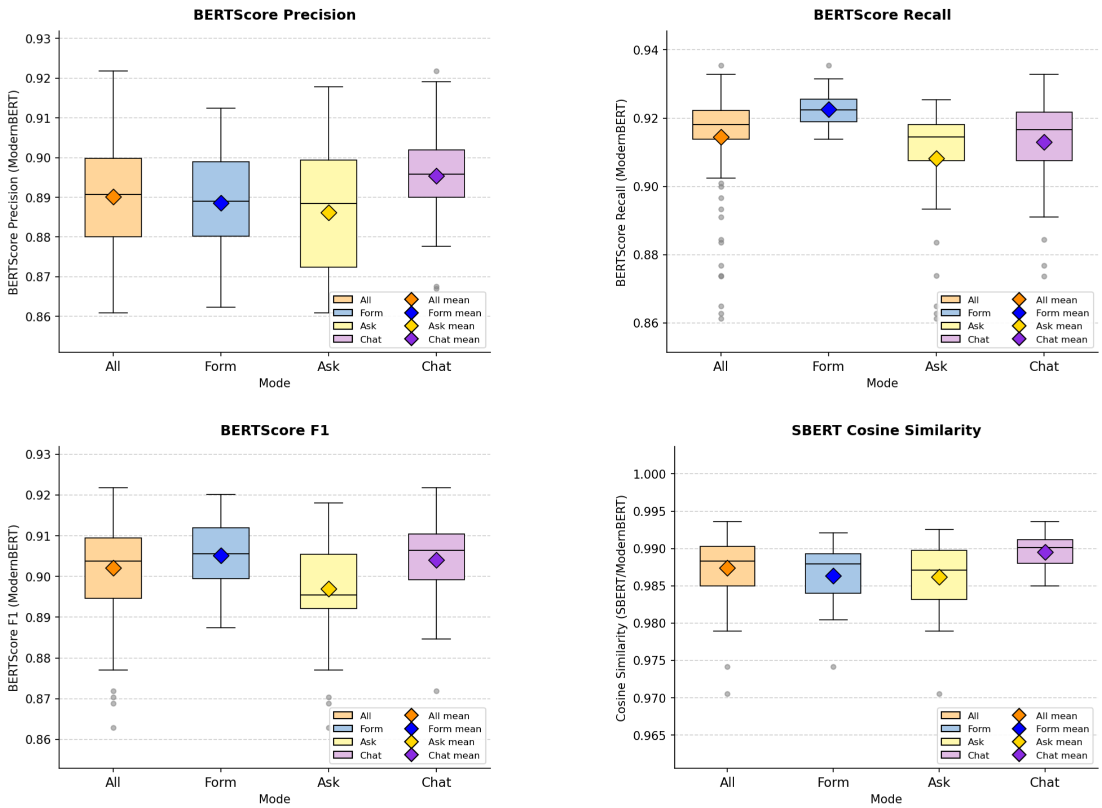


Figure B.11: Document-level distributions of BERTScore Precision, Recall, F1, and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$), one plot per metric.

B.8 Chapter-Level NLP Metric Distributions

The chapter-level box plots are reproduced inline with the chapter-level NLP results in Section 5.1.2; the per-chapter, per-mode means (with SDs in parentheses) for every metric are reproduced here.

B.8.1 Lexical Metrics

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.181 (0.080)	0.179 (0.077)	0.200 (0.085)	0.166 (0.076)
Purposes & Legal Bases	0.121 (0.060)	0.145 (0.057)	0.098 (0.038)	0.121 (0.070)
Personal Data Categories	0.081 (0.050)	0.110 (0.041)	0.077 (0.052)	0.058 (0.043)
Data Subjects & Recipients	0.097 (0.075)	0.123 (0.076)	0.081 (0.082)	0.088 (0.063)
Retention & Deletion	0.044 (0.076)	0.038 (0.044)	0.023 (0.073)	0.068 (0.094)
Technical & Org. Measures	0.080 (0.066)	0.078 (0.068)	0.070 (0.080)	0.091 (0.047)

Table B.10: Chapter-level BLEU mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.571 (0.114)	0.550 (0.097)	0.605 (0.129)	0.559 (0.109)
Purposes & Legal Bases	0.414 (0.109)	0.457 (0.094)	0.373 (0.093)	0.410 (0.122)
Personal Data Categories	0.467 (0.164)	0.600 (0.084)	0.366 (0.120)	0.437 (0.174)
Data Subjects & Recipients	0.437 (0.116)	0.491 (0.111)	0.393 (0.128)	0.426 (0.086)
Retention & Deletion	0.260 (0.131)	0.252 (0.092)	0.212 (0.132)	0.314 (0.143)
Technical & Org. Measures	0.331 (0.130)	0.313 (0.137)	0.330 (0.157)	0.350 (0.090)

Table B.11: Chapter-level ROUGE-1 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.386 (0.145)	0.363 (0.131)	0.430 (0.158)	0.365 (0.138)
Purposes & Legal Bases	0.269 (0.109)	0.332 (0.093)	0.216 (0.077)	0.261 (0.119)
Personal Data Categories	0.298 (0.148)	0.425 (0.097)	0.198 (0.089)	0.272 (0.149)
Data Subjects & Recipients	0.250 (0.109)	0.317 (0.105)	0.215 (0.113)	0.220 (0.077)
Retention & Deletion	0.131 (0.113)	0.134 (0.071)	0.087 (0.107)	0.168 (0.137)
Technical & Org. Measures	0.176 (0.118)	0.166 (0.120)	0.163 (0.141)	0.197 (0.089)

Table B.12: Chapter-level ROUGE-2 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.534 (0.129)	0.518 (0.110)	0.576 (0.140)	0.511 (0.128)
Purposes & Legal Bases	0.351 (0.111)	0.404 (0.096)	0.299 (0.083)	0.351 (0.125)
Personal Data Categories	0.426 (0.168)	0.567 (0.085)	0.309 (0.107)	0.403 (0.179)
Data Subjects & Recipients	0.372 (0.121)	0.433 (0.114)	0.330 (0.131)	0.355 (0.096)
Retention & Deletion	0.229 (0.136)	0.218 (0.087)	0.183 (0.141)	0.283 (0.152)
Technical & Org. Measures	0.257 (0.135)	0.237 (0.140)	0.246 (0.158)	0.285 (0.100)

Table B.13: Chapter-level ROUGE-L mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.471 (0.128)	0.456 (0.120)	0.506 (0.136)	0.453 (0.124)
Purposes & Legal Bases	0.400 (0.113)	0.452 (0.075)	0.348 (0.101)	0.401 (0.131)
Personal Data Categories	0.315 (0.120)	0.415 (0.074)	0.275 (0.111)	0.258 (0.104)
Data Subjects & Recipients	0.355 (0.109)	0.440 (0.098)	0.312 (0.101)	0.314 (0.077)
Retention & Deletion	0.314 (0.160)	0.332 (0.114)	0.245 (0.160)	0.362 (0.177)
Technical & Org. Measures	0.293 (0.113)	0.295 (0.108)	0.280 (0.145)	0.305 (0.079)

Table B.14: Chapter-level METEOR mean (SD) by mode (full sample $n = 113$).

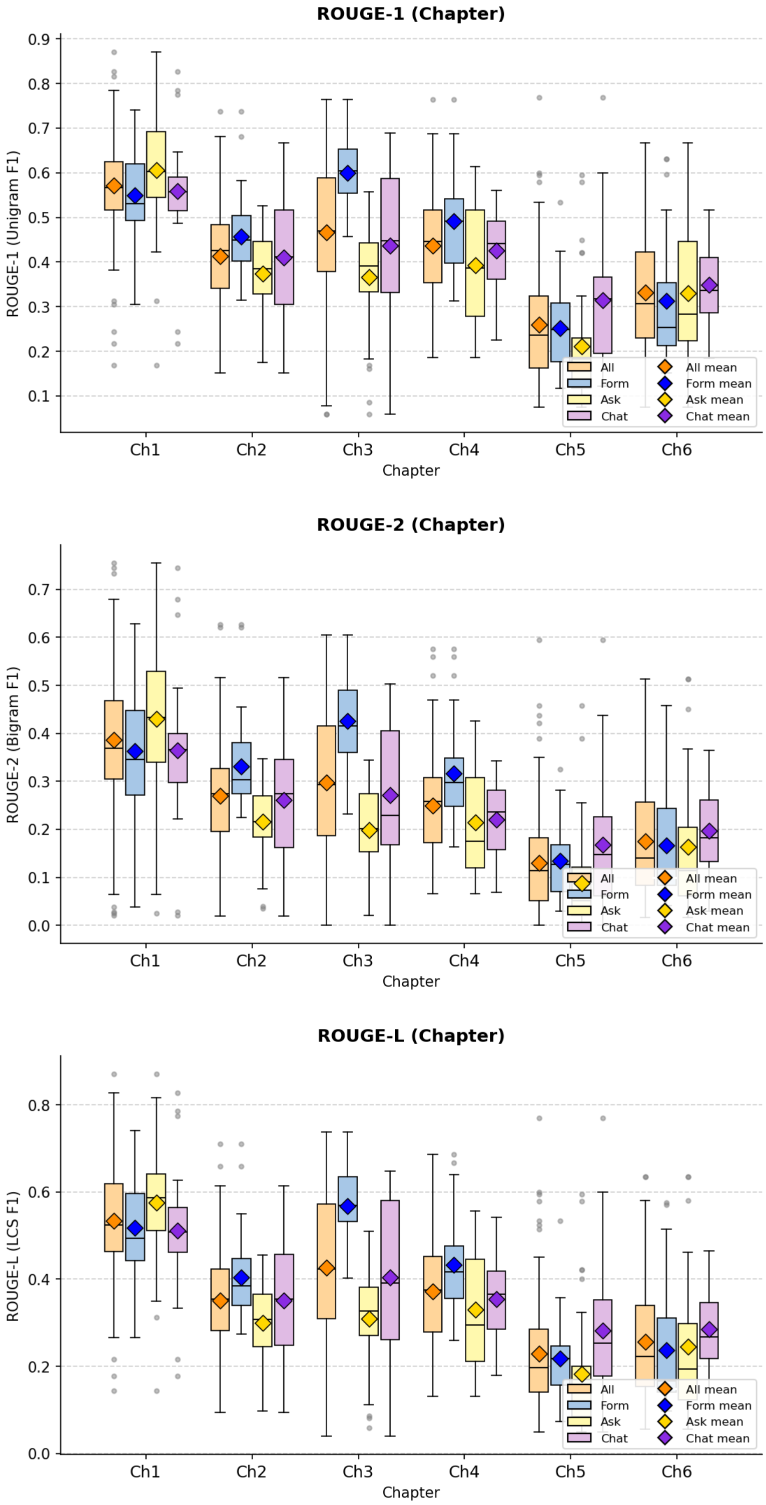


Figure B.12: Chapter-level distributions of ROUGE-1, ROUGE-2, and ROUGE-L by mode (full sample $n = 113$), one plot per metric showing all six ROPA chapters side by side.

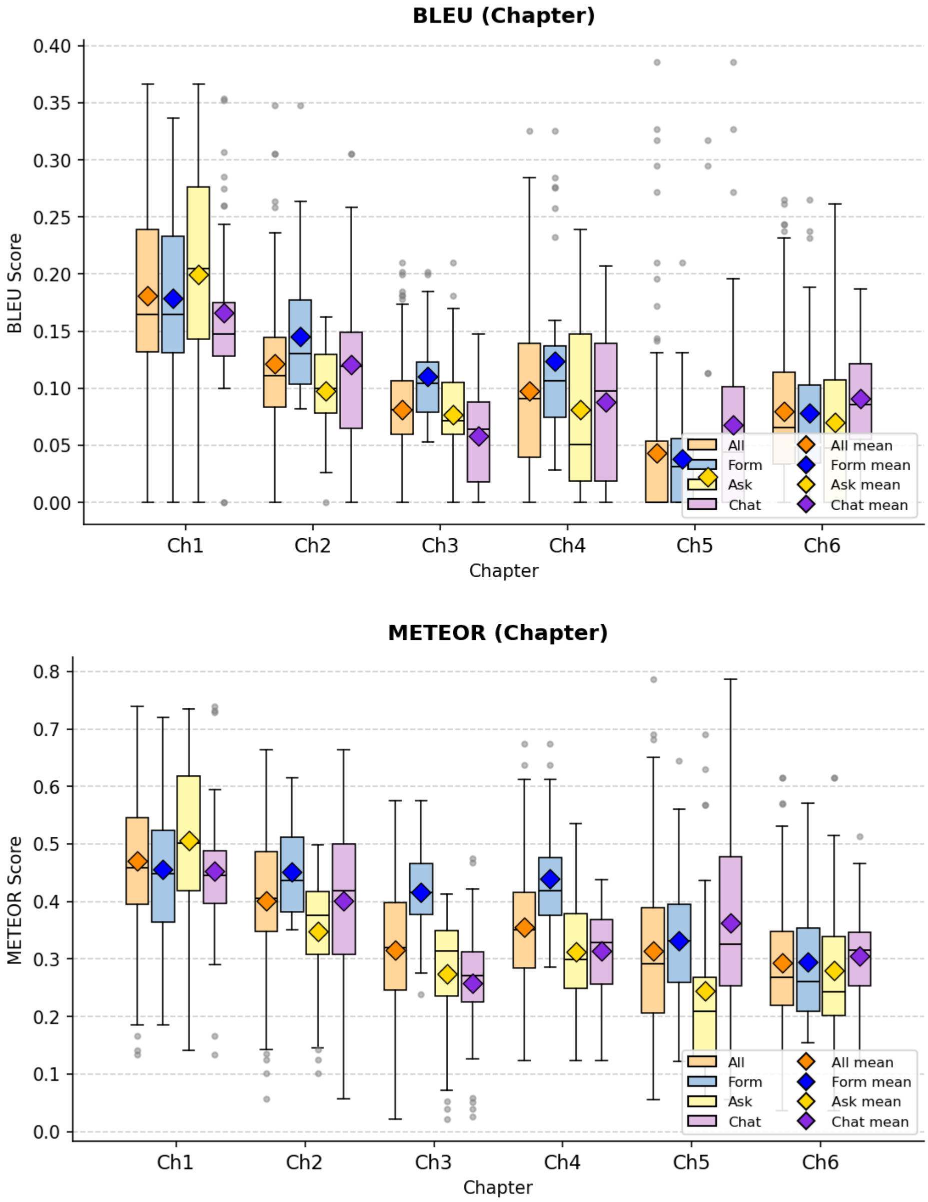


Figure B.13: Chapter-level distributions of BLEU and METEOR by mode (full sample $n = 113$), one plot per metric showing all six ROPA chapters side by side.

B.8.2 BERT-Based and Sentence-Embedding Metrics

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.857 (0.017)	0.855 (0.015)	0.862 (0.020)	0.855 (0.016)
Purposes & Legal Bases	0.873 (0.021)	0.872 (0.020)	0.869 (0.020)	0.877 (0.022)
Personal Data Categories	0.883 (0.022)	0.897 (0.015)	0.868 (0.019)	0.884 (0.022)
Data Subjects & Recipients	0.857 (0.028)	0.856 (0.025)	0.853 (0.035)	0.864 (0.023)
Retention & Deletion	0.824 (0.042)	0.814 (0.033)	0.817 (0.050)	0.840 (0.037)
Technical & Org. Measures	0.854 (0.033)	0.841 (0.038)	0.858 (0.031)	0.863 (0.027)

Table B.15: Chapter-level BERTScore Precision mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.882 (0.022)	0.879 (0.019)	0.887 (0.025)	0.879 (0.022)
Purposes & Legal Bases	0.902 (0.024)	0.911 (0.012)	0.892 (0.028)	0.904 (0.024)
Personal Data Categories	0.878 (0.041)	0.910 (0.016)	0.862 (0.042)	0.861 (0.039)
Data Subjects & Recipients	0.880 (0.026)	0.904 (0.014)	0.873 (0.024)	0.864 (0.022)
Retention & Deletion	0.889 (0.024)	0.890 (0.018)	0.879 (0.023)	0.897 (0.027)
Technical & Org. Measures	0.888 (0.025)	0.889 (0.018)	0.882 (0.032)	0.893 (0.022)

Table B.16: Chapter-level BERTScore Recall mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.869 (0.019)	0.867 (0.016)	0.875 (0.022)	0.867 (0.017)
Purposes & Legal Bases	0.887 (0.018)	0.891 (0.015)	0.880 (0.018)	0.890 (0.020)
Personal Data Categories	0.880 (0.029)	0.903 (0.012)	0.865 (0.025)	0.872 (0.029)
Data Subjects & Recipients	0.868 (0.020)	0.879 (0.017)	0.862 (0.022)	0.863 (0.017)
Retention & Deletion	0.855 (0.031)	0.850 (0.024)	0.847 (0.035)	0.867 (0.031)
Technical & Org. Measures	0.871 (0.025)	0.864 (0.027)	0.870 (0.027)	0.878 (0.019)

Table B.17: Chapter-level BERTScore F1 mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.976 (0.005)	0.975 (0.004)	0.977 (0.006)	0.975 (0.005)
Purposes & Legal Bases	0.979 (0.008)	0.979 (0.006)	0.977 (0.009)	0.981 (0.008)
Personal Data Categories	0.973 (0.013)	0.982 (0.003)	0.969 (0.015)	0.970 (0.014)
Data Subjects & Recipients	0.971 (0.008)	0.973 (0.008)	0.970 (0.008)	0.971 (0.009)
Retention & Deletion	0.949 (0.016)	0.945 (0.013)	0.946 (0.018)	0.955 (0.014)
Technical & Org. Measures	0.971 (0.010)	0.967 (0.012)	0.972 (0.010)	0.975 (0.007)

Table B.18: Chapter-level SBERT (ModernBERT) cosine similarity mean (SD) by mode (full sample $n = 113$).

Chapter	Aggregated	Form	Ask	Chat
Controller & Contact	0.687 (0.122)	0.663 (0.119)	0.736 (0.133)	0.662 (0.101)
Purposes & Legal Bases	0.741 (0.074)	0.763 (0.053)	0.740 (0.081)	0.721 (0.080)
Personal Data Categories	0.713 (0.122)	0.768 (0.063)	0.705 (0.149)	0.669 (0.118)
Data Subjects & Recipients	0.706 (0.094)	0.691 (0.087)	0.692 (0.115)	0.733 (0.072)
Retention & Deletion	0.519 (0.141)	0.516 (0.129)	0.474 (0.150)	0.566 (0.131)
Technical & Org. Measures	0.710 (0.095)	0.711 (0.073)	0.701 (0.110)	0.717 (0.099)

Table B.19: Chapter-level SBERT (`all-MiniLM-L6-v2`) cosine similarity mean (SD) by mode (full sample $n = 113$). Reported as a robustness check on the ModernBERT pass.

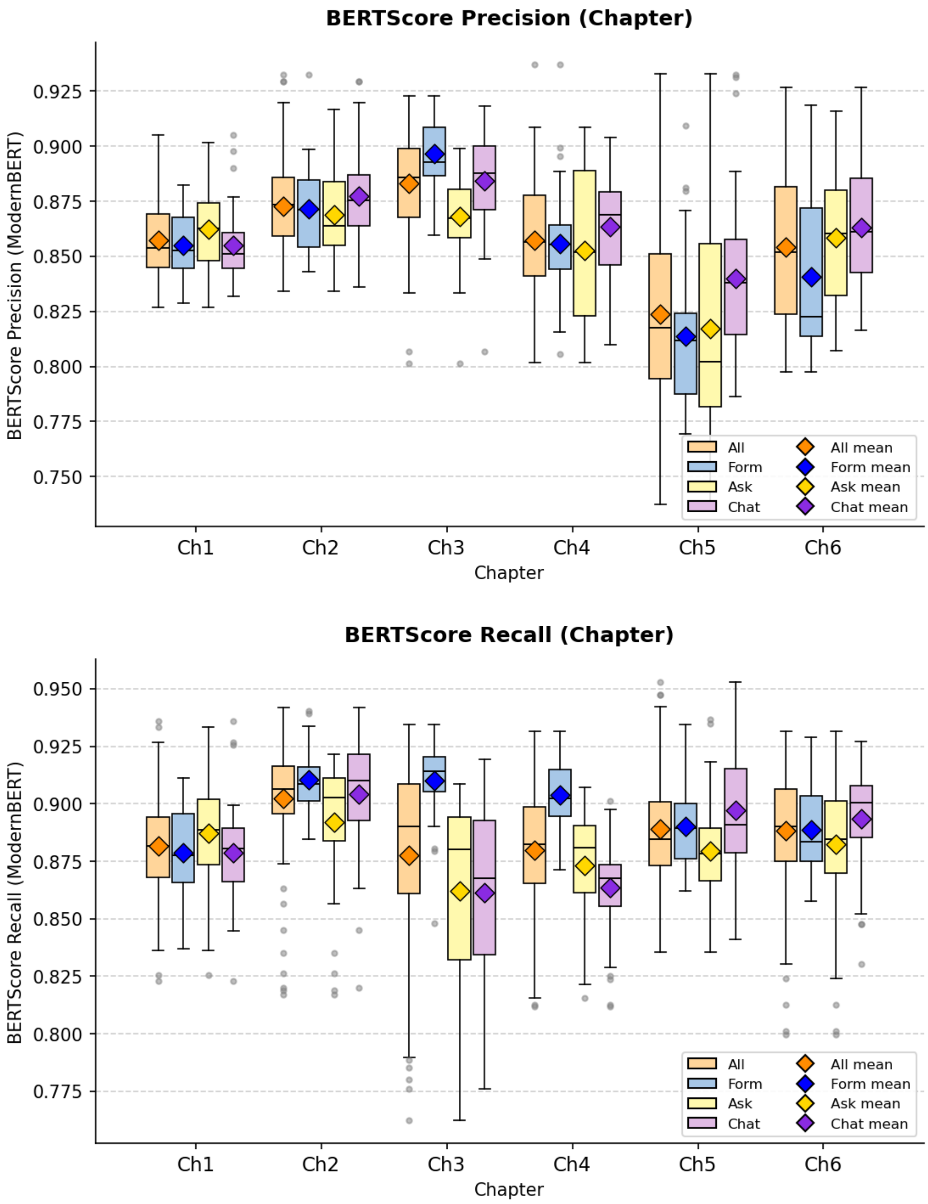


Figure B.14: Chapter-level distributions of BERTScore Precision and Recall by mode (full sample $n = 113$). Continues in Figure B.15.

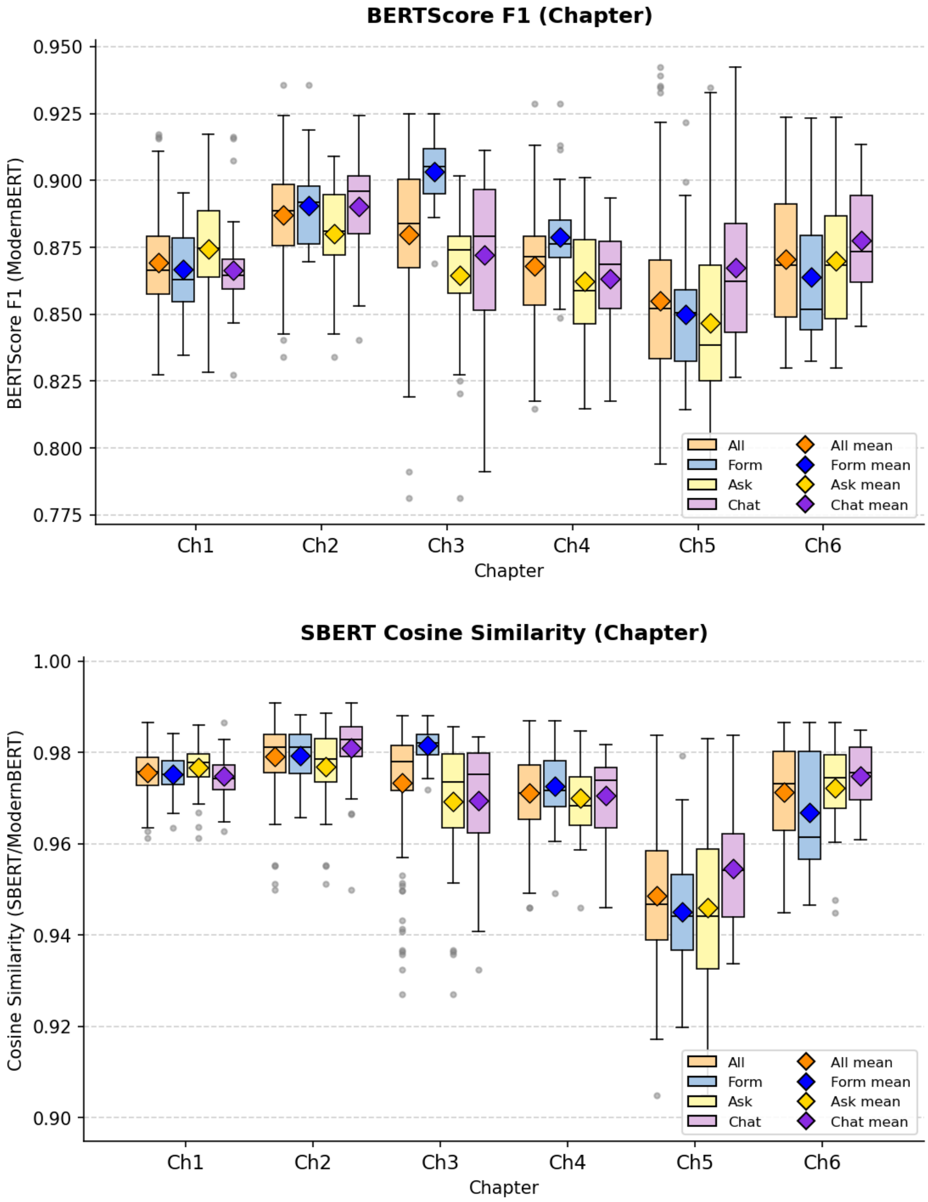


Figure B.15: Chapter-level distributions of BERTScore F1 and SBERT cosine similarity (ModernBERT) by mode (full sample $n = 113$). Continued from Figure B.14.

C Inferential Statistics in Full

This appendix reproduces the complete inferential test output referenced in Chapters 4, 5, and 6. The within-subjects user-study tests use $n = 69$ (matched within-subjects, Friedman + Wilcoxon); the document-quality and timing analyses use $n = 113$ (between-groups, Kruskal–Wallis + Mann–Whitney U); the confidence–quality correlations supporting the RQ3 discussion use the joined document-survey sample $n = 102$. The full sample-size derivation is reproduced first, followed by the user-study inferential tests, the NLP inferential tests, the confidence–quality correlations, and the timing inferential tests.

C.1 Sample-Size Derivation

All sample sizes used in the thesis are derived from the raw data files committed to the repository: `python/survey/docs/data.csv` for the user study (one row per participant) and `python/metrics/output/all_scores_combined.csv` for the NLP and judge scores (one row per LLM-generated document).

User-study subset	**n**
Recruited participants (raw)	73
Participants with at least one valid session ($p\_0001 \geq 1$)	69
Form responses (mode-level)	69
Ask responses (mode-level)	69
Chat responses (mode-level)	69
Matched-triple participants (valid response in all three modes)	69

Table C.1: User-study sample sizes after consent and quality filtering.

NLP / judge subset	**n**
Total documents with parseable NLP scores and judge ratings	113
Form documents	37
Ask documents	37
Chat documents	39
Unique participants represented	47
Participants with at least one document in each mode	28
Strict matched triple (exactly one document per mode)	26

Table C.2: NLP and judge sample sizes. The full $n = 113$ sample is used for the document-quality inferential tests (Kruskal–Wallis, Mann–Whitney U).

User-study $\times$ NLP join	**n**
Raw joined rows (document with matching user-study response)	106
Deduplicated joined rows (one row per participant–mode)	102
Form joined rows (deduplicated)	34
Ask joined rows (deduplicated)	33
Chat joined rows (deduplicated)	35
Unique participants in the joined sample	44
Participants with user-study $\times$ NLP coverage in all three modes	27

Table C.3: Triangulated samples used for confidence–quality analyses. The canonical RQ3 sample is the deduplicated join $n = 102$. Four participants had multiple submissions per mode in `document_results.csv`; the canonical row per participant–mode was selected by highest `id` (latest submission). Three participants in the documents file had no matching survey row and were dropped. The within-subjects $n = 27$ is used for paired confidence $\times$ quality correlations.

C.2 User-Study Inferential Tests

C.2.1 Normality (Shapiro–Wilk, $n = 69$)

Measure	Mode	W	p	Normal?
SUS	Form	0.944	0.004	No
SUS	Ask	0.976	0.210	Yes
SUS	Chat	0.927	0.001	No
R-TLX composite	Form	0.940	0.003	No
R-TLX composite	Ask	0.926	0.001	No
R-TLX composite	Chat	0.945	0.005	No
AI Support (6-item)	Form	0.923	< 0.001	No
AI Support (6-item)	Ask	0.965	0.048	No
AI Support (6-item)	Chat	0.952	0.010	No
AI Confidence (5_{ai})	Form	0.846	< 0.001	No
AI Confidence (5_{ai})	Ask	0.856	< 0.001	No
AI Confidence (5_{ai})	Chat	0.864	< 0.001	No
AI Reuse (6_{ai})	Form	0.799	< 0.001	No
AI Reuse (6_{ai})	Ask	0.872	< 0.001	No
AI Reuse (6_{ai})	Chat	0.861	< 0.001	No

Table C.4: Shapiro–Wilk normality tests for user-study measures by mode.

Non-normal distributions are detected for nearly every measure. Non-parametric tests (Friedman omnibus and Wilcoxon signed-rank post-hoc) are used throughout.

C.2.2 Friedman Omnibus ($n = 69$ matched triple)

Measure	χ^2	df	p	W	Significant
SUS (0–100)	5.51	2	0.064	0.040	No
R-TLX composite (1–21)	5.24	2	0.073	0.038	No
R-TLX Mental Demand	3.13	2	0.209	0.023	No
R-TLX Physical Demand	2.08	2	0.354	0.015	No
R-TLX Temporal Demand	12.77	2	0.002	0.093	**Yes**
R-TLX Effort	3.78	2	0.151	0.027	No
R-TLX Frustration	8.93	2	0.012	0.065	**Yes**
R-TLX Performance (pos.)	2.33	2	0.311	0.017	No
AI Confidence (5_{ai})	3.76	2	0.152	0.027	No
Perceived Confidence ($\overline{4{+}5}$)	4.50	2	0.105	0.033	No
AI Support (6-item composite)	1.05	2	0.592	0.008	No
AI Reuse intention (6_{ai})	11.12	2	0.004	0.081	**Yes**

Table C.5: Friedman omnibus tests across the three modes for user-study measures. Kendall's W: 0–0.1 weak, 0.1–0.3 moderate, 0.3–0.5 strong.

C.2.3 Pairwise Wilcoxon Signed-Rank Post-hoc (Bonferroni $\alpha = 0.0167$)

Post-hoc tests are reported for measures with a significant Friedman result and, for completeness, for *SUS* and the R-TLX composite. Sign convention: positive r means the first-named mode ranks higher. Effect-size labels: $|r| < 0.1$ negligible, 0.1–0.3 small, 0.3–0.5 medium, ≥ 0.5 large.

Measure	Pair	n	W	p	r	Sig (Bonf)
SUS	Form vs Ask	60	578.5	0.013	$+0.368$ medium	**Yes**
SUS	Form vs Chat	58	549.5	0.018	$+0.358$ medium	No
SUS	Ask vs Chat	61	885.5	0.666	-0.063 negligible	No
R-TLX composite	Form vs Ask	65	684.0	0.011	-0.362 medium	**Yes**
R-TLX composite	Form vs Chat	65	636.0	0.004	-0.407 medium	**Yes**
R-TLX composite	Ask vs Chat	63	995.5	0.932	$+0.012$ negligible	No
Temporal Demand	Form vs Ask	43	237.0	0.004	-0.499 medium	**Yes**
Temporal Demand	Form vs Chat	46	233.0	< 0.001	-0.569 large	**Yes**
Temporal Demand	Ask vs Chat	46	427.0	0.211	-0.210 small	No
Frustration	Form vs Ask	52	403.5	0.009	-0.414 medium	**Yes**
Frustration	Form vs Chat	57	519.5	0.014	-0.371 medium	**Yes**
Frustration	Ask vs Chat	52	629.0	0.584	$+0.087$ negligible	No
AI Reuse (6_{ai})	Form vs Ask	49	402.5	0.034	$+0.343$ medium	No
AI Reuse (6_{ai})	Form vs Chat	43	216.0	0.002	$+0.543$ large	**Yes**
AI Reuse (6_{ai})	Ask vs Chat	45	454.5	0.468	$+0.122$ small	No

Table C.6: Pairwise Wilcoxon signed-rank post-hoc tests for user-study measures.

C.2.4 Mode Preference Ranking

A chi-square goodness-of-fit test on the rank-1 vote distribution against a uniform null gives $\chi^2(2, N = 69) = 14.17$, $p < 0.001$, Cramér's $V = 0.320$ (medium effect). The Friedman test on the full rank scores (1–3) is non-significant: $\chi^2(2) = 5.07$, $p = 0.079$, $W = 0.037$.

C.3 NLP Metric Inferential Tests

C.3.1 Kruskal–Wallis Omnibus ($n = 113$)

Metric	H	p	Sig ($\alpha = 0.05$)	ε^2
BLEU	4.48	0.107	No	—
ROUGE-1	5.57	0.062	No	—
ROUGE-2	9.56	0.008	Yes	0.069 (moderate)
ROUGE-L	7.79	0.020	Yes	—
METEOR	9.80	0.007	Yes	0.071 (moderate)
BERTScore Precision	9.11	0.011	Yes	—
BERTScore Recall	32.38	< 0.001	Yes	0.276 (large)
BERTScore F1	9.75	0.008	Yes	0.071 (moderate)
SBERT (ModernBERT)	17.43	< 0.001	Yes	0.140 (large)

Table C.7: Kruskal–Wallis H tests on the unbalanced full document set ($n = 113$). Primary omnibus for document-quality inferential testing.

C.3.2 Pairwise Mann–Whitney U ($n = 113$)

Pairwise Mann–Whitney U tests on the unbalanced full sample, serving as the post-hoc family where the Kruskal–Wallis omnibus is significant. Bonferroni-corrected $\alpha = 0.0167$. Only significant comparisons are listed; non-significant comparisons are omitted for brevity.

Metric	Comparison	p	r	Sig
BERTScore Recall	Form > Ask	< 0.001	0.756 (large)	Yes
BERTScore Recall	Form > Chat	< 0.001	0.512 (large)	Yes
BERTScore F1	Form > Ask	0.004	0.386 (medium)	Yes
BERTScore F1	Chat > Ask	0.011	0.340 (medium)	Yes
SBERT (ModernBERT)	Chat > Form	< 0.001	0.505 (large)	Yes
SBERT (ModernBERT)	Chat > Ask	< 0.001	0.453 (large)	Yes
ROUGE-2	Form > Ask	0.006	0.369 (medium)	Yes
ROUGE-2	Chat > Ask	0.011	0.342 (medium)	Yes
METEOR	Form > Ask	0.003	0.407 (medium)	Yes

Table C.8: Pairwise Mann–Whitney U post-hoc tests on the unbalanced full sample of $n = 113$ documents.

Ask is the significantly lowest-scoring mode (Ask < Form, Ask < Chat) on the metrics that reach significance, most clearly on BERTScore Recall, F1, ROUGE-2, and METEOR; Form and Chat trade leadership depending on whether the metric emphasizes lexical recall or document-level semantic alignment.

C.4 Confidence–Quality Correlations

This section reproduces the full pooled and per-mode descriptive means, Spearman rank correlations, and judge cross-checks used in the RQ3 triangulation reported in Chapter 6.

C.4.1 Joined Sample and Deduplication

The triangulation is computed on the joined document-survey subset: $n = 102$ document-mode rows from 44 unique participants (Form 34, Ask 33, Chat 35). The join required a row in both the NLP/judge file `all_scores_combined.csv` (under `python/metrics/output/`) and the user-study file. Four participants had multiple submissions per mode in `document_results.csv`; the canonical row was selected by highest `id` (latest submission). Three participants present in the documents file had no matching survey row and were dropped. The resulting $n = 102$ is the source of the sample size used throughout the RQ3 discussion and differs from the unmatched join ($n = 106$) reported in earlier drafts. The underlying tables are available on disk as `rq3_confidence_nlp_corr.csv`, `rq3_per_mode_ranks.csv`, and `rq3_judge_vs_confidence_nlp.csv` under `python/metrics/output/` and `python/judge/output/`.

C.4.2 Per-Mode Descriptive Means ($n = 102$)

Measure	Form ($n = 34$)	Ask ($n = 33$)	Chat ($n = 35$)
Item 5_{ai} (legal-basis confidence, 1–5)	3.38	3.67	3.77
Perceived Confidence ($\overline{4{+}5}$, 1–5)	3.34	3.76	3.77
`AI_support` mean (items 1–6, 1–5)	3.93	3.93	3.88
BERTScore F1	0.9059	0.8955	0.9036
SBERT (ModernBERT)	0.9866	0.9857	0.9895
Judge legal correctness (1–5)	3.59	3.58	3.23
Judge overall (1–5)	2.38	2.55	2.69

Table C.9: Per-mode means for the confidence, NLP, and judge measures on the joined RQ3 sample.

C.4.3 Pooled Spearman Correlations ($n = 102$)

Primary confidence-quality correlations against the two reference-based NLP metrics. The pooled correlations are all essentially zero and none are significant.

	vs BERTScore F1		vs SBERT	
Confidence measure	ρ	p	ρ	p
Item 5_{ai}	-0.013	0.895	$+0.004$	0.964
Perceived Confidence ($\overline{4{+}5}$)	$+0.037$	0.715	$+0.062$	0.538
`AI_support` mean	$+0.066$	0.510	-0.001	0.995

Table C.10: Pooled Spearman rank correlations between confidence and reference-based NLP metrics ($n = 102$).

C.4.4 Judge Cross-Check ($n = 102$)

The reference-free LLM judge enters as a third lens. Its overall score correlates strongly with BERTScore F1 and moderately with SBERT, confirming it tracks the reference-based picture on holistic quality. The judge's legal-correctness rubric provides a dimension the reference-based metrics structurally cannot score.

Comparison	ρ	p
BERTScore F1 vs judge overall	$+0.566$	$< 10^{-9}$
SBERT vs judge overall	$+0.328$	0.0008
Item 5_{ai} vs judge legal correctness	$+0.163$	0.10
Item 5_{ai} vs judge overall	$+0.143$	0.15
Perceived Confidence ($\overline{4+5}$) vs judge legal corr.	$+0.201$	0.042
Perceived Confidence ($\overline{4+5}$) vs judge overall	$+0.230$	0.020
`AI_support` vs judge overall	$+0.212$	0.03

Table C.11: Pooled Spearman rank correlations involving the LLM judge ($n = 102$).

The confidence-correctness inversion identified in Chapter 6 survives this triangulation at the mode-rank level: Chat is tied at the top with Ask on the perceived confidence composite and uniquely highest on item 5_{ai}, yet lowest on judge legal correctness; Form is lowest on both confidence measures and tied with Ask for highest on judge legal correctness.

C.5 Timing Inferential Tests

Inferential tests on the per-mode completion time recorded in the `Time elapsed` line emitted by ROPAgen at document submission (one value per participant–mode pairing). Sample sizes match the NLP document-level analysis: full $n = 113$ (Form 37, Ask 37, Chat 39); matched-triple within-subjects subset $n = 26$ participants. Non-parametric tests are used throughout, consistent with the NLP and user-study pipelines, as Shapiro–Wilk rejects normality on every mode (Form $W = 0.780$, $p < 0.001$; Ask $W = 0.912$, $p = 0.030$; Chat $W = 0.601$, $p < 0.001$). Source: `python/metrics/output/statistical_tests.md` (*Per-Mode Completion Time* section).

C.5.1 Kruskal–Wallis Omnibus ($n = 113$)

Measure	H	p	Sig ($\alpha = 0.05$)	ε^2
Completion time (s)	16.455	< 0.001	Yes	0.147 (large)

Table C.12: Kruskal–Wallis H test on per-mode completion time across the unbalanced full sample of $n = 113$ documents.

C.5.2 Pairwise Mann–Whitney U ($n = 113$)

Pairwise Mann–Whitney U tests on the unbalanced full sample, serving as the post-hoc family where the Kruskal–Wallis omnibus is significant. Bonferroni-corrected $\alpha = 0.0167$.

Pair	U	p	r	Median A (s)	Median B (s)	Sig (Bonf)
Form vs Ask	334.0	< 0.001	$+0.512$ large	420.0	858.0	**Yes**
Form vs Chat	420.0	0.002	$+0.418$ medium	420.0	717.0	**Yes**
Ask vs Chat	793.5	0.457	-0.100 negligible	858.0	717.0	No

Table C.13: Pairwise Mann–Whitney U post-hoc tests on per-mode completion time on the unbalanced full sample of $n = 113$ documents. Effect-size r is positive when the first-named mode is *slower*.

Form is significantly faster than both Ask and Chat (large and medium effects); Ask and Chat are statistically indistinguishable from one another.

D Acronyms

This appendix lists the acronyms used throughout the thesis, together with their expansions, for ease of reference.

Acronym	Expansion
API	Application Programming Interface
BDSG	Bundesdatenschutzgesetz (German Federal Data Protection Act)
BERTScore	Bidirectional Encoder Representations from Transformers Score
BLEU	Bilingual Evaluation Understudy
DBIS	Institute of Databases and Information Systems (University of Ulm)
DPO	Data Protection Officer
DSGVO	Datenschutz-Grundverordnung (German term for GDPR)
EU	European Union
GDPR	General Data Protection Regulation
HGB	Handelsgesetzbuch (German Commercial Code)
IfSG	Infektionsschutzgesetz (German Infection Protection Act)
LCS	Longest Common Subsequence
LLM	Large Language Model
METEOR	Metric for Evaluation of Translation with Explicit ORdering
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
ROPA	Record of Processing Activities
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
R-TLX	Raw NASA Task Load Index
RQ	Research Question
SBERT	Sentence-BERT
SD	Standard Deviation
SUS	System Usability Scale
UX	User Experience

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Declaration

I hereby declare that I have written this thesis independently and have used no sources or aids other than those indicated.

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